

Installation and Configuration Guide for SAP RDP Integration Accelerator

Contents

SUPPORT AND MAINTENANCE LIFECYCLE **5**
PURPOSE OF THIS DOCUMENT **5**
HIGH LEVEL SCOPE AND APPROACH **5**
TECHNICAL PREREQUISITE FOR INSTALLING THE PACAKGE **5**
SECURITY **6**
COMPONENT FOR CUSTOMER INCIDENTS **6**
USER ASSISTANCE - SAP RESPONSIBLE DESIGN AND PRODUCTION **6**
What's New in RDP Integration Accelerator V6 ..... **7**
What's New in RDP Integration Accelerator V7 ..... **8**
INSTALLATION OF TRANSPORT REQUESTS ON SYSTEM **12**
STMS to import transport files 12
Copy provided files 13
Upload Transport Files 13
Install Transport Requests 15
CONTENT AND PACKAGE HIEARCHY **17**
STEPS AND PREREQUISITES FOR CONFIGURATIONS OF INTEGRATION ACCELERATOR **18**
Choosing the API for Data Replication (RDP API vs. SDF API) 19
Prerequisites 21
RFC Connections for RDP replication reports 21
RFC Connections for RDP Packaging Data reports 25
RFC Connections for RDP Packaging fee reports 26
Configuration settings for services 26
Maintain the path prefix and package size 27
Path prefix list for RDP API: 28
Path prefix list for SDF API: 29
Maintain Application Parameters 31
MATERIAL_TYPE 31
MAX_LINES_COUNT 31
SOURCE_ID 31
DB_CONNECTION (Database connection) 32
ENABLE_TRACE 33
DLV_CATCH_WEIGHT_ON 33
Maintain Email Notification Recipients - optional 34
Create and activate Change Pointers 34
CONTENT OF RDP INTEGRATION PACKAGE **36**
HOW TO REPLICATE DATA TO RDP **37**
RDP ERP Integration reports 38
General Approach for Loading Data 38
Reports with Specific Load Functionality 39
Loading Configuration data 40
Loading Incoterms 41

Loading Organization data 42
Loading Batch data 42
Loading Supplier data 43
Loading Customer and Customer Extension data 45
Loading Product and Product extension data 46
Loading Deliveries 48
Loading Material documents 50
Loading Billing Document Items and Supplier Invoice Items 55
Loading Inventory 58
Extensibility for RDP ERP Integration reports 60
Sample Implementation 63
Badi Implementation Guidelines 64
Product and Customer extensions 66
Email notification adaptation 71
Excel documents processing adaptation 71
Resending failed payloads 72
Trace Maintenance Report 73
Report for RunId maintenance 75
RDP Packaging Data Integration reports 75
Creating recommended classes and characteristics 76
Upload Packaging Elements data from PLM 77
Class overview 78
Extensibility 78
Last change information 79
Upload Packaging Compositions from PLM 79
Class overview 79
Access the Specifications data 80
Upload Packaging Compositions from Recipe Management 80
Class overview 80
Access the Recipe Management data 81
Uploading Packaging Compositions from Bills of Material 81
Report parameters: 81
Class Overview 84
Uploading Packaging elements from Material Classifications 85
Report parameters 85
Class Overview 87
Uploading Packaging Composition Items from Handling Units 87
Report Parameters 89
Class Overview 91
Extensibility in packaging data reports 91

VPCOE/BADI_UPH_CUSTOM – technical documentation 92
Common class overview 95
Messages 95
API Version Selection for Packaging Data Integration 95
Packaging fee replication and posting 96
Packaging fee retrieval 97
Packaging fee posting 99
Replication of Customer Exemptions 100
Replication of CompanyCode Exemptions 100
Replication of Customer Roles 100
Extensibility in Packaging fee reports 100
/VPCOE/BADI_PCKF_CUSTOM – technical documentation 101
HOW TO CHECK REPLICATION RESULTS 102
Results on the screen 102
Application log results 103
KNOWN TECHNICAL ISSUES AND TROUBLESHOOTING 104
Troubleshooting Guide – RDP Integration Accelerator 104
Known Technical Issues 107

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SUPPORT AND MAINTENANCE LIFECYCLE

Important Note: RDP Integration Accelerator Support and Maintenance

Please consider the following support and maintenance timelines when planning your use of the RDP Integration Accelerator:

- Extended technical support: Technical support for the Integration Accelerator, including ticket support, will remain available through December 2027.
- End of feature development: No new RDP features or code enhancements will be added to the Integration Accelerator content after December 2026.
- End of support and maintenance: From January 2028, SAP will no longer support or maintain the Integration Accelerator content. Customers who continue using the solution beyond this date will be responsible for maintaining and updating the integration content themselves.

PURPOSE OF THIS DOCUMENT

This document provides details of the installation and configuration steps for the Integration accelerator consulting package for RDP integration into ECC EHP7 or S/4 HANA 1909/2021. This document is not a replacement for standard RDP product scope. For detailed product information, please refer to the standard help documentation of SAP Responsible Design and Production.

If you are interested in the details of scope of the Integration Accelerator package, please refer to the Detailed Scope Document which comes with this package.

HIGH LEVEL SCOPE AND APPROACH

The SAP Responsible Design and Production Integration Accelerator aims to speed-up the implementation of SAP RDP with ECC 6.0 EhP7 or with S/4 HANA 1909/2021.

The offering provides an ABAP based integration package, which comes with prebuilt integration content, which can be deployed to act as a fast-starting point for customers to integrate their RDP solution to their backend SAP system. The integration content helps in selecting, extracting and pushing the relevant data from the backend tables into the cloud based RDP solution via the RDP push API interfaces.

This document describes the steps for installing and configuring the RDP Integration Accelerator Package.

TECHNICAL PREREQUISITE FOR INSTALLING THE PACAKGE

The below technical software component is the minimum version prerequisite for successfully installing and using the accelerator package:

SAP ERP	ECC 6.0 EHP 7
SAP_BASIS	740 Package Level 0019

Below is the minimum version for PLM software component required for using the PLM package:

SPLMWUI	747 SP Level 0016 Support Package: SAPK-74716INPLMWUI

In case Packaging Composition from HU report is used from a logistics system, please consider that RDP Integration Accelerator can be deployed on customer logistics system if it is a NetWeaver systems with ABAP application server.

SECURITY

All replication endpoints are secured with TLS (https) to ensure encryption of the customer data being sent.

Authorization is using XSUAA and the OAuth 2.0 client credentials grant.

Using the RDP service broker on BTP, customers can create a service instance that allows to get an access token for replication.

SSL certificates/handshake to build the trust between the systems is a prerequisite for data replication.

COMPONENT FOR CUSTOMER INCIDENTS

For any technical assistance for RDP Integration Package, you can create support tickets accessing SAP ONE Support Launchpad using Ticket Component: SV-COE-VPT-RDP.

USER ASSISTANCE - SAP RESPONSIBLE DESIGN AND PRODUCTION

For the user assistance for SAP Responsible Design and Production, see SAP Responsible Design and Production - SAP Help Portal.

SAP Responsible Design and Production provides public API for data replication. To be able to push data from your system you need to configure the replication in SAP Responsible Design and Production. Please refer to this link to get more information on the steps to be performed.

To access the RDP API documentation, go to API Reference | Replication | SAP API Business Hub (ondemand.com) .

For more information on the API elements, see API for data Replication and API References

SAP API Business HUB RDP Overview page: Overview | Replication | SAP API Business Hub

Note:

To get started with SAP Responsible Design and Production use the standard procedures for the SAP BTP, Cloud Foundry environment. For an overview of the required steps, see Getting Started in the Cloud Foundry Environment.

The following prerequisites must be fulfilled

- An SAP BTP global account
- Members are added to the SAP BTP account
- An SAP BTP Cloud Foundry subaccount is created and enabled

Subscribing to Services

Subscribe to the following services in the SAP BTP cockpit.

- SAP Responsible Design and Production

For instruction, see Subscribe to Multitenant Applications Using the Cockpit.

For more information on the steps to be performed on BTP global account for setting up the SAP

What's New in RDP Integration Accelerator V6

The **V6 release** of the SAP RDP Integration Accelerator introduces enhancements to improve data replication, performance, and additional attributes added or removed. This version includes new features and refinements across multiple integration areas:

1. Customer Data Enhancements

- Updated sample **BAdI implementation** and **Excel upload functionality** to filter tax numbers based on SAP-defined tax categories, eliminating unsupported entries.
- **New selection filter:** Added **Company Code** as a filtering criteria.

2. Supplier Data Enhancements

- **New selection filter:** Added **Company Code** as a filtering criteria.

3. Product Data Replication Enhancements

- Introduced additional attributes for **Quantity Dimensions**, including:
  - **Net Weight**
  - **Gross Weight**
  - **Weight Unit of Measure**

4. Batch Replication Enhancements

- Implemented additional logic to determine **packaging composition** in batch replication.

5. Delivery Data replication Enhancements

- Performance optimizations for faster and more efficient data replication.
- Support for **Delivery Batch Split** scenarios.
- Handling of **Catch Weight Material** scenarios.

7. Packaging Composition Enhancements

- Added support for ending validity of a product in a packaging composition - **Valid To Date assignments**. Sample implementation is offered in the sample BAdI implementation.
- Improved handling of material number **selection criteria restrictions** to enhance flexibility.
- **New exclusion rule:** Data with a **deletion flag** will now be excluded from replication.

8. Packaging Elements Enhancements

- **New exclusion rule:** Data with a **deletion flag** will now be excluded from replication.
- Introduction of **Reusability** functionality (isReusable API attribute)

9. Packaging Data V2 APIs

- Enhanced support for **Version 2 RDP APIs** for both Packaging elements and Packaging compositions.
- Customers can choose if they want to use V2 or V1 of APIs, by adjusting the services paths
- Sample BAdI implementations are available for both V1 and V2 of the APIs

10. Packaging Fee Replication Enhancements

- **New Sample BAdI Implementation** for **Packaging Fee Replication for CONAI**.
    - Support for retrieving **Customer Exemptions** and **Company Code Exemptions**.
- **Customer-Dependent Pricing Condition Replication**:
    - **New parameter: Customer Role Dependent Fees** included in the packaging fee retrieval report.
    - Retrieval of **Customer Role Replication**.

11. **New functionality: RDP Classes & Characteristic Generation**

- A new report to create recommended **RDP classes and characteristics**, supporting:
    - Packaging Elements in Material Classification
    - Packaging Compositions in BOM, classifying Pckg. Elmnts. into Main/Subordinate
    - Packaging Compositions and Main/Subordinate Packaging Elements in PLM SpecDB
    - Packaging Fee Data Classification
    - Customer Extension Data Classification

What's New in RDP Integration Accelerator V7

1. The **V7 release** of the SAP RDP Integration Accelerator introduces two alternative APIs for replicating sustainability-related data:

   ▶ RDP Push API

   This API continues to support:

   - Master and transactional data replication
   - Packaging composition and packaging fee replication
   - All existing integration scenarios covered in previous accelerator versions

   Note:
   If the customer chooses to continue using the RDP Push API, **no configuration changes** and **no BAdI implementation adjustments** are required.

   ▶ SAP Sustainability Solutions (SDF API) - New

A new API provided through the SAP Sustainability Data Foundation (SDF) portfolio: https://api.sap.com/package/SAPSustainabilitySolutions/

The new API introduces **Replication RunId** to manage and track replication sessions.

This enhancement allows customers to choose which API to use for data replication per entity type.

Note: replication of packaging fees is not supported as of now in SDF API.

All new attributes will be supported/introduced only in SDF API.

⚠️ **Important note:**

Switching from the RDP API to the SDF API **may require adjustments** in customer BAdI implementations, because:

- Attribute names may differ
- Payload structures may differ
- Mappings for entities may need to be revised in badi implementations

Before switching APIs, customers should **carefully analyze their custom BAdI implementations** and adjust mappings where necessary.

Configuration Requirements for Using the SDF API

Before you use the SDF API, make the following configuration:

1. In the SAP BTP cockpit, subscribe to the *sustainability data integration service* in your subaccount for the SAP Sustainability solution. You need to create a service instance with an *integration* plan, and then create a service key to get credentials for the API.

   More details on this step: Initial Setup | SAP Help Portal
2. In the SAP RDP solution, define the settings for push replication:

   1. Define source systems with a unique identifier and of type *SAP Sustainability Solutions APIs*.
   2. When you create replications for all the objects you want to import using the API, assign the relevant source system, give the replication a unique name, and activate the *Push Replication* option.

How the API Selection Works in the Accelerator

API usage is controlled via:

- **RFC Destination** VPCOE_RDP_GENERIC
- **Path Prefix** maintained in accelerator customizing

Customers can configure path prefix:

- /api/replication/<entity_name>/v1/... → to use the RDP Push API

- /api/<entity_name>/v1/<entity_group>... → to use the SDF API

The accelerator will route the replication to the chosen API.

You can find here more details on choosing the API for data replication.

2. **New Report: Close Replication Sessions / Run Id**

   Report: /VPCOE/SUMA_REPLICATION_MAINT

   The R*eplication Run Id* uniquely identifies the scope of a replication run: the source system, the object to be replicated, and the target solution.

   When a replication encounters errors, the replication RunId it has not been properly finished/closed.
   If a session is still open, it prevents further replications for the same entity, resulting in:
   - Stuck replications
   - Inability to replicate initial load or delta updates
   - Data inconsistency risks

   To address this, Release V7 introduces a new administration report for Replication Session Maintenance / RunId, to allow customers to:
   - View active replication sessions
   - Identify sessions opened / blocked
   - **Manually close a replication session** to unlock replication for that entity

What's new in release V7.1

⚠ **Important Note:** The new PPWR Compliance Requirements attributes introduced in this release are supported only when using the SAP Sustainability Solutions API (SDF API). These attributes are not supported by the RDP Push API.

Customers using the **RDP API** might not see these attributes replicated.

Customers using custom BAdI implementations should review and adjust their mapping logic accordingly. Sample badi implementation is provided including for

Release V7.1 introduce compliance requirements attributes in following reports:

1. Product replication report
   The **Product replication report** has been enhanced to support additional PPWR-related attributes.

   New Product Attributes
   - mfrPartNumber
   - mfrNumber

New Subnode: globalTradeItemNumbers
The following attributes are now supported under the new subentity:

- unitOfMeasure
- isMain
- globalTradeItemNumber

These attributes allow replication of additional product identification data such as manufacturer identifiers and GTIN-related information.

2. Material Document report

Two additional variants has been introduced starting V7.1.3. for statistical goods receipts:

Group 6: Statistical Goods Receipt with Purchase Order reference imported in specific country

All statistical goods receipts with a purchase order reference imported in the selected country will be replicated. Statistical goods receipts are used for third-party drop shipment scenarios and do not result in a physical stock movement.

Group 7: Statistical Goods Receipt with Purchase Order reference in specific country (domestic)

All statistical goods receipts with a purchase order reference within the same country will be replicated. Statistical goods receipts are used for domestic third-party drop shipment scenarios and do not result in a physical stock movement.

3. Packaging data reports

The following packaging data replication reports have been enhanced:

- Packaging Composition from BOM
- Packaging Element from Material Classification
- Packaging Composition from PLM / Spec DB
- Packaging Element from PLM

New Attribute

- hasProductContact (Boolean)

New Subnode: complianceRequirements
The following attributes are supported under the compliance requirements node:

- requirementCode
- complianceStatusCode
- validFrom
- validTo

This enhancement enables replication of compliance-related PPWR data for packaging compositions and packaging elements.

4. The report for generating recommended classes and characteristics has been enhanced to support Compliance Requirements data.

Packaging Composition in BOM

New Class

- ZRDP_PACKCOMP_CR

New Characteristics

- ZRDP_PACKCOMP_CR_CODE
- ZRDP_PACKCOMP_CR_STATUS
- ZRDP_PACKCOMP_CR_VALID_FROM
- ZRDP_PACKCOMP_CR_VALID_TO

Packaging Elements in Material Classification

New Classes

- ZMCL_PACKELEM_CR01
- ZMCL_PACKELEM_CR02
- ZMCL_PACKELEM_CR03
- ZMCL_PACKELEM_CR04
- ZMCL_PACKELEM_CR05

New Characteristics

For each class (CR01 to CR05), a dedicated set of characteristics is generated using the same naming pattern.

Example for CR01:

- ZRDP_PACKELEM_CR01_CODE
- ZRDP_PACKELEM_CR01_STATUS
- ZRDP_PACKELEM_CR01_VALID_FROM
- ZRDP_PACKELEM_CR01_VALID_TO

The same structure applies for:

- CR02
- CR03
- CR04
- CR05

This enables customers to maintain multiple compliance requirement records per packaging element.

INSTALLATION OF TRANSPORT REQUESTS ON SYSTEM

Prerequisites:
Authorizations for the following transaction codes:

- CG3Z in client 000 or any other needed client to upload transport files (or authorization to execute Function Module ARCHIVFILE_CLIENT_TO_SERVER)
- STMS

STMS to import transport files

As a first step, the provided packages must be imported into your ECC or S/4 HANA system. Follow the below steps to import the necessary RDP Integration Accelerator packages.

Copy provided files

For installing RDP development package on the backend ECC or S/4 HANA system it is necessary to copy the provided files on the system on which SAP is installed. (Ex. ***K****900446.BP7 &* ***R****900446.BP7*). Copy all the files we have provided to you, for General objects, RDP and/or PLM objects.

Please note the following sequence to install the transport requests (if otherwise stated in ReadMe.txt file when provided):

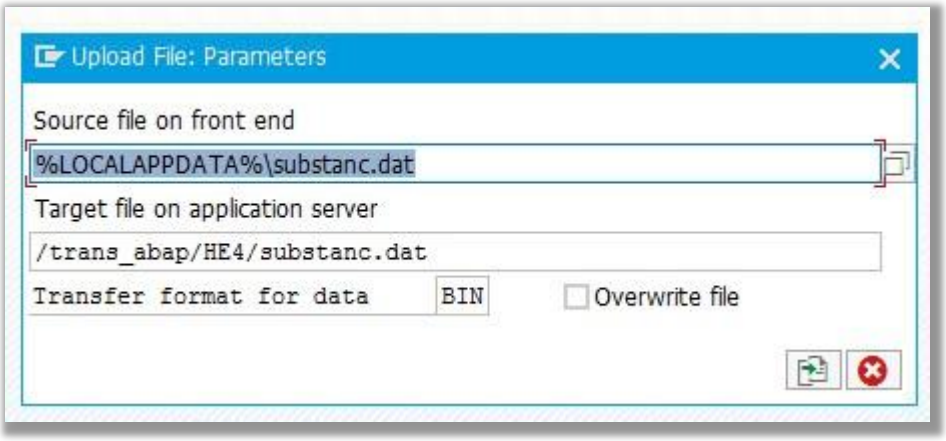
1. TR for General Objects
2. TR for RDP Objects
3. TR for PLM Objects

Please read the file ReadMe.txt shared together with the TRs files, explaining the TRs shared and sequence to be installed in more particular way.

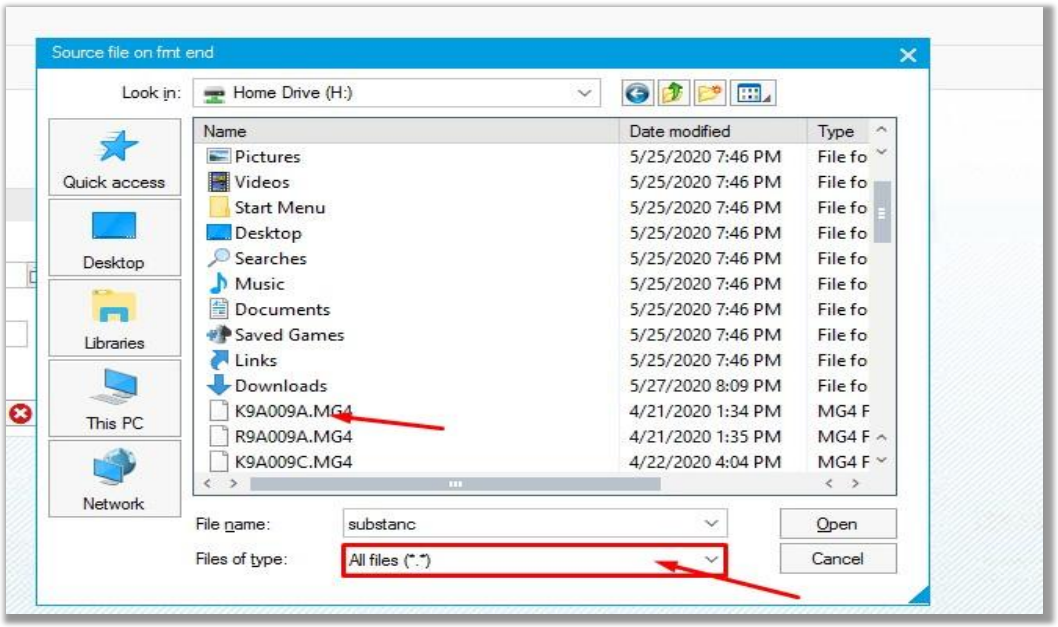
Note: the screenshots for uploading and installing the TRs are using sample TRs and are for your reference only.

Upload Transport Files

In your system, open necessary client and upload transport files on your system server. Open transaction '*CG3Z*' (If transaction doesn't exist on the system see next step).



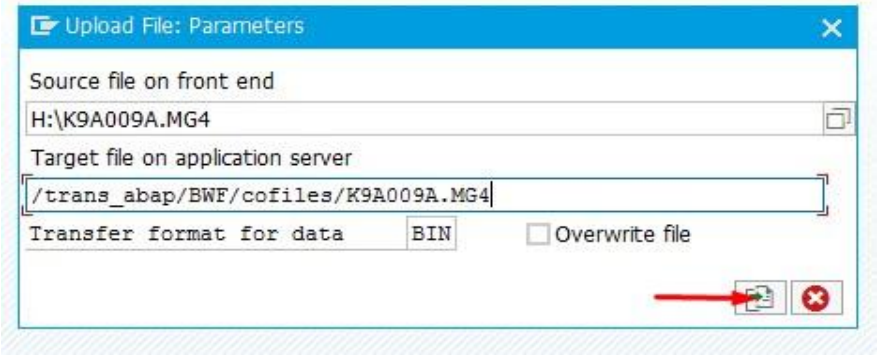
Choose transport files provided to you:



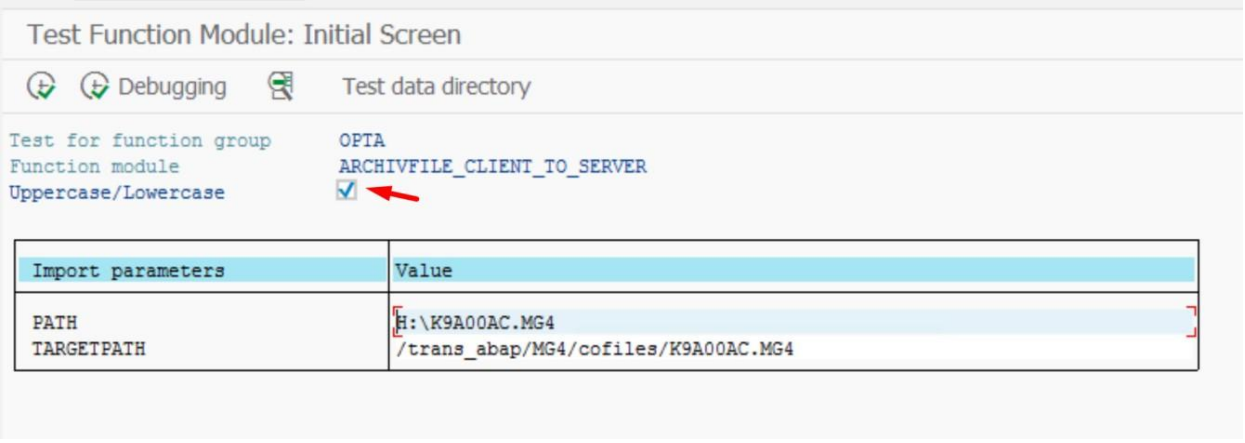
Enter details on source file and target file on your application server:

1. /trans_abap/{system}/cofiles/{name transport file} (for file having first later ‘K’)
2. /trans_abap/{system}/data/{name transport file} (for file having first later ‘R’)

Note: For details on path configured on your application server, use transaction ‘AL11’.



Press upload. (Ex. upload cofiles on BWF system).



If ‘*cg3z’* transaction doesn’t exist on your system, please follow below step:

- Use Function Group: **OPTA**, function module: **ARCHIVFILE_CLIENT_TO_SERVER (execute FM using transaction code SE37)**
  Add the path where source files will be saved, and also the server path (target file on application server). Use Uppercase/Lowercase flag.
- Enter details on source file and target file on application server:
  1. /trans_abap/{system}/cofiles/{name transport file} (for the file having first later 'K')
  2. /trans_abap/{system}/data/{name transport file} (for the file having first later 'R')
- Execute above steps for all provided files, for general objects and for RDP and PLM objects.

  Note: For details on path configured on your application server, use transaction 'AL11'.

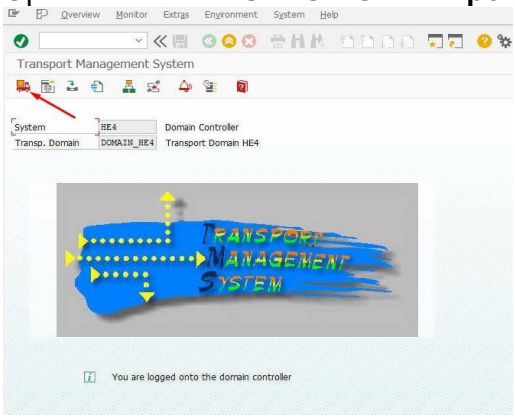
Install Transport Requests

Please follow below steps to install all transport requests provided.

Note: Please note the order to install the transport request (check also the ReadMe.txt file shared to you together with TRs, for any particular cases):

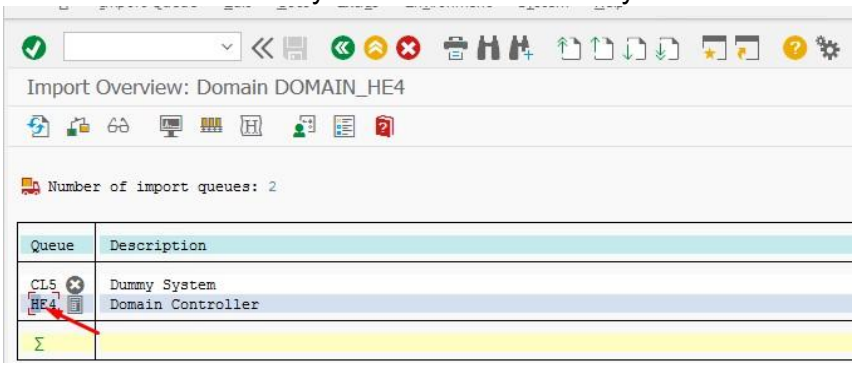
1. TR for General Objects (mandatory pre-requisite for all other packages)
2. TR for RDP Objects
3. TR for PLM Objects (only needed if you are integrating SAP PLM to RDP)

1. Open transaction '*STMS*'. Click '**Import Overview** (F5)'

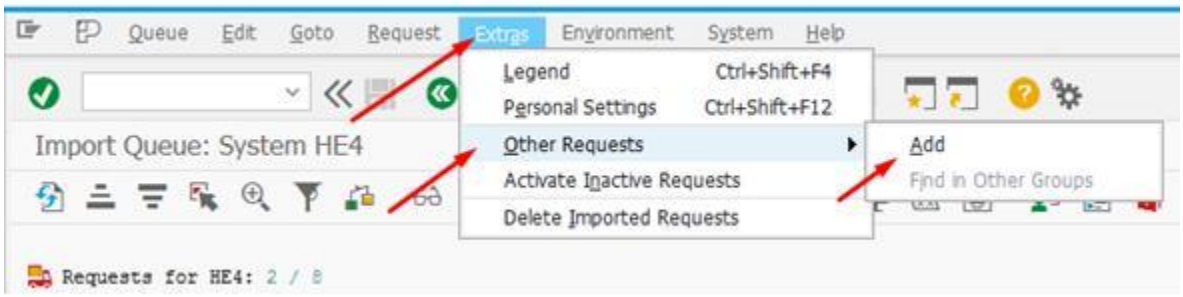


s

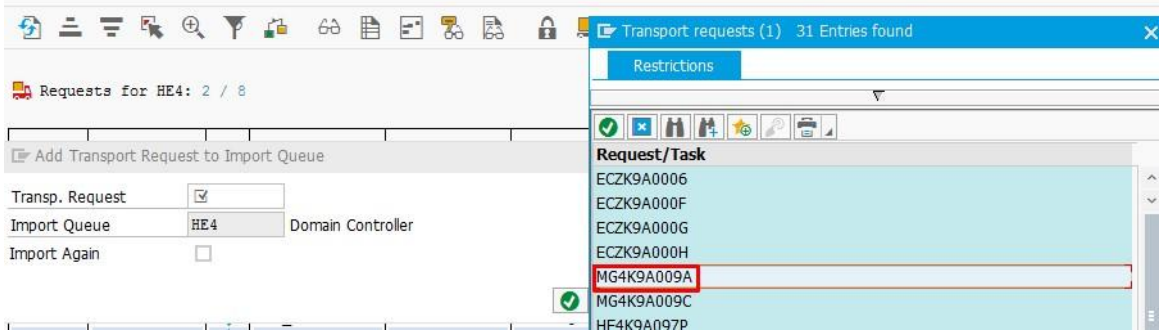
2. Double-click on the entry with the name of the system.

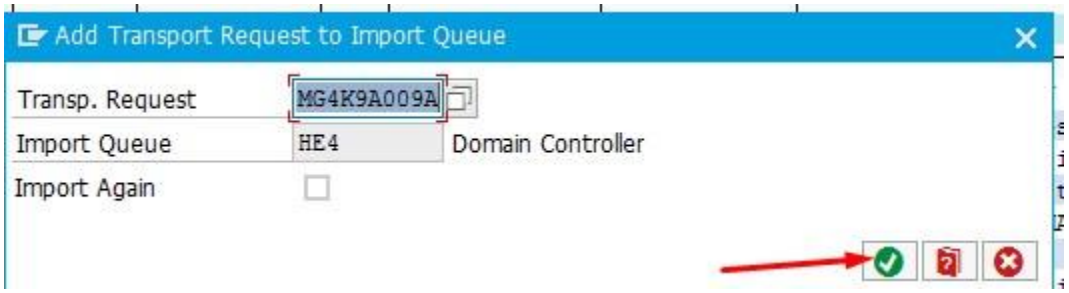


3. Add new transport request on the queue.

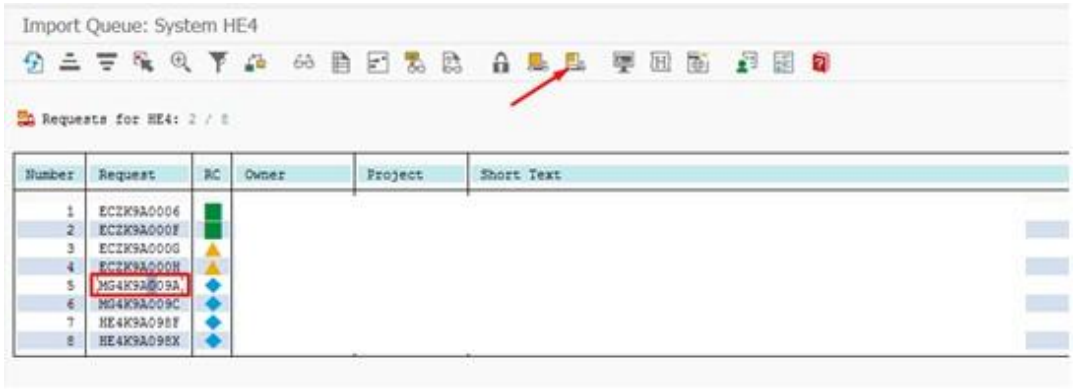


4. Choose necessary transport request (since ‘K9A009A.MG4‘ file has been uploaded, ‘MG4K9A009A’ entry should be selected).

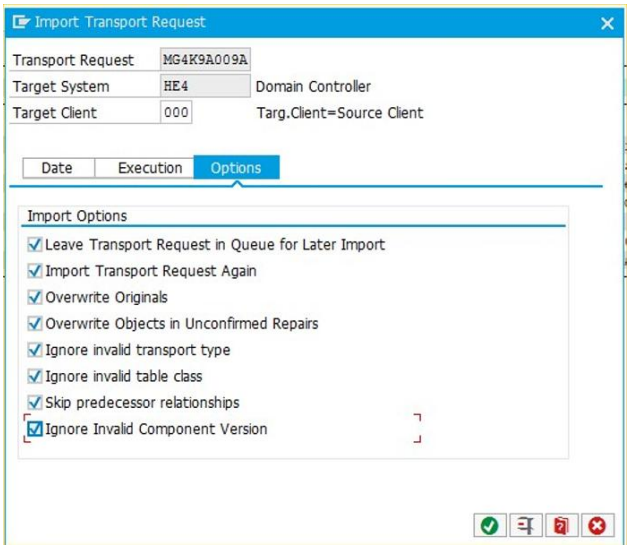




5. Choose the entry with the name of the transport request and click ‘Import request’:



6. Enter target client and select all check boxes and press Continue. On the next screen click ‘Yes’:



CONTENT AND PACKAGE HIEARCHY

The package provides the relevant code to select the required data from SAP backend system and replicating it to the RDP tenant, using standard HTTP endpoints.

The package includes the following integration components:

ERP Integration Reports

- **Organizational Data** – Replication of company structures, plants, sales organizations, distribution channels, and divisions.
- **Configuration Data** – Replication of unit of measure definitions, currency details, product hierarchies and other configuration elements.
- **Master Data Replication**
    - Supplier
    - Customer
    - Product
    - Batches

Transactional Data Replication

- **Delivery Documents** including the **Billing Document Items**
- **Material Movements** including the **Billing Document Items & Supplier Invoice Items**
- **Inventory**
- **Incoterms**

Packaging Data Integration

- **Packaging Composition Replication**
    - Extracted from PLM, Recipe Development, and Bill of Materials (BOM).
- **Packaging Elements Replication**
    - Retrieved from PLM and Material Classification.
- **Partial Packaging Composition Item Replication**

- Handling Units replication.

- **Creation of RDP recommended classes and characteristics**
  - create example classes and characteristics for storing packaging data in your SAP ECC / S/4HANA system based on Material Classification or Specification Database.

Packaging Fee Replication

- **Synchronization of Packaging Fees** – Replication from RDP back to ECC/S4H for creating pricing conditions and compliance reporting.
- **Customer Role Replication** – Replication of customer-specific role attributes from RDP to ECC/S4H for creating customer dependent/independent pricing conditions and material classifications
- **Customer Exemptions and Company Code Exemptions** – Replication of customer-specific and company code-specific exemptions from RDP to ECC/S4H for creating customer dependent/independent pricing conditions and material classifications.

Analytical and Configuration Reports

- **Report for Generating Recommended RDP Classes and Characteristics**
  - Packaging Elements in Material Classification
  - Packaging Compositions in BOM, classifying Pckg. Elmnts. into Main/Subordinate
  - Packaging Compositions and Main/Subordinate Packaging Elements in PLM SpecDB
  - Packaging fee data classification
  - Customer extension data classification

Error Handling and Logging

- **Error Monitoring and Troubleshooting Reports** – Comprehensive logging and troubleshooting capabilities for monitoring replication status and resolving data inconsistencies.

Package hierarchy and structures:

/VPCOE/CLOUD_INTEGRATION is the base package containing following sub-packages:

/VPCOE/COMMON_INTEGRATION containing common Integration objects

/VPCOE/RDP_INTEGRATION containing *RDP ERP* Integration objects/reports

/VPCOE/PLM_INTEGRATION containing RDP Packaging Data Integration objects/reports (only for PLM component)

/VPCOE/OBJ_INTEGRATION containing RDP Packaging Data Integration objects/reports from BOM and Material classification. It also includes the Handling Units replication report.

/VPCOE/RDP_PCKGFEE_INTEGRATION containing packaging fee Integration objects/reports and also customer role/ customer exemptions and company code exemptions replications from RDP back to ECC/S4H

STEPS AND PREREQUISITES FOR CONFIGURATIONS OF INTEGRATION ACCELERATOR

Once you have followed the above steps to import the transport packages, the next step is to configure the technical connectivity between the backend system and RDP tenant.

Before configuring or executing any replication in the accelerator, customers must decide **which API will be used to retrieve sustainability-related data**.

Choosing the API for Data Replication (RDP API vs. SDF API)

Starting with Release **V7**, the accelerator supports two alternative API providers:

- **RDP Push API** (existing; used in all previous versions)
- **SAP Sustainability Solutions API (SDF API)** (new in V7)

This chapter explains the differences between the two APIs, required setup steps, and important considerations when choosing one API or the other.

1. Available API Options

1.1 RDP Push API

This is the default and existing API used to replicate:

- Master data
- Transactional data
- Packaging composition
- Packaging fee data
- Customer role–dependent fees

Characteristics:

- Stable and widely used
- No changes required for existing customers
- No RunId lifecycle management

When to choose:

Use the RDP API if you want the same behavior as in previous accelerator releases.If your replication includes packaging fee data, which is not yet supported by SDF API, you can still choose the SDF API, and keep the RDP Push API for packaging fee replication in the accelerator configuration.

1.2 SDF API (SAP Sustainability Solutions API)

A modern API provided through the **SAP Sustainability Data Foundation (SDF)** portfolio: https://api.sap.com/package/SAPSustainabilitySolutions/

Characteristics:

- Uses **Replication RunId** to manage session lifecycle
- Structured for future expansion into broader sustainability topics
- For customers already using a previous accelerator version, it requires new configuration and may require updates to BAdI implementation mappings
- Does not support packaging fee replication yet

When to choose:

Use the SDF API if:

- You want future-proof integration aligned with SAP's Sustainability Solutions
- Your scenario does not currently include packaging fee replication. If it does, you can still opt for SDF API and configure RDP API only for packaging fee replication.
- You prefer the enhanced RunId-based replication control

2. What Changes When You Choose an API?

Replication behavior, data structures, and some field names **differ** depending on the API. Customers need to be aware of the following:

2.1 API Payload Structure

- The SDF API used different attribute names and JSON structures for some entities then RDP API
- Customer BAdI implementations may require adjustments
- Sample BAdI classes have been updated accordingly in V7

2.2 Replication Session Handling

- The **RDP API** does *not* require explicit session management
- The **SDF API** requires RunId lifecycle handling
- A replication session must be closed before the next run
- If a RunId remains open due to an error, replication becomes blocked
- V7 introduces report /VPCOE/SUMA_REPLICATION_MAINT to maintain (close) RunIds

2.3 API Configuration

API selection is controlled via:

- **RFC destination:** VPCOE_RDP_GENERIC
- **Path Prefix** in accelerator customizing

Examples:

- /api/replication/<entity_name>/v1/... → **RDP API**
- /api/<entity_name>/v1/... → **SDF API**

The chosen prefix determines which API will be used for the corresponding entity.

3. Required Configuration Before Using SDF API

To enable SDF-based replication, customers must complete the following steps:

3.1 In SAP BTP Cockpit

- Subscribe to the **Sustainability Data Integration** service
- Create a **service instance** (integration plan)
- Create a **service key** to obtain credentials for the SDF API

3.2 In SAP RDP Solution (for both RDP and SDF API)

- Define **source systems** of type *SAP Sustainability Solutions APIs*
- Create replications for the objects to be imported
- Assign the source system to each replication
- Activate **Push Replication** for SDF

4. Compatibility and Limitations

4.1 When Switching APIs

Switching from RDP API → SDF API requires:

- Reviewing and adjusting BAdI mappings
- Ensuring entity support (packaging fee replication not yet available in SDF)
- Understanding RunId session handling
- Updating customizing to adjust the path prefix
- Testing payload mappings based on entity-specific JSON formats

5. Summary for Users

Before replicating data, users must:

✓ Decide which API will be used (RDP or SDF)
✓ Verify that their BAdI mapping supports the selected API
✓ Ensure the correct path prefix is maintained in customizing
✓ Understand whether RunId control applies (SDF API only)
✓ Validate that the selected API supports the required entity type

Once the API choice is made, all replication reports will automatically route requests through the correct API endpoints based on the configuration.

Prerequisites

- Setting up the replication services in SAP BTP Cloud Cockpit
  Please follow these steps to set up the replication services and subscribe to RDP service
- Getting credentials of the SAP RDP service from your tenant:
  - Client Id
  - Client Secret
  - token endpoint URL
- Establish the SSL connection to SAP BTP
  - follow this link to establish connection: Establish the SSL Connection to SAP BTP - SAP Help Portal
  - Follow the instruction in note for downloading and importing BTP certificates: 2853519 - How to download the SSL certificate of SAP BTP - SAP for Me

    Overview steps:
    1. Open the RDP API gateway (the endpoint for RFC destination VPCOE_RDP_OAUTH)
    2. Choose the lock icon before the URL in the address bar.
    3. Select Connection is secure from the context menu and then Certificate is valid.
    4. Switch to the Details tab of the Certificate dialog box.
    5. Choose Export selected certificate.
       Specify pem as the certificate type and then save the certificate

  The required certificates to be imported in ERP:

  DigiCert Global Root G2

  DigiCert Global G2 TLS RSA SHA256 2020 CA1

- Authorizations for the following transactions:
  - SM59 to create an RFC destination

For more information on all the steps required to be performed on BTP Cockpit and on token endpoint URL and client credentials, see Push Data Replication.

The integration package invokes the RDP APIs via HTTPS connection from the backend system. To support this, several RFC connections has to be created on the backend system for RDP reports. The below steps have to be done to create and configure the RFC connections.

Note: For RDP Integration package there is no need to configure OAuth Client 2.0

RFC Connections for RDP replication reports

The following RFC destinations of type 3 – ABAP Connection must be created in the backend system. (Transaction code '*SM59*')

- `VPCOE_RDP_GENERIC`
- `VPCOE_RDP_OAUTH`
- `VPCOE_RDP_PLM`

- VPCOE_RDP_PLM_PKG_ELMNT

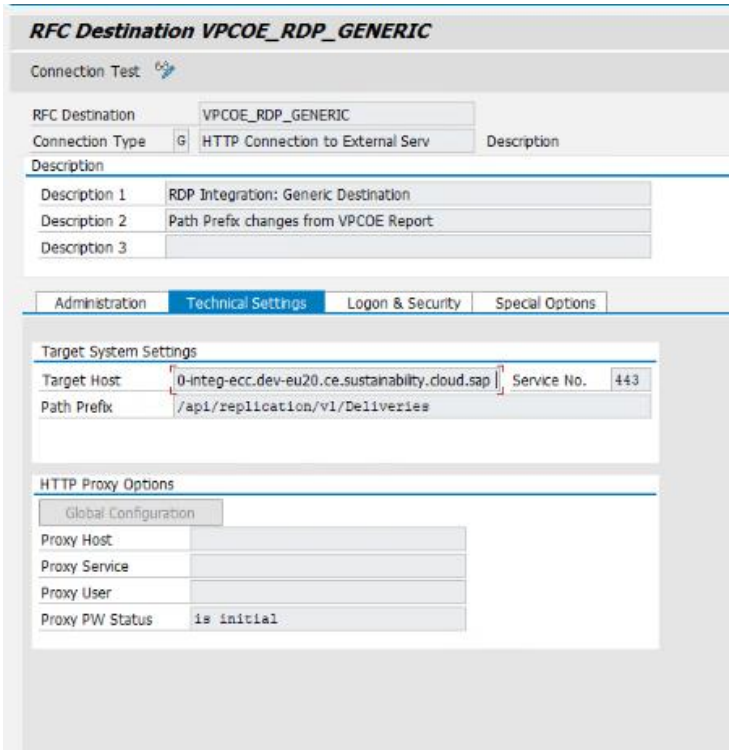
- VPCOE_RDP_PCKF

To create a new RFC destination, carry out the following steps:

1. Open transaction SM59
2. Choose *Create*.
3. Enter name VPCOE_RDP_GENERIC for the RFC destination.
4. Select connection type G (HTTP Connection to External Server).
5. On the **Technical Settings** tab, enter the following data:
    1. **Target Host**: server endpoint URL

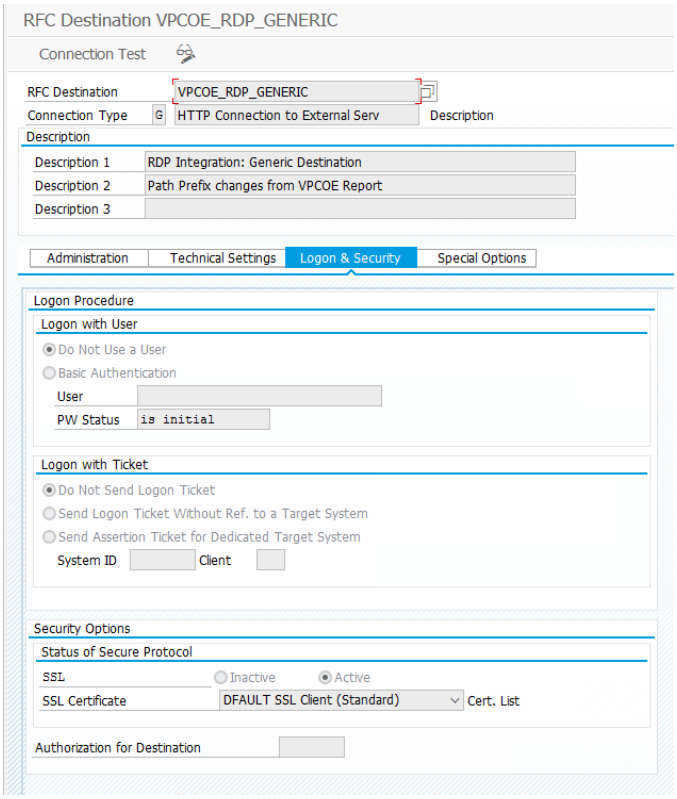
       For RDP API (e.g. {tenant-subdomain}.eu20.ce.sustainability.cloud.sap

       For SDF API e.g. {tenant-subdomain}.eu20.sdi.sustainability.cloud.sap
    2. **Port**: 443



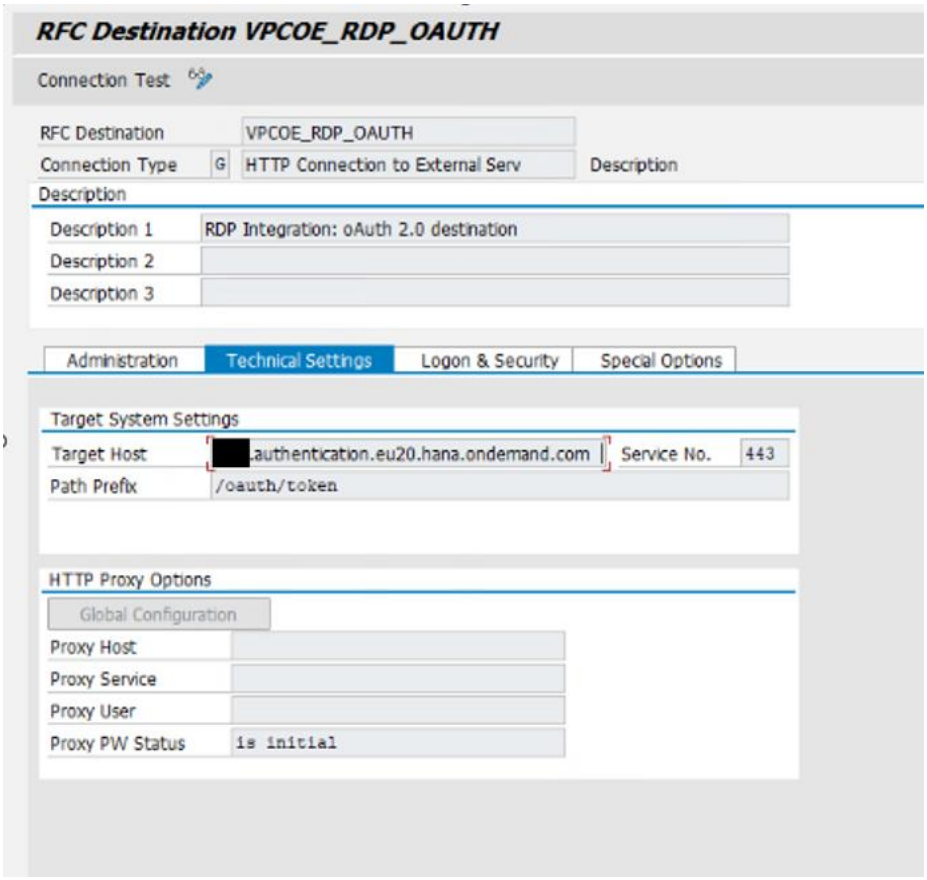
1. On the **Logon & Security** tab
    1. Set **Do not use a user**.

       Set **SSL to Active**.



Create another RFC destination for OAuth token following below steps:

1. Open transaction `SM59`
2. Choose *Create*.
3. Enter name `VPCOE_RDP_OAUTH` for the RFC destination.
4. Select connection type G (HTTP Connection to External Server).
5. On the **Technical Settings** tab, enter the following data:
    1. **Target Host**: `authentication endpoint URL (ex:` {tenant-subdomain}.authentication.eu20.hana.ondemand.com)
    2. **Path prefix**: `/oauth/token`
    3. **Port**: `443`

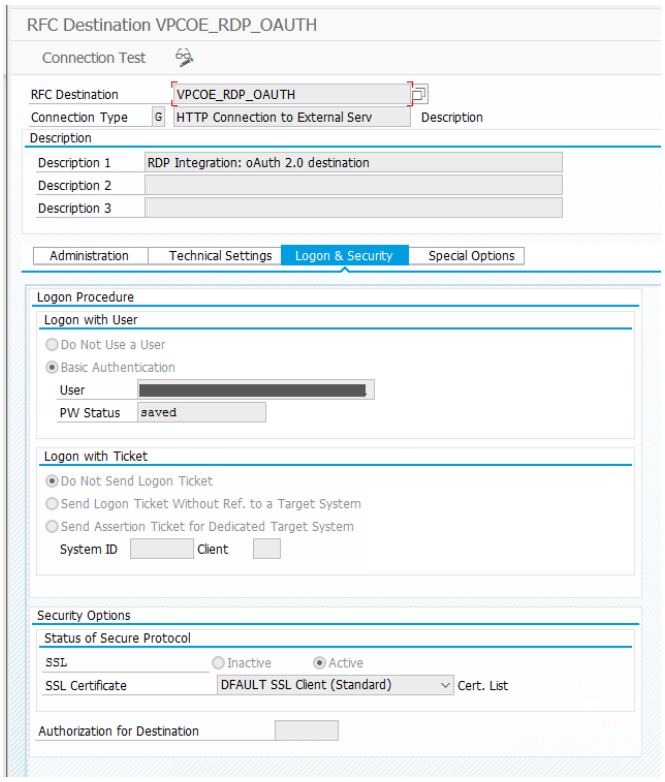


6. On the **Logon & Security** tab

    1. Set **Basic Authentication**

    2. Enter **username** and **password (**client id and client secret from RDP or SDF instance key)

    3. Set **SSL to Active** and choose SSL certificate to be used.



7. Save

Note: The token URL, username, and password required for authentication can be retrieved from the SAP Business Transformation Platform (BTP) Cockpit, as detailed in the official documentation.

When testing the RFC connection **VPCOE_RDP_OAUTH**, you may encounter the error message: `{"error":"invalid_request","error_description":"Missing grant type"}`
This message can be disregarded, as the RDP Integration package handles the grant type

RFC Connections for RDP Packaging Data reports

Create a new RFC destination, carry out the following steps:

1. Open transaction `SM59`
2. Choose *Create*.
3. Enter name `VPCOE_RDP_PLM` for the RFC destination.
4. Select connection type G (HTTP Connection to External Server).
5. On the **Technical Settings** tab, enter the following data:
    1. **Target Host**: `token endpoint URL`

       For RDP API `(e.g.` tenant-subdomain}.eu20.ce.sustainability.cloud.sap

       For SDF API `e.g.` {tenant-subdomain}.eu20.sdi.sustainability.cloud.sap
    2. **Path Prefix:**
        1. **For RDP API:** `/api/replication/v2/PackagingCompositions`
        2. **For SDF API:** `/api/packaging/v1/packagingCompositions`
    3. **Port**: `443`
6. On the **Logon & Security** tab
    1. Set **Do not use a user**.
    2. Set **SSL to Active** and choose SSL certificate to be used.
7. Save

Create another RFC destination following the same steps above:

1. Enter name `VPCOE_RDP_PLM_PKG_ELMNT` for the RFC destination
2. On the **Technical Settings** tab, enter the following data:

1. **Target Host**: `token endpoint URL (e.g.` {tenant-subdomain}.eu20.ce.sustainability.cloud.sap)
2. **Path Prefix:**
    1. **For RDP API:** `/api/replication/v2/PackagingElements`
    2. **For SDF API:** `/api/packaging/v1/packagingElements`
3. **Port**: `443`

3. Save

Note: the name of RFC destinations created for Packaging Data reports can differ, as the RFC destination is a selection parameter that has to be specified when executing the reports.

For more information on creating the RFC connections for Packaging Data Integration, please refer to this link.

RFC Connections for RDP Packaging fee reports

Create a new RFC destination, carry out the following steps:

8. Open transaction `SM59`
9. Choose *Create*.
10. Enter the name `VPCOE_RDP_PCKF` for the RFC destination.
11. Select connection type G (HTTP Connection to External Server).
12. On the **Technical Settings** tab, enter the following data:
    1. **Target Host**: `token endpoint URL`

    For RDP API `(e.g.` {tenant-subdomain}.eu20.ce.sustainability.cloud.sap

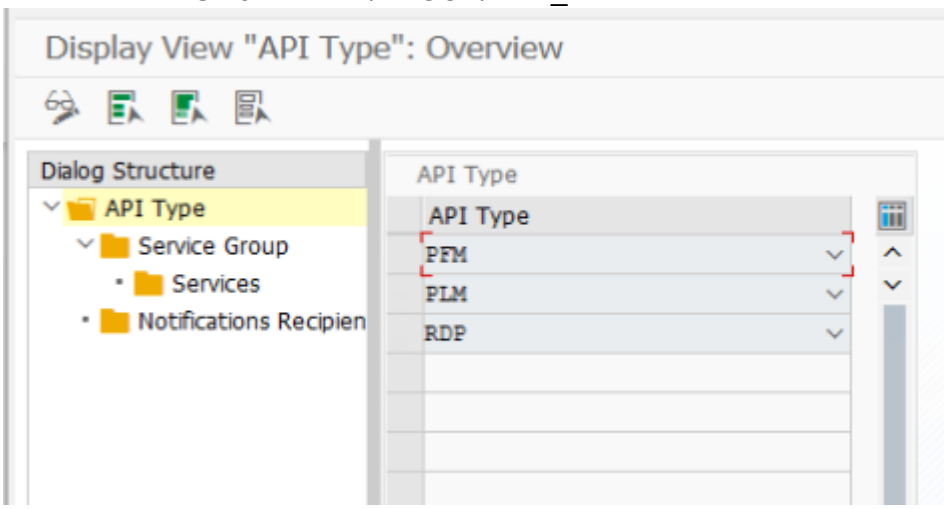
    For SDF API `e.g.` {tenant-subdomain}.eu20.sdi.sustainability.cloud.sap

    2. **Path Prefix:**
        1. **For RDP API:** `/api/packaging-report/v1/`
        2. **For SDF API:** SDF API is not supported for packaging fee
13. On the **Logon & Security** tab
    1. Set **Do not use a user**.
    2. Set **SSL to Active** and choose SSL certificate to be used.
14. Save

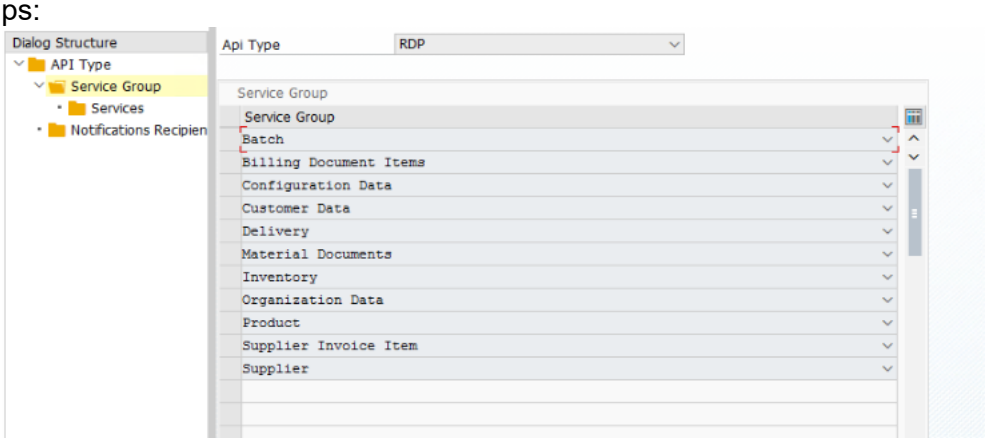
Configuration settings for services

In the following steps you will add the required configuration settings to the service groups/ service IDs .

Open transaction '*SM34*'. Enter /VPCOE/RDP_SRV view cluster and click on 'Maintain' button.



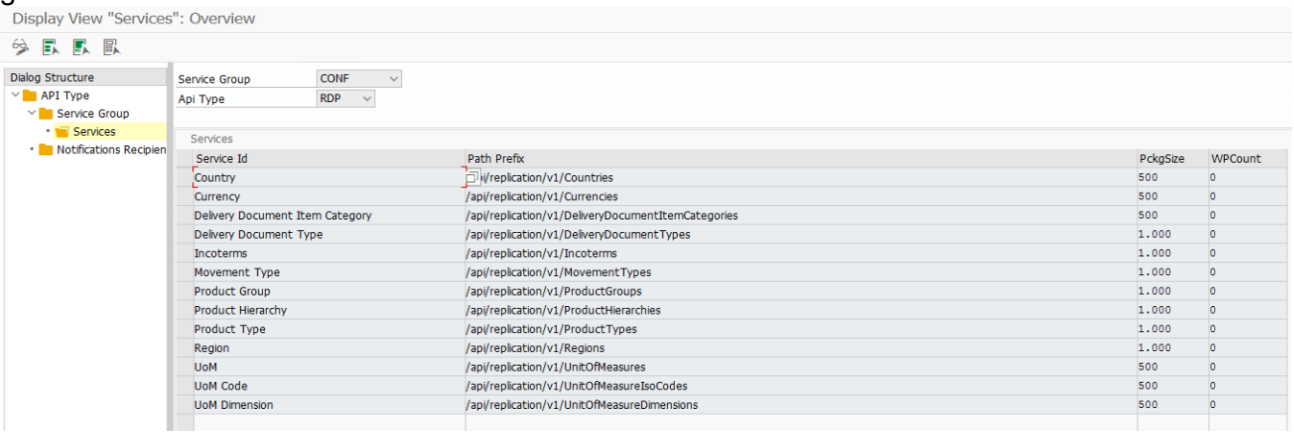
Select API type 'RDP' and double-click on '*Service Group'*. You will get a list with all RDP service groups:

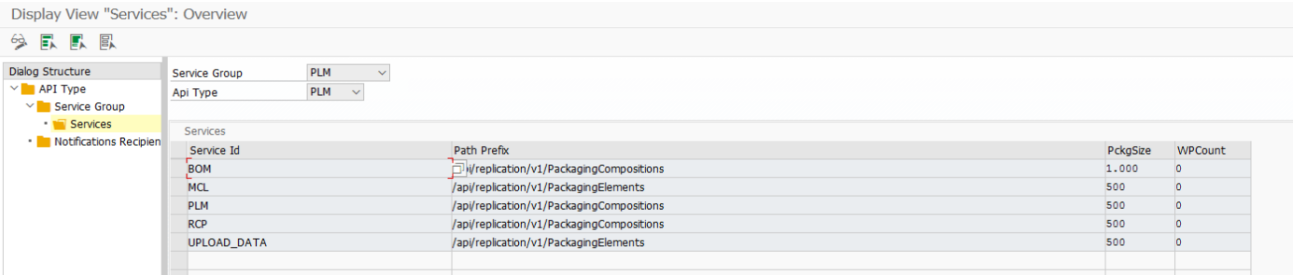


Maintain the path prefix and package size

Click on necessary Service Group and choose *Services* to maintain the path prefix or package size for it.

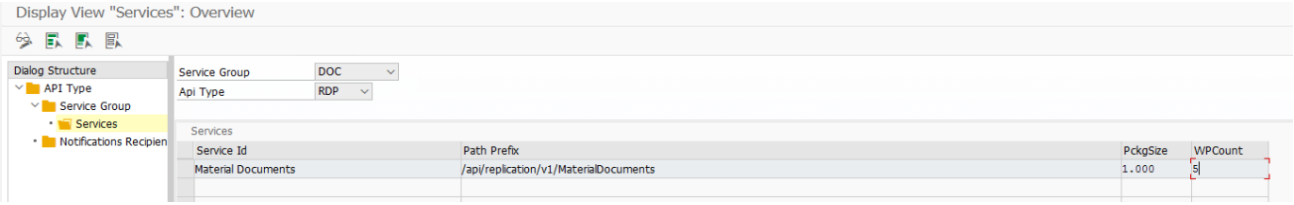
The maximum recommended package size to be used for all services is 1000 (except Material Documents where maximum recommended package size is 2000). Click on Save button to save all configurations.





WPCount represent the number of Work Processes to be used in parallel when data are sent to RDP for a specific service. WPCount is not editable for all the services except Deliveries and Material Documents.
Please note that transactional data reports (Material document, Delivery) are executing using multiple parallel work processes. The number of work processes to be used can be configure for those services. The maximum recommended number of work processes to be used in parallel is 5:





Path prefix list for RDP API:

Batch	
Batch	/api/replication/v1/Batches/
Configuration Data	
Country	/api/replication/v1/Countries
Currency	/api/replication/v1/Currencies
Delivery Document Item Category	/api/replication/v1/DeliveryDocumentItemCategories
Delivery Document Type	/api/replication/v1/DeliveryDocumentTypes
Incoterms	/api/replication/v1/Incoterms
Movement Type	/api/replication/v1/MovementTypes
Product Group	/api/replication/v1/ProductGroups
Product Type	/api/replication/v1/ProductTypes
Product Hierarchy	/api/replication/v1/ProductHierarchies
Region	/api/replication/v1/Regions
UoM	/api/replication/v1/UnitOfMeasures
UoM Code	/api/replication/v1/UnitOfMeasureIsoCodes
UoM Dimension	/api/replication/v1/UnitOfMeasureDimensions
Customer Data	
Customers	/api/replication/v1/Customers
Customer Extensions	/api/replication/v1/CustomerExtensions
Delivery	
Delivery	/api/replication/v1/Deliveries
Material Documents	
Material Documents	/api/replication/v1/MaterialDocuments
Organization Data	

Company Code	/api/replication/v1/CompanyCodes
Plant	/api/replication/v1/Plants
Sales Organization	/api/replication/v1/SalesOrganizations
Distribution Channel	/api/replication/v1/DistributionChannels
Division	/api/replication/v1/Divisions
Batch	
Batch	/api/replication/v1/Batches
Product	
Product	/api/replication/v1/Products
Product Extensions	/api/replication/v1/ProductExtensions
Supplier	
Suppliers	/api/replication/v1/Suppliers
Supplier Extensions	/api/replication/v1/SupplierExtensions
Inventory	
Inventory	/api/replication/v1/Inventories
Incoterms	
Incoterms	/api/replication/v1/Incoterms
Billing Document Items	
Billing Document Items	/api/replication/v1/BillingDocumentItem
Supplier Invoice Items	
Supplier Invoice Items	/api/replication/v1/SupplierInvoiceItem

Path prefix list for Packaging Data APIs:

Packaging Data	
BOM	/api/replication/v1/PackagingCompositions
MCL	/api/replication/v1/PackagingElements
PLM	/api/replication/v1/PackagingCompositions
RCP	/api/replication/v1/PackagingCompositions
UPLOAD_DATA	/api/replication/v1/PackagingElements

In case user wants to use V2 of the packaging data API, the path prefix should be adjusted accordingly:

Packaging Data	
BOM	/api/replication/v2/PackagingCompositions
MCL	/api/replication/v2/PackagingElements
PLM	/api/replication/v2/PackagingCompositions
RCP	/api/replication/v2/PackagingCompositions
UPLOAD_DATA	/api/replication/v2/PackagingElements

Path prefix list for SDF API:

Batch	
Batch	/api/batch/v1/batches
Configuration Data	
Country	/api/config/v1/countries
Currency	/api/config/v1/currencies
Delivery Document Item Category	/api/config/v1/deliveryDocumentItemCategories

Delivery Document Type	/api/config/v1/deliveryDocumentTypes
Incoterms	/api/config/v1/Incoterms
Movement Type	/api/config/v1/movementTypes
Product Group	/api/config/v1/productGroups
Product Type	/api/config/v1/productHierarchies
Product Hierarchy	/api/config/v1/productTypes
Region	/api/config/v1/regions
UoM	/api/config/v1/unitOfMeasures
UoM Code	/api/config/v1/unitOfMeasureIsoCodes
UoM Dimension	/api/config/v1/unitOfMeasureDimensions
Customer Data	
Customers	/api/masterdata/v1/customers
Customer Extensions	/api/materdata/v1/ /customerEPRInformation
Delivery	
Delivery	/api/delivery/v1/deliveries
Material Documents	
Material Documents	/api/materialDocument/v1/materialDocuments
Organization Data	
Company Code	/api/orgdata/v1/companies
Plant	/api/orgdata/v1/plants
Sales Organization	/api/orgdata/v1/salesOrganizations
Distribution Channel	/api/orgdata/v1/distributionChannels
Division	/api/orgdata/v1/divisions
Product	
Product	/api/masterdata/v1/products
Product Extensions	/api/masterdata/v1/productEPRInformation
Supplier	
Suppliers	/api/masterdata/v1/suppliers
Inventory	
Inventory	/api/inventory/v1/inventories
Incoterms	
Incoterms	/api/config/v1/incoterms
Billing Document Items	
Billing Document Items	/api/billingDocument/v1/billingDocumentItems
Supplier Invoice Items	
Supplier Invoice Items	/api/supplierInvoice/v1/supplierInvoiceItems

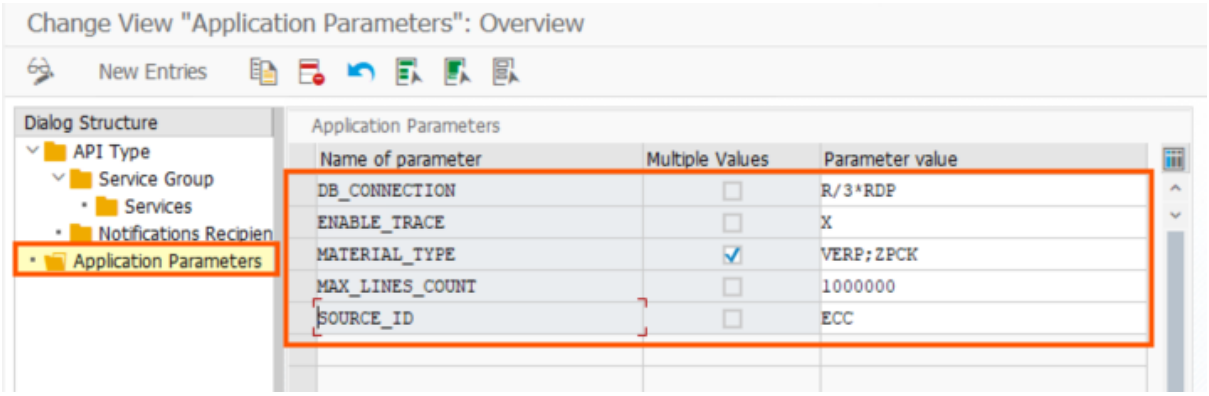
Path prefix list for Packaging Data APIs:

Packaging Data	
BOM	/api/packaging/v1/packagingCompositions
MCL	/api/packaging/v1/packagingElements
PLM	/api/packaging/v1/packagingCompositions
RCP	/api/packaging/v1/packagingCompositions
UPLOAD_DATA	/api/packaging/v1/packagingElements
End of validity for a product	/api/packaging/v1/EndPackagingCompositionValidityForProduct
Get Packaging Composition	/api/replication/v2/GetPackagingCompositionsForProduct

Path for Packaging composition items (from handling units) is defined in the entity processor class `/VPCOE/CL_UPH_TM_HU` attribute MC_HU, value '/PackagingCompositionItems'. This value can be overwritten if it's necessary.

Maintain Application Parameters

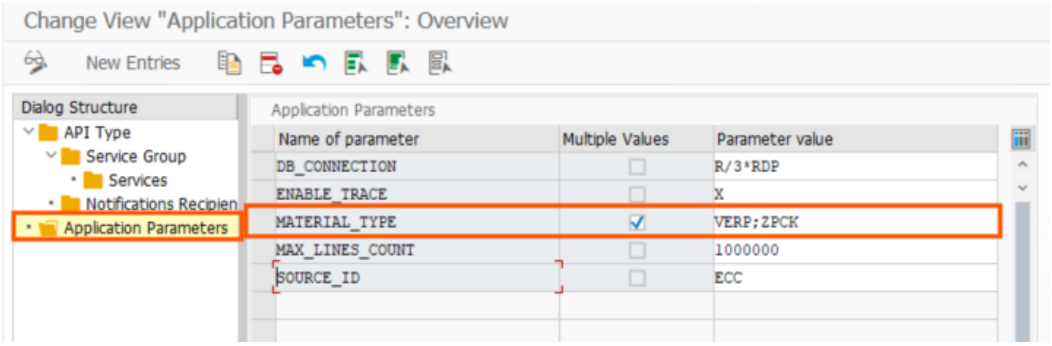
This section is intended to allow users to maintain different application parameters in one single place:



Application parameters to be maintained:

MATERIAL_TYPE

This value will be used in selection of packaging products, in product replication report, to be replicated to RDP.



The product report replicates materials with the Material Type maintained in this table, as well as all materials which are extended to any Sales Organizations (regardless of their types) - indicating they are sold.

If multiple Material Types need to be maintained, ensure the **Multiple Values** checkbox is selected, and separate each Material Type with a semicolon (';').

MAX_LINES_COUNT

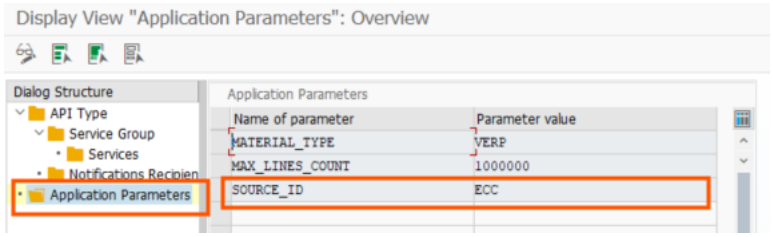
User can specify within this parameter the maximum value of records to be selected for each variant for material documents replication. Each customer needs to decide the maximum value based on their system parameters. If no value is provided or the parameter is not maintained in the system, integration accelerator will limit the data selection up to 1.000.000 (1 million) records per variant at a time.

SOURCE_ID

Users can maintain Source Id for data to be replicated, in one of the following places:

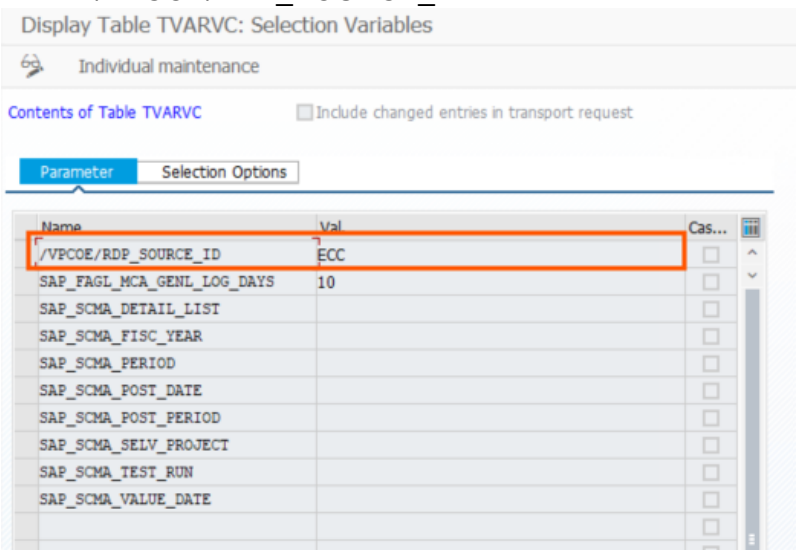
1. In 'Configuration settings for accelerator services', tcode *SM34*'.

Change the parameter value as per your needs:



2. In TCode STVARV

Open transaction '*stvarv*'. Create new entry:
Name: /VPCOE/RDP_SOURCE_ID



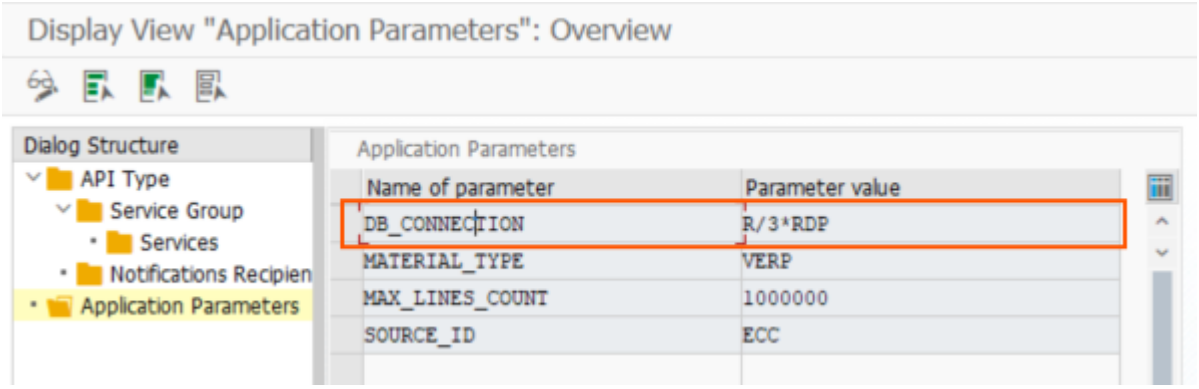
Important Note: Value for the source ID should match the value of the source system created using **Configuration Cockpit** application as described here.

The source ID defined in this section it's not valid for handling units replication also. The source id for Handling Units replication it is specified in the selection parameters defined in the report. For more details, please read handling units replication report selection parameters details.

DB_CONNECTION (Database connection)

This parameter allows the user to specify the secondary database connection to be used when retrieving data for the *Delivery and Material Documents* report. Using a dedicated secondary connection can improve performance, especially when handling large data volumes.

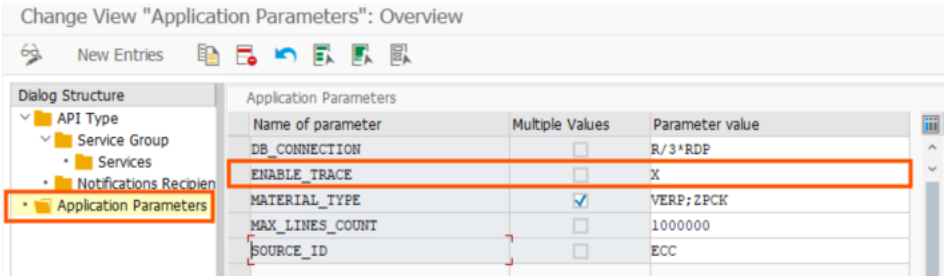
If not specified, the default value is the standard SAP service connection: *R/3*RDP*.



ENABLE_TRACE

This parameter allows the user to specify whether trace logging should be activated for replication reports.

When set to X, the system will record trace information in a temporary table for each object selected for replication — regardless of whether it was successfully sent or skipped.



Usage Considerations:

- This trace can be reviewed using the **/VPCOE/TRACE_MAINTENANCE** report.
- Tracing is useful for troubleshooting and validation of replication runs.
- To avoid performance impact and data buildup, it is recommended to **only activate tracing temporarily** when needed, and to clean up trace data regularly using the maintenance report.

DLV_CATCH_WEIGHT_ON

This parameter allows the user to specify whether catch weight quantities and the delivery catch weight unit of measure should be used instead of the base quantity and base unit of measure for delivery replication reports.

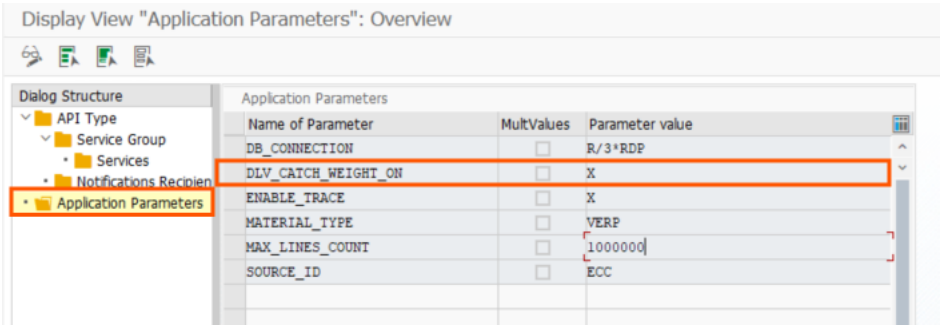
By default, the value is empty (false), meaning the base unit of measure will be used.

When set to **X**, the delivery replication will replace the base quantity and base unit of measure in the delivery with the sales quantity and sales unit of measure:

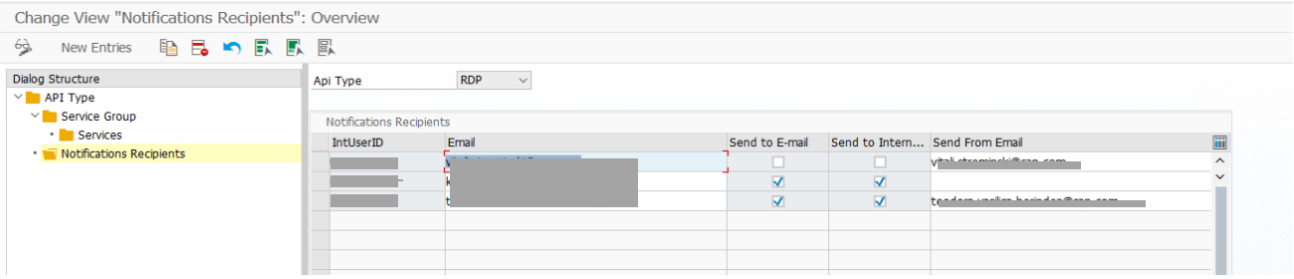
- **base_quantity** will be set to the sales quantity (field **LFIMG** in the delivery item).
- **base_unit** will be set to the sales unit (field **VRKME** / **LFIME** in the delivery item in ECC/S4H).

Important notes:

- Packaging compositions in RDP are defined either in the product base unit of measure or in one of the alternative UoMs that can be converted to the base UoM. Therefore, catch weight quantities are not relevant for RDP (RDP needs the quantity in the base unit of measure).
- Unless there is a specific business reason to send catch weight materials to RDP, the quantities in the base unit of measure should be sent.



Maintain Email Notification Recipients - optional



In case of any failed replication in background mode, if email addresses are maintained, emails with the status of failed replications are sent to maintained users. There are two options for sending emails: to ECC internal box or directly to email addresses. Please note that in order emails to be sent out from ECC system, all basis configurations need to be done to allow this.

Create and activate Change Pointers

Through Change Pointer's mechanism if any master data field value is changed in SAP like Material master, Vendor master etc. through standard t-code like MM01, MM02, VA01, VA02, the corresponding records will be created in change pointers tables (BDCP, BDCPS).

RDP Integration package installation does not create the change pointers automatically. Change pointers needs to be created and activated manually.

Services using change pointers mechanism:

- Batch
- Customer
- Supplier
- Product

By default, RDP Integration reports are using specific change pointers listed below. These separate message types below need to be created and it's not recommended to re-use existing message types used in other processes also:

- /VPCOE/BATCH
- /VPCOE/CUSTOMER
- /VPCOE/SUPPLIER
- /VPCOE/PRODUCT

Also please make sure the minimum change items described below are assigned to the corresponding message types.

Follow the steps below to create the change pointers for each of the services and activate them:

1. Create message type (tcode WE81)

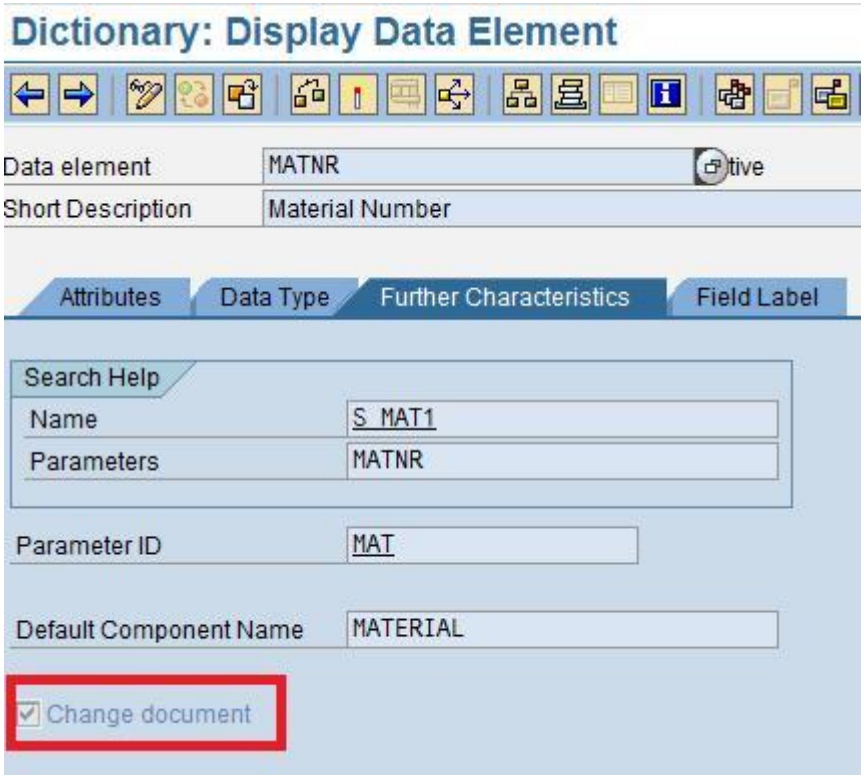
Message Type	**Short Text**
/VPCOE/BATCH	Batch
/VPCOE/CUSTOMER	Customer
/VPCOE/PRODUCT	Product
/VPCOE/SUPPLIER	Supplier

2. Assign change items to message type (tcode BD52)
   Assign the table fields to message types which you want to be considered for change pointers generations.
   Following are the minimum fields to be assigned for each message type:

Message type:	**Object**	**Table name**	**Filed name**
/VPCOE/BATCH	CHARGE	MCHA	CHARG
	CHARGE	MCHA	HSDAT
	CHARGE	MCHA	KEY
		MCHA	WERKS
	ESSUB	ESTRH	SUBID
	MATERIAL	MARA	MATNR
/VPCOE/PRODUCT	MATERIAL	DMAKT	MAKTX
	MATERIAL	DMARM	KEY
	MATERIAL	MARA	KEY
	MATERIAL	MARA	LVORM
/VPCOE/CUSTOMER	DEBI	KNA1	KEY
	DEBI	KNA1	LOEVM
	DEBI	KNA1	NAME1
	DEBI	KNA1	NAME2
	DEBI	KNA1	NODEL
/VPCOE/SUPPLIER	KRED	LFA1	KEY
	KRED	LFA1	LOEVM
	KRED	LFA1	NAME1
	KRED	LFA1	NODEL

	KRED	LFB1	KEY

3. Activate change pointers generally (tcode BD61)
4. Activate change pointers corresponding to fields in message types (transaction code BD50).
   Set Active flag for each of the message type created and Save it.
5. Also check if the data elements of all the required fields have 'change Document' field checked (Below screen shot for Material Number)



CONTENT OF RDP INTEGRATION PACKAGE

Once you have done all the above steps successfully, your RDP Integration Accelerator package is ready to use.

Report list and assigned transaction codes included in the integration accelerator package:

Development Package	Report Name	Transaction Code
/VPCOE/COMMON_INTEGRATION	/VPCOE/REPROCESS_PAYLOAD	/VPCOE/REPROCESS
/VPCOE/RDP_INTEGRATION	/VPCOE/BATCH_TO_RDP	/VPCOE/RDP_BATCH
	/VPCOE/CONFIGURATION_TO_RDP	/VPCOE/RDP_CONF
	/VPCOE/ORG_DATA_TO_RDP	/VPCOE/RDP_ORG_DATA
	/VPCOE/PRODUCT_TO_RDP	/VPCOE/RDP_PRODUCT
	/VPCOE/CUSTOMER_TO_RDP	/VPCOE/RDP_CUSTOMER
	/VPCOE/SUPPLIER_TO_RDP	/VPCOE/RDP_SUPPLIER
	/VPCOE/DELIVERY_TO_RDP	/VPCOE/RDP_DELIVERY
	/VPCOE/MAT_DOC_TO_RDP	/VPCOE/RDP_MAT_DOC
	/VPCOE/BILLDOCITEMS_TO_RDP	-
	/VPCOE/SUPPLINVITEMS_TO_RDP	-
	/VPCOE/INVENTORY_TO_RDP	/VPCOE/RDP_INVENTORY
	/VPCOE/INCOTERMS_TO_RDP	/VPCOE/RDP_INCOTERMS
	/VPCOE/TO_RDP	/VPCOE/RDP
/VPCOE/PLM_INTEGRATION	/VPCOE/R_UPH_PCKG_CMP_PLM_LOAD	/VPCOE/UPH_PC_PLM
	/VPCOE/R_UPH_PCKG_CMP_RCP_LOAD	/VPCOE/UPH_PC_RCP
	/VPCOE/R_UPH_PCKG_DATA_LOAD	/VPCOE/UPH_PE_PLM

/VPCOE/OBJ_INTEGRATION	/VPCOE/R_UPH_PCKG_MCL_LOAD	/VPCOE/UPH_PE_MCL
	/VPCOE/R_UPH_PCKG_CMP_BOM_LOAD	/VPCOE/UPH_PC_BOM
	/VPCOE/R_PCKG_CMP_HU_LOAD	/VPCOE/HU
/VPCOE/RDP_PCKGFEE_INTEGRATION	/VPCOE/R_PCKF_POSTING	/VPCOE/PCKF_POSTING
	/VPCOE/R_PCKF_RETRIEVAL	/VPCOE/PCKF_RETRIEVE
	/VPCOE/R_PCKF_CUSTOMER_ROLE	/VPCOE/PCKF_CUS_ROLE
	/VPCOE/R_PCKF_CUSTOM_EXEMPT	/VPCOE/PCKF_CUS_EXMP
	/VPCOE/R_PCKF_COMP_CODE_EXEM	/VPCOE/PCKF_CC_EXMP

Transaction code /VPCOE/RDP has been created to run the general report /VPCOE/TO_RDP. If it's necessary, please create your own custom role with following authorizations and assign it to specific users to allow them to execute the transaction.

Authorization needed to execute TCode /VPCOE/RDP (in foreground mode):

Object	Field Name	From	To Activity
S_TCODE	TCD	/VPCOE/RDP	
S_ADMI_FCD	S_ADMI_FCD		ST0M
S_GUI	ACTVT		61
S_ALV_LAYO	ACTVT		23
W_BETR_USR	ACTVT		03
W_BETR_USR	WERKS	*	
M_MSEG_WMB	ACTVT		03
M_MSEG_WMB	WERKS	*	
M_MATE_MAR	ACTVT		03
M_MATE_MAR	BEGRU		
M_MATE_MAT	ACTVT		03
M_MATE_MAT	BEGRU		
M_MATE_WGR	ACTVT		03
M_MATE_WGR	BEGRU		
K_KEKO	ACTVT		03
K_KEKO	BUKRS	*	
K_KEKO	KLVAR		

Please note that exact authorization setup can vary from customer to customer, and security team can determine and setup the roles with all authorizations required.

Authorization to run the reports in background mode is needed for the transaction code for each specific report.

HOW TO REPLICATE DATA TO RDP

RDP Integration reports will extract needed data from your system, pack the data in the required format, and send them to RDP.

RDP Integration interfaces will support two scenarios to load the data (excepting organization and configuration data which only supports Initial load):

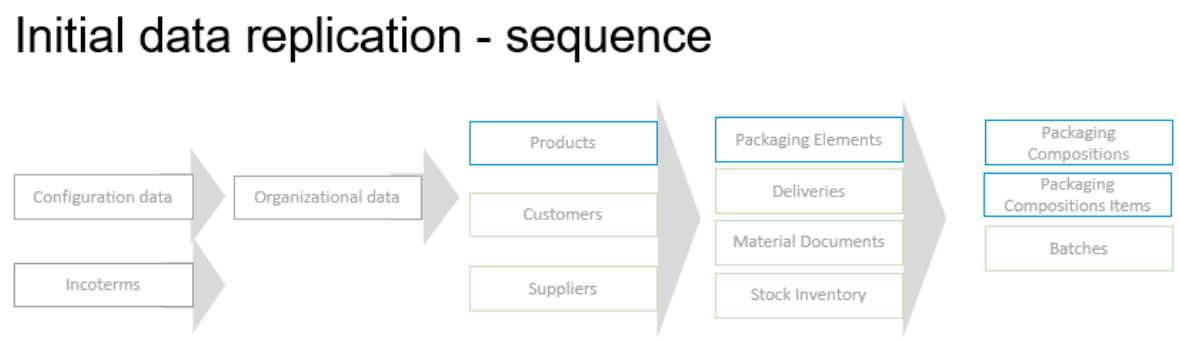
1. **Initial Load** – In this case, all the objects selected based on filters criteria will be sent to RDP
2. **Delta Load** – In this case, all changed (new, changed) business object instances from the last load will be sent to RDP.

The below steps describe how you can use the integration accelerator to load data into the RDP.

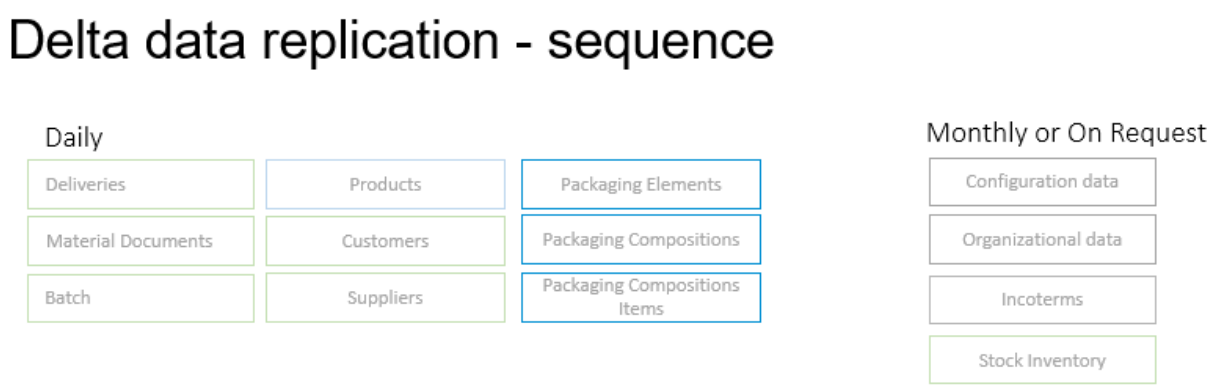
RDP ERP Integration reports

Each report in the package, for each service id, can be executed individually using transaction code SE38 or in background mode.

For a proper replication we propose to follow the extraction of the data in the following order for an initial replication:



For delta load replication, you can consider daily or monthly replication or depending on your needs:



Also, one general report for RDP ERP is available, /VPCOE/TO_RDP which helps you to load the data to RDP for all services available from one single place. Transaction code /VPCOE/RDP has been created to execute /VPCOE/TO_RDP report.

General Approach for Loading Data

The RDP Integration Accelerator provides multiple options for loading data into **SAP Responsible Design and Production (RDP)**. The approach varies depending on the type of data being processed and the desired load method.

Initial Load

- All objects selected based on the defined filter criteria are sent to RDP.
- For all entities, a new import overwrites and deletes the existing data in RDP.

Get Only Delta Flag

- Extracts delta data based on **change pointers** or **specific report logic**.
- Only newly created or modified objects are selected and sent to RDP.

Preview Mode

- Displays the data that would be sent to RDP based on the selection criteria without transferring the data.
- The results are shown on the screen after executing the report.

Test Mode

- **Available for packaging data reports.**
- When the **Test Mode** checkbox is selected, the data extraction and mapping processes remain the same, but the transfer of data to the RDP API is skipped.
- Test Mode is supported in both **initial load** and **delta load** scenarios.

Export to XLSX File

- Allows the generation of an Excel file based on the selected criteria.
- The exported data can be used with the **Data Import** application in RDP.
- Note: The number of records exported to the XLSX file is limited to **50,000**.

Selection ID

- Whenever an **initial load** or **delta load** is successfully performed, the selection parameters along with the current date are saved.
- This saved **Selection ID** is used to determine the next delta load.
- The last usage date of the Selection ID is also recorded for reference.

Reports with Specific Load Functionality

Reports Without Delta Load Functionality

The following reports do not support **delta load** functionality. If data changes occur, a re-execution of the **initial load** is required:

- **/VPCOE/CONFIGURATION_TO_RDP** – Configuration data replication.
- **/VPCOE/ORG_DATA_TO_RDP** – Organization data replication.
- **/VPCOE/INCOTERMS_TO_RDP** – Incoterms data replication.

Reports Using Change Pointers for Delta Load

The following reports use **change pointers** to identify and replicate all new or changed objects since the last replication:

- **/VPCOE/BATCH_TO_RDP** – Batch data replication.
- **/VPCOE/PRODUCT_TO_RDP** – Product data replication.
- **/VPCOE/CUSTOMER_TO_RDP** – Customer data replication.

- **/VPCOE/SUPPLIER_TO_RDP** – Supplier data replication.

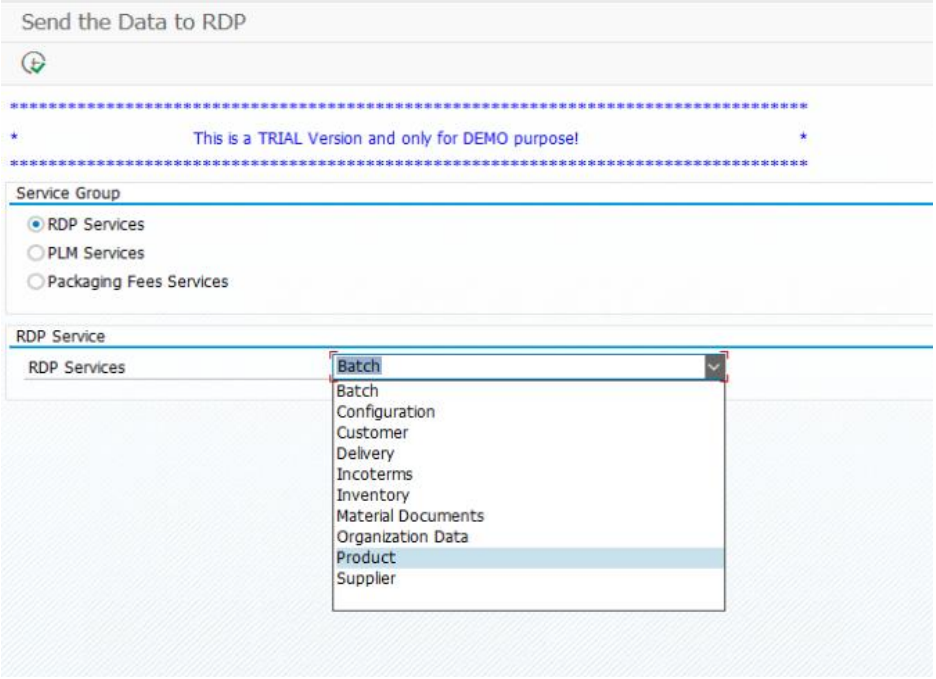
Reports with Delta Load Logic

The following reports do not rely on change pointers and instead apply their own logic to determine delta data:

- **/VPCOE/DELIVERY_TO_RDP** – Delivery data replication.
- **/VPCOE/MAT_DOC_TO_RDP** – Material documents replication.
- **/VPCOE/INVENTORY_TO_RDP** – Inventory data replication.

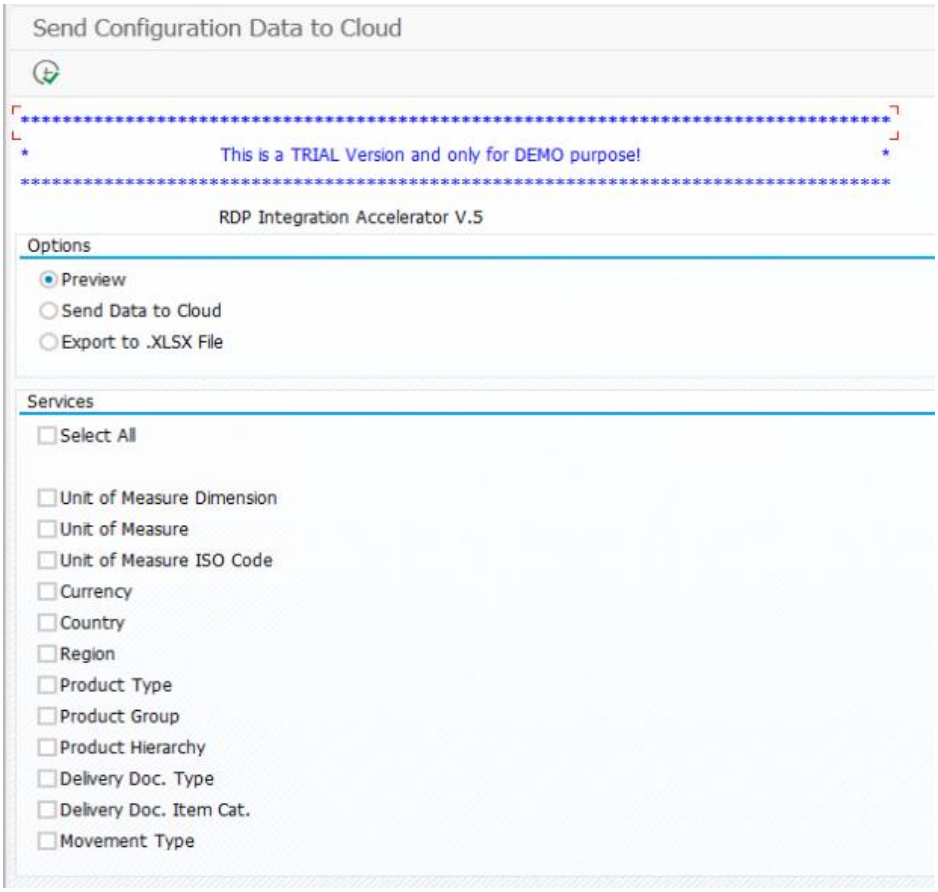
Open transaction code /VPCOE/RDP.

Select necessary service group and service name and execute the report:



Loading Configuration data

Open transaction code VPCOE/RDP and select service ID ‘Configuration’ (alternatively you can run the report /VPCOE/CONFIGURATION_TO_RDP):



Executing the Configuration report you can load specific configuration data related to:

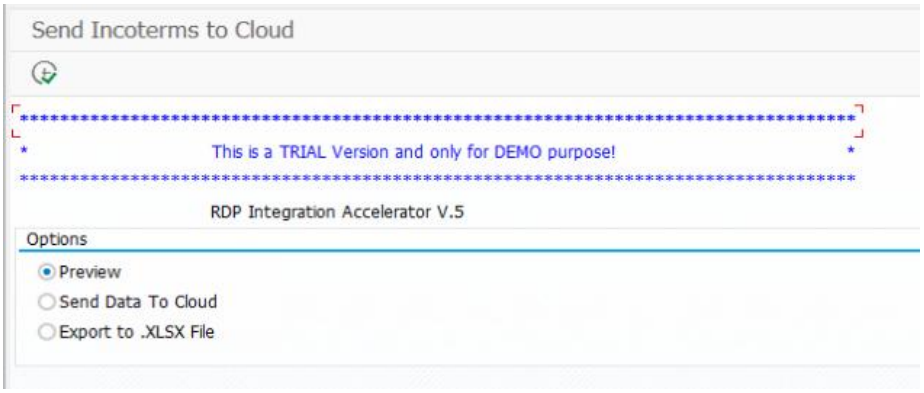
- Unit of Measure
- Unit of Measure Dimension
- Unit of Measure ISO Code
- Currency
- Country
- Region
- Product Type
- Product Group
- Product Hierachy
- Delivery Document Type
- Delivery Document Item Category
- Movement Type

Note: You can select all the services or only needed ones in order to send the data to RDP.

Note: Internal units of measure will be sent to RDP.

Loading Incoterms

Open transaction code VPCOE/RDP and select service ID ‘Incoterms’ (alternatively you can run the report /VPCOE/INCOTERMS_TO_RDP):



Loading Organization data

Open transaction code VPCOE/RDP and select service ‘Organization data’ (alternatively you can run the report /VPCOE/ORG_DATA_TO_RDP):



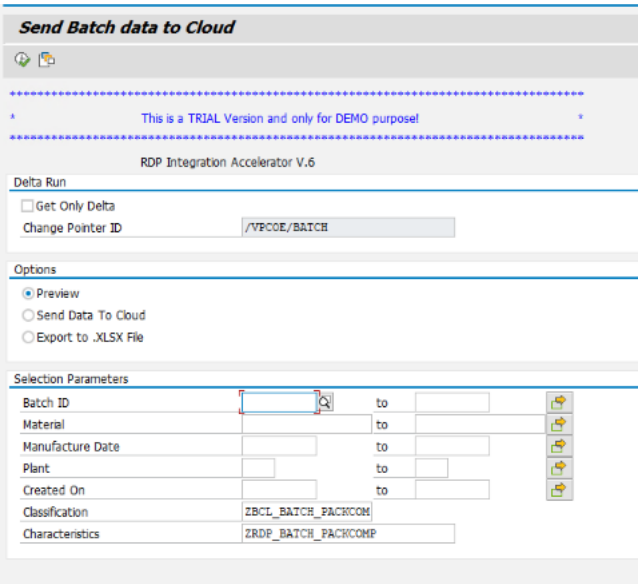
Executing the Organization data report you can load specific organization data related to:

- Company Code
- Plant
- Sales Organization
- Distribution Channel
- Division
- Sales Area

Note: You can select all the services or only needed ones in order to send the data to RDP.

Loading Batch data

Open transaction code VPCOE/RDP and select the service ‘Batch data’ (alternatively you can run the report /VPCOE/BATCH_TO_RDP):



Initial Load Approach

The initial load extracts and transfers all relevant batch data based on the defined selection criteria. The process includes:

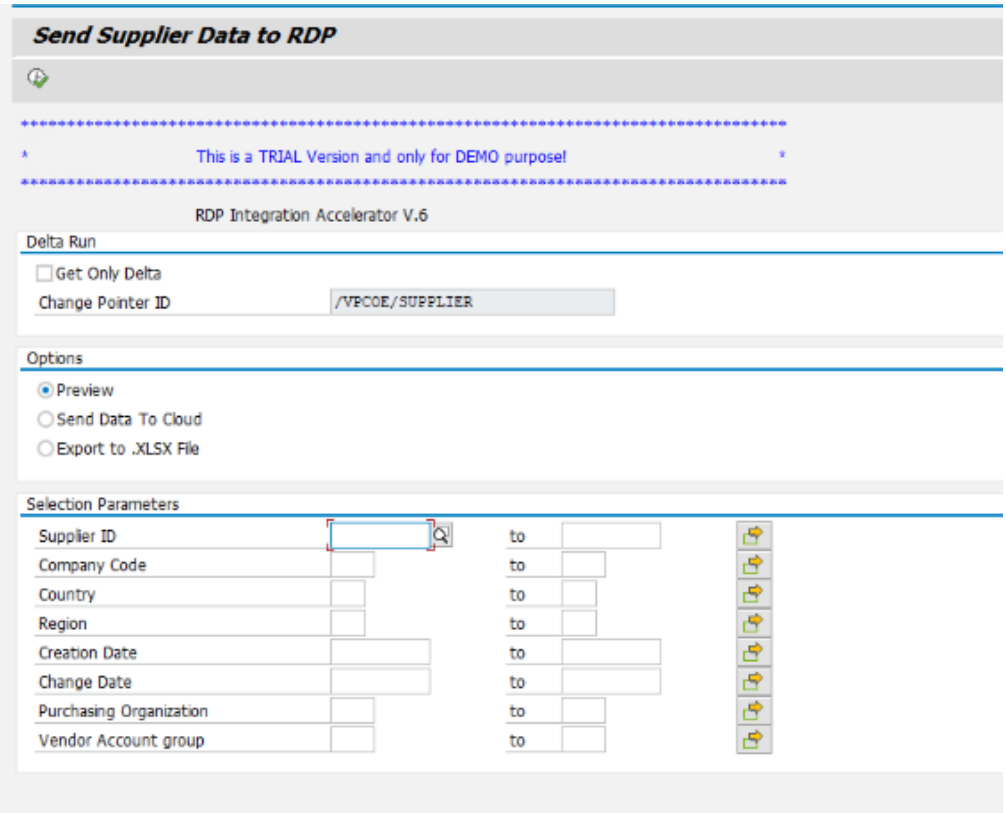
- Selection of all batch data based on specified filters.
- Determination of Packaging Composition in Batch:
  - The report identifies all batches matching the existing filter criteria for classes and characteristics.
  - If a batch contains a characteristic entry related to Packaging Composition ID, the corresponding characteristic value is retrieved.
  - For batches with an entry in classes and characteristics, the Packaging Composition ID is additionally transferred to RDP.
  - The identified Packaging Composition is additionally replicated.

Delta Load Approach

- The Delta Load functionality operates based on Change Pointers, ensuring that only newly created or modified batch data is selected.
- The system identifies changes using the defined change pointer logic and replicates the relevant updates to RDP.

Loading Supplier data

Open transaction code VPCOE/RDP and select service ‘Supplier’ (alternatively you can run the report /VPCOE/SUPPLIER_TO_RDP):



Initial Load Approach

During the initial load, all relevant supplier data is extracted and transferred to RDP based on the specified selection criteria:

- All suppliers matching the defined filters are selected and sent to RDP.
- Suppliers **marked as deleted** are also included in the replication to ensure data consistency. This is a requirement/recommendation on RDP.

Delta Load Approach

- The **Delta Load** functionality is based on **Change Pointers**, ensuring that only newly created, modified, or deleted suppliers are selected for replication.
- All selection parameters remain available in delta load mode, allowing users to **restrict the scope of data replication** as needed.

Tax Number and Tax Code Selection Logic for Supplier Data

During supplier (vendor) replication, tax number and tax code are selected and transferred to RDP. The selection logic depends on the system configuration and whether **Customer Vendor Integration (CVI)** is active:

Logic to Select Tax Number:

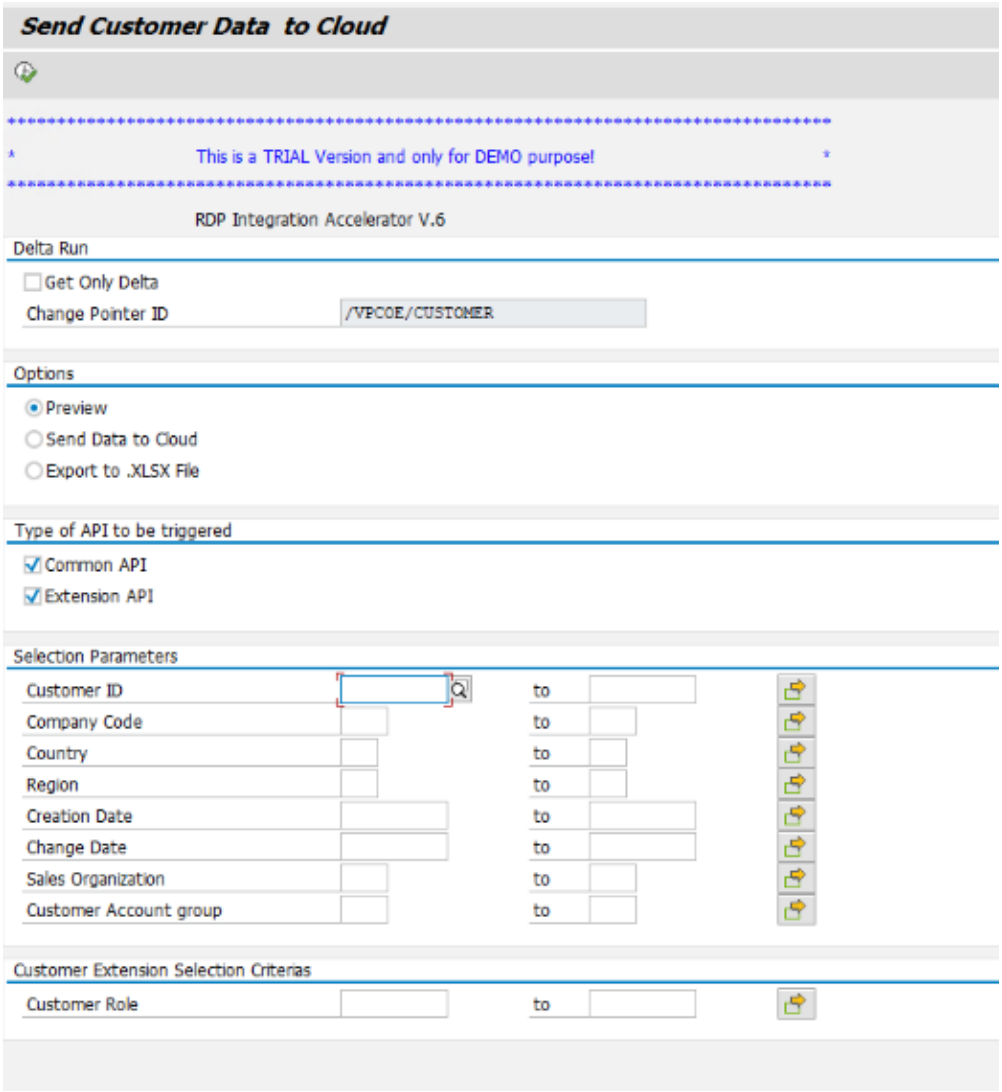
- **If CVI is active**:
  - Tax number is selected from:
    - **DFKKBPTAXNUM-TAXNUM** or
    - **DFKKBPTAXNUM-TAXNUMXL**
- **If CVI is not active:**
  - Tax number is selected from:
    - **LFA1-STCEG** or
    - **LFAS-STCEG**

Note:

- The usage of **VAT registration codes and tax numbers** in customer/vendor master data may vary depending on country-specific tax regulations and the organization's specific requirements.
- If the default selection logic does not meet the organization’s requirements, a **BAdI implementation** is available to **overwrite the logic** for tax number selection.

Loading Customer and Customer Extension data

Open transaction code VPCOE/RDP and select service ‘Customer’ (alternatively you can run the report /VPCOE/CUSTOMER_TO_RDP):



You can choose between two types of APIs based on the required data:

- **Common API** – Select this checkbox to send **customer data** based on the defined selection criteria to RDP.
- **Extension API** – Select this checkbox to send **additional attributes** based on the selection criteria, as implemented in the BAdI.
    - For details on extending functionality, refer to the **Extensibility** section.

Initial Load Approach

During the initial load, all relevant customer data is extracted and transferred to RDP:

- All customers matching the defined filters are selected and sent to RDP.

- Customers **marked as deleted** are also included in the replication to ensure data consistency.

Delta Load Approach

- The **Delta Load** functionality is driven by **Change Pointers**, ensuring that only newly created, modified, or deleted customer records are selected.
- Selection can be further refined using selection filters to optimize data extraction.
- All selection fields are available in delta load mode, allowing users to **restrict data selection** as needed.

Loading Customer extension data

- To load **Customer Extension Data**, select the **Extension API** checkbox.
- For further information, refer to the Extensibility capabilities for RDP Integration reports.

Sample BAdI Implementation for Customer Extensions

The sample BAdI implementation follows RDP recommendations for defining classes and characteristics as follows:

Class	ZCCL_RDP_CUST_EXT
Description	RDP specific Customer Attributes
Class Type	011

Characteristics	Description	Data Type	Length	Decimal	UoM / Values
ZRDP_CUST_EXT_ROLE	Customer Role	CHAR	10	0	Code List
ZRDP_PROD_EXT_TAG	Customer Tag	CHAR	10	0	Customer defined code list

Tax Number and Tax Code Selection Logic

During customer replication, tax number and tax code are selected and transferred to RDP. The selection logic depends on the system configuration:

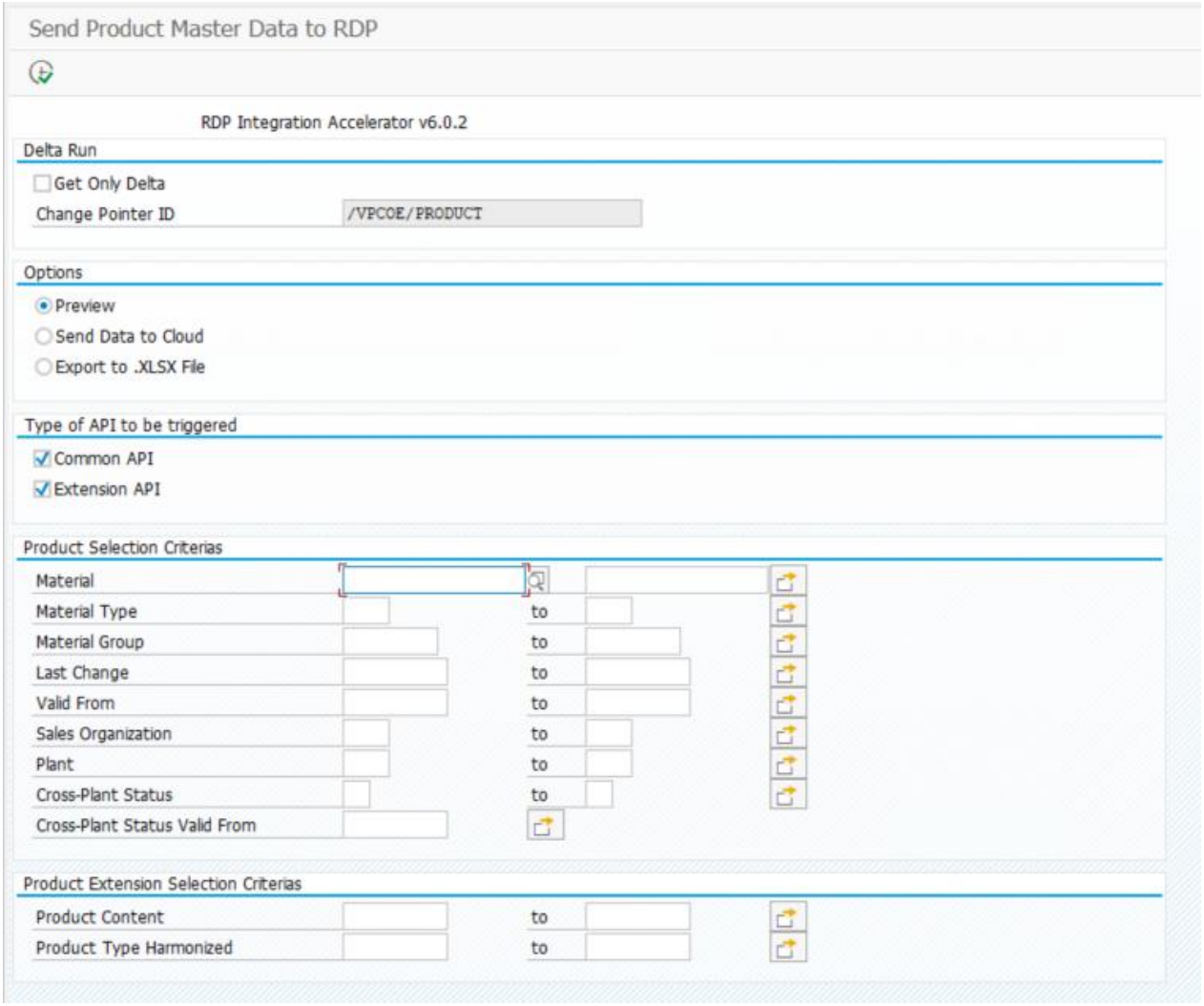
- **If CVI is active** in ECC (and in S/4H):
  - Tax number is selected from **DFKKBPTAXNUM-TAXNUM** or **DFKKBPTAXNUM-TAXNUMXL**.
- **If CVI is not active:**
  - Tax number is selected from **KNA1-STCEG** or **KNAS-STCEG**.

Note:
The usage of VAT registration codes and tax numbers in customer/vendor master data can vary depending on country-specific tax regulations and organizational requirements.
If customization is required, use the **available BAdI** to overwrite the default logic for tax number selection.

Please use the BAdI available if you need to overwrite the above logic for tax number information based on your specific needs.

Loading Product and Product extension data

Open transaction code VPCOE/RDP and select service ‘Product’ (alternatively you can run the report /VPCOE/PRODUCT_TO_RDP):



You can choose between two types of APIs based on the required data:

- **Common API** – Select this checkbox to send **product data** based on the defined selection criteria to RDP.
- **Extension API** – Select this checkbox to send **additional attributes** based on the selection criteria, as implemented in the BAdI.
    - For more information on extending functionality, refer to the Extensibility section

Initial Load Approach

During the initial load, all relevant product data is extracted and transferred to RDP:

- All products matching the defined filters are selected and sent to RDP.

Delta Load Approach

- The **Delta Load** functionality is based on **Change Pointers**, ensuring that only newly created, modified, or deleted product records are selected.
- The system identifies changes using the change pointers mentioned on the selection screen and applies the **Last Change Date** filters defined in the **Product Selection Criteria**.
- **All products, including deleted products**, are included in the replication to ensure data consistency.

Note:

- **Packaging materials** are sent to RDP if either of the following conditions is met:

- The material's **type matches** an entry in the **customizing table** for packaging material types.
- The customizing table is **empty**, and the material type is equal to **"VERP"** (standard packaging material type).

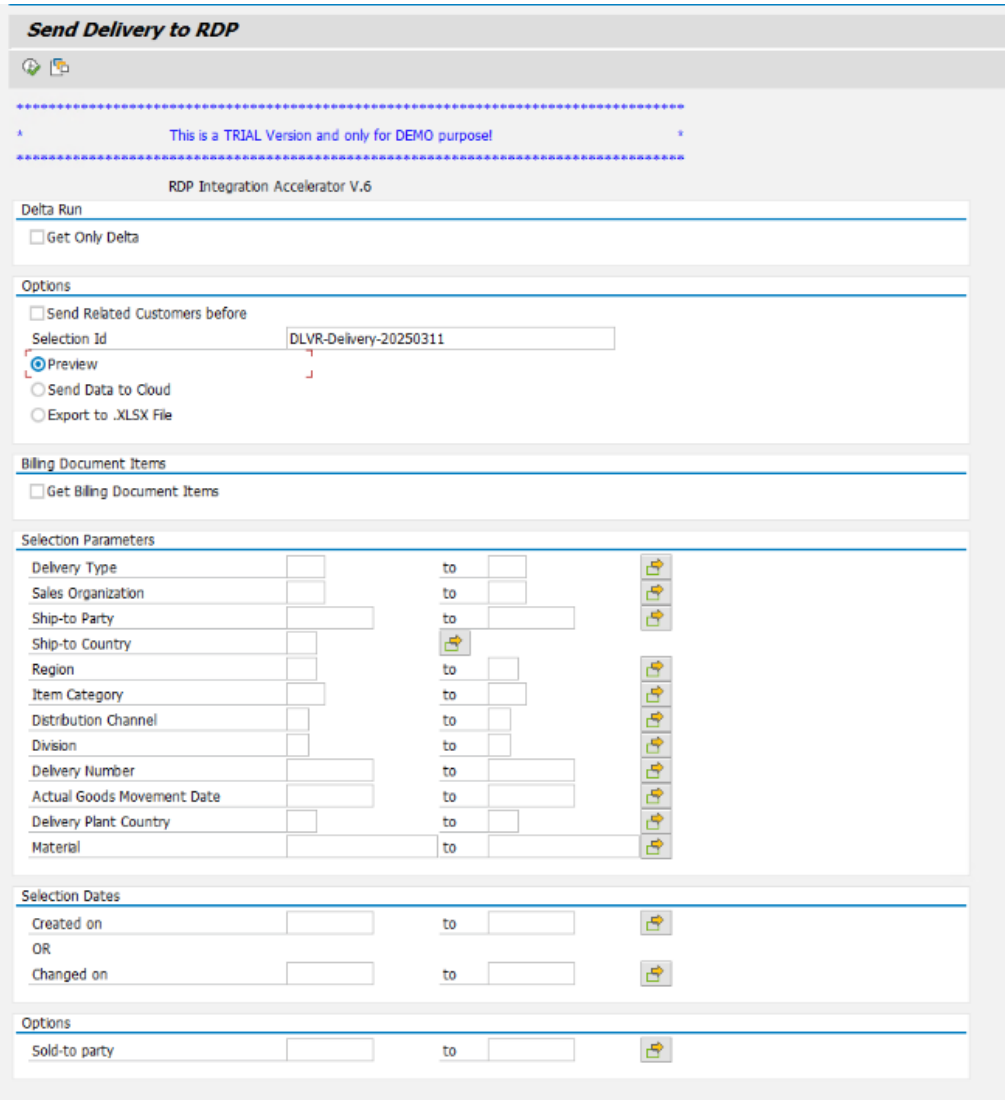
- **Sold materials** are always sent to RDP, regardless of their material type, **if they are maintained in any Sales Organization** — i.e., if the material has an entry in the **MVKE table**.

Loading Product extension data

- to load **Product Extension Data**, select the **Extension API** checkbox.
- For further information, refer to the Extensibility capabilities for RDP Integration reports.

Loading Deliveries

Open transaction code VPCOE/RDP and select service ‘Delivery’ (alternatively you can run the report /VPCOE/DELIVERY_TO_RDP):



Initial load approach:

- all the deliveries selected based on filters criteria will be sent to RDP. Please note only deliveries with status ‘Completely Process’ are considering for replication.
- by selecting ‘Send related customer before’ checkbox, all the related customers according to selection criteria and for the change pointers specified will be also sent to RDP
- all deleted deliveries will be sent to RDP

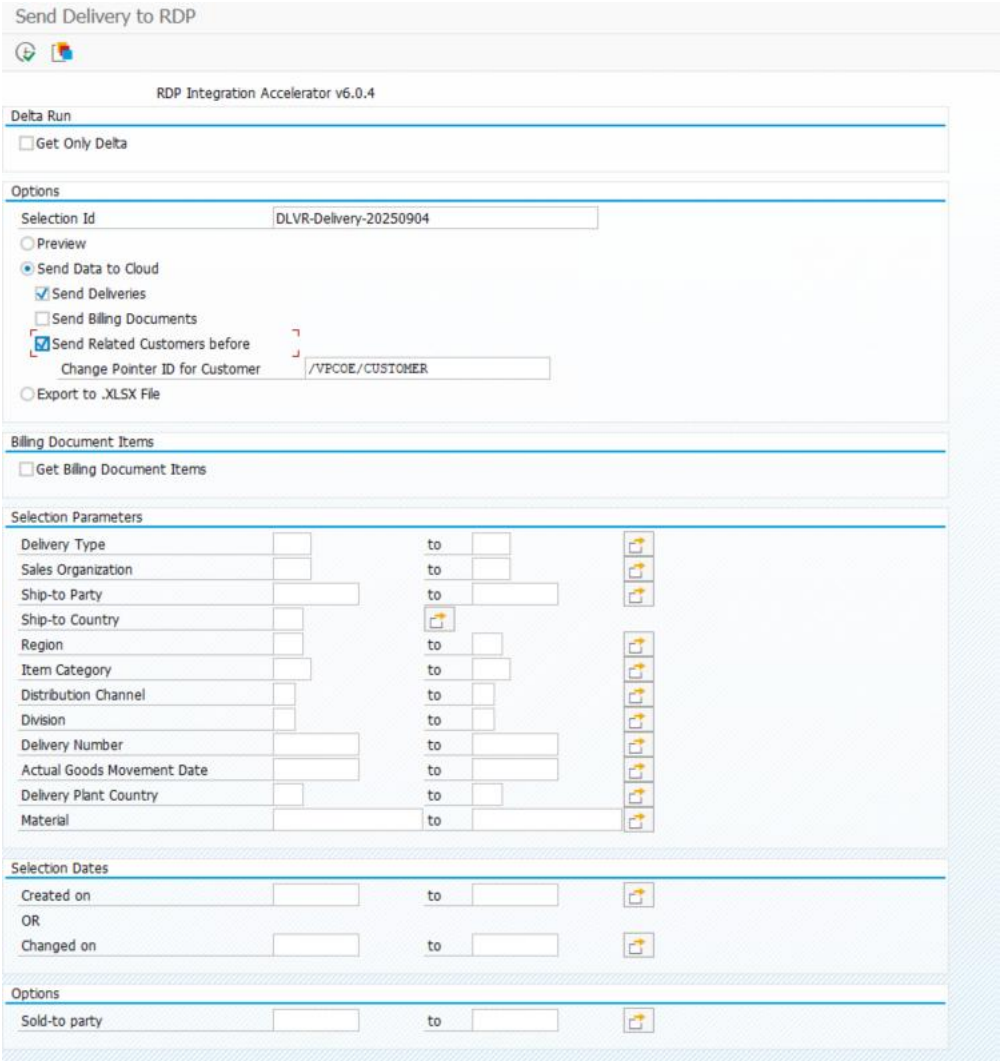
‘Send related customer before’ option behavior:

When the **“Send related customers before”** option is selected in the Delivery replication report, only **delta customers** (i.e., customers created or changed since the last replication) that are **directly related to the selected deliveries** will be included and sent prior to the delivery data.

This option is **not intended** to serve as a replacement for full customer replication. Users should continue to use the **Customer report** to replicate all in-scope customer master data. The purpose of this option is to ensure that any new or updated customers linked to the deliveries being replicated are also sent, avoiding data dependency errors during delivery processing.

With this behavior:

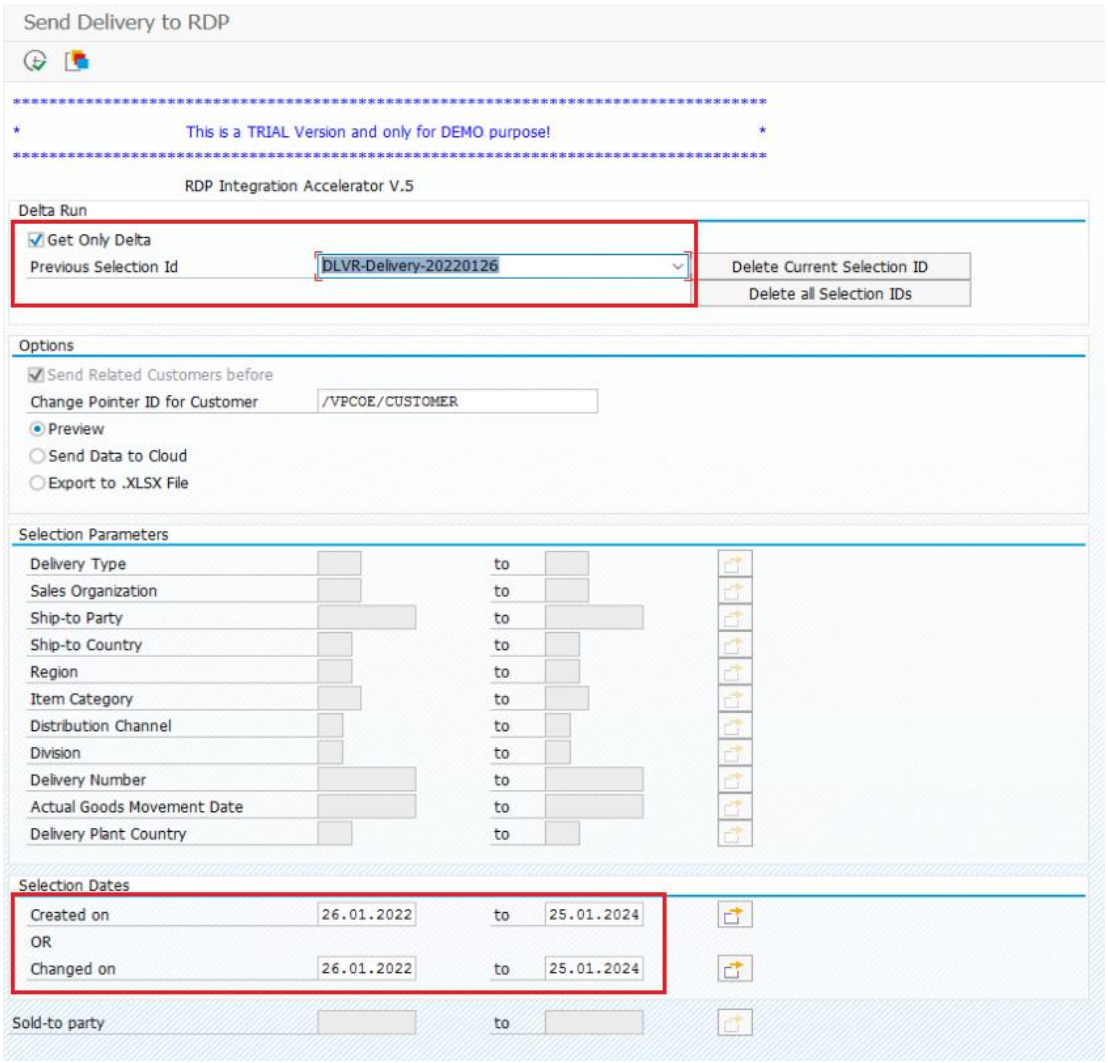
- Newly created or updated customers since the last customer replication, **and** who are involved in the selected deliveries, will be picked up and replicated.
- No unrelated or unchanged customers will be included, helping to keep the replication scope precise and efficient.



Delta load approach for Delivery documents report:

- All the objects selected from last successful replication, will be sent to RDP.
- Some filter criteria become unchangeable, to assure delta load consistency for all filters combinations used before
  When an initial load is successfully executed, the selection parameters used for initial load are saved into so called ‘Selection ID’ together with selection parameters and creation date/change date.
- The **Delivery Document Created Date/Changed Date** is predefined programmatically in this case. These values contain the date since the next Delta can be retrieved up to current

date. In this way, all the new created/posted documents after last successful replication till current date will be selected for delta load.

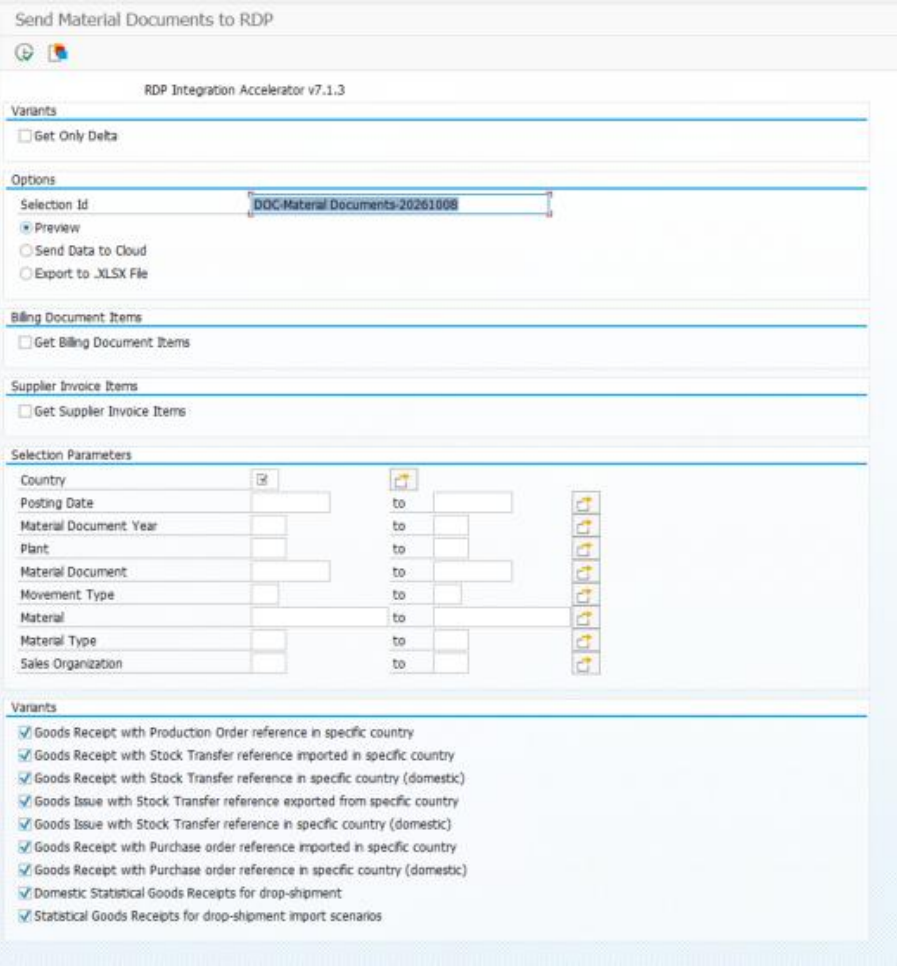


Catch Weight Materials:
When Catch Weight is enabled in system, the quantity may vary from product to product. In such cases, the parallel unit of measure can be used. Therefore, the quantity will differ from standard materials.

For Catch Weight materials, the following mapping will be applied for delivery items:

- The base_quantity will be set to the sales quantity (field LFIMG in delivery item).
- The base_unit will be set to the sales unit (field VRKME/LFIME in delivery item in ECC/S4H).

Loading Material documents
Open transaction code VPCOE/RDP and select service ‘Material Documents’ (alternatively you can run the report /VPCOE/MAT_DOC_TO_RDP):



All the goods movements according to selection criteria will be selected. Country is a mandatory selection filter. Selected country can represent Ship From and/or Ship To country, depending on the variant selected.

Group 1: Goods Receipt with Production Order reference in specific country

All goods receipts will be replicated with manufacturing order reference, but no supplier and customer reference and not from a stock transfer.

Group 2: Goods Receipt with Stock Transfer reference imported in specific country

All goods receipt with stock transfer reference imported in selected country will be replicated.

Group 2b: Goods Receipt with Stock Transfer reference in specific country (domestic)

All goods receipt with stock transfer reference within same country will be replicated.

Group 3: Goods Issue with Stock Transfer reference exported from specific country

All goods issue with stock transfer reference exported from selected country will be replicated.

Group3b: Goods Issue with Stock Transfer reference in specific country (domestic)

All goods issue with stock transfer reference within same country will be replicated.

Group 4: Goods Receipt with Purchase order reference imported in specific country

All goods receipt with purchased order reference imported in specific country and not from a stock transfer.

Group 5: Goods Receipt with Purchase order reference in specific country (domestic)

All goods receipt with reference to Purchase order within the same country (Domestic procurement).

Group 6: Statistical Goods Receipt with Purchase Order reference imported in specific country

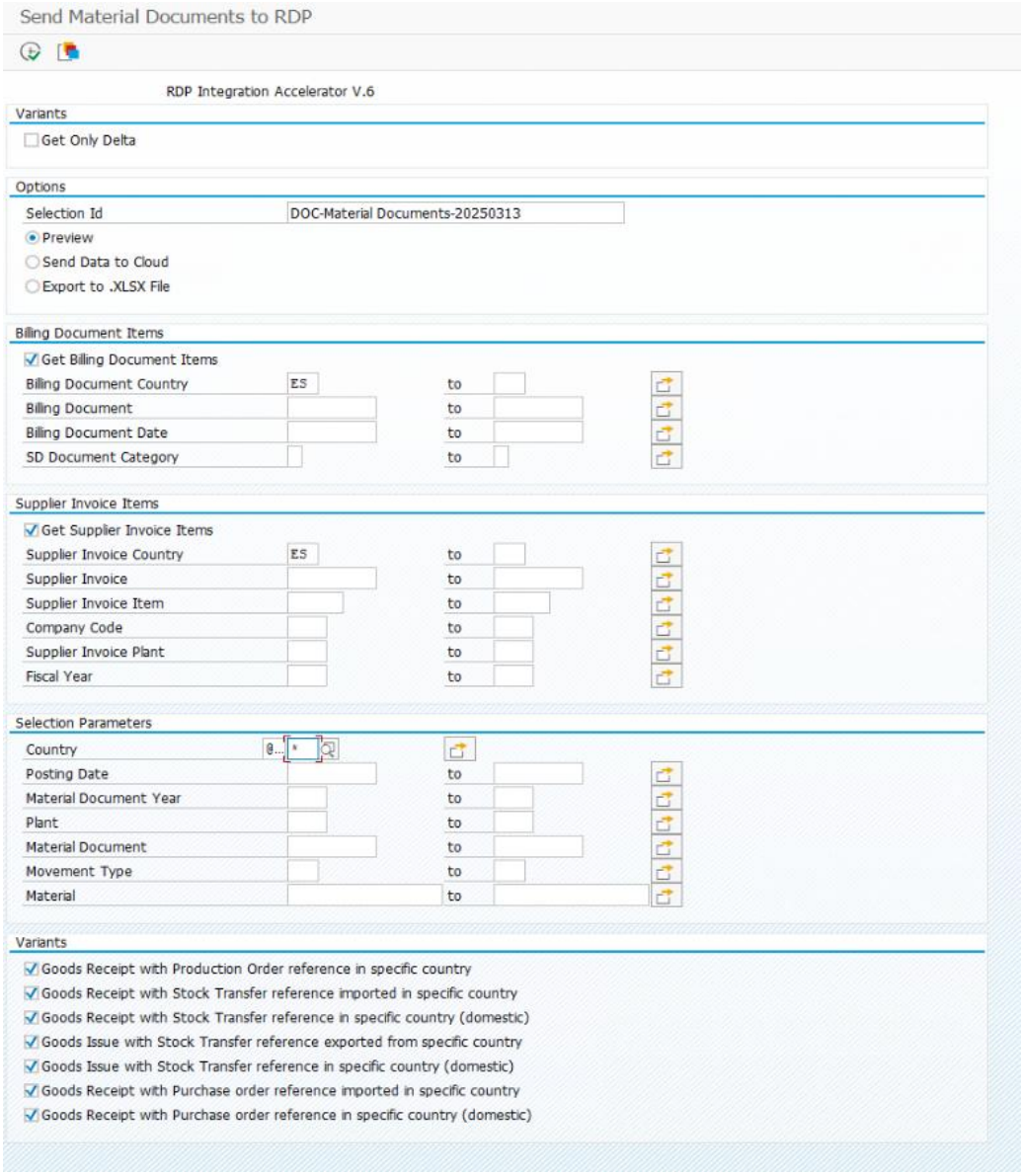
All statistical goods receipts with a purchase order reference imported in the selected country will be replicated. Statistical goods receipts are used for third-party drop shipment scenarios and do not result in a physical stock movement.

Group 7: Statistical Goods Receipt with Purchase Order reference in specific country (domestic)

All statistical goods receipts with a purchase order reference within the same country will be replicated. Statistical goods receipts are used for domestic third-party drop shipment scenarios and do not result in a physical stock movement.

Initial load approach for Material documents report:

- All the objects selected based on filters criteria and for variant(s) selected will be sent to RDP
- All goods movements according to the selection criteria will be selected. Country is a mandatory selection filter. The selected country can represent the Ship From and/or Ship To country, depending on the variant selected.
- If the **Billing Documents** and/or **Invoice Documents** checkboxes are selected, the system retrieves the respective documents based on the defined selection criteria.
- **Country Restriction:** Billing documents and invoice item documents are **selected only for Spain country**.



Delta load approach for Material documents report:

- All the objects selected (based on posting date) from last load, will be sent to RDP
- Some filter criteria become unchangeable, to assure delta load consistency for all filters combinations used before
  When an initial load is successfully executed, the selection parameters used for initial load are saved into so called ‘Selection ID’ together with selection parameters and creation date/posting date.

- The **Material Document Posting Date** is predefined programmatically in this case. These values contain the date since the next Delta can be retrieved up to current date. In this way, all the new posted documents after last successful replication till current date will be selected for delta load.



Important Notes for Material Documents Loading

- **Stock Type and Specialized Stock Type:**
  Goods movement data are not restricted based on stock type or specialized stock type. All relevant data is extracted and transferred without applying stock-specific restrictions.
- **Emergency Corrections:**
  If an emergency correction is performed after data extraction, it is recommended to re-execute the initial load to ensure that corrected data is transferred to RDP.
- **Data Overwrite During Import:**
  For all material documents, a new import will overwrite and delete the existing data in RDP. User needs to ensure that the most accurate data is reloaded when necessary.
- No link between the documents of a **two step stock transfer** (with movement types 303/313 and 305/315).
  - According to SAP Note 1800451, **there is no direct system link to a referenced material document** in standard SAP logic/tables.

- Even using the **movement type directly** does not help in determining the reference document.
- SAP recommends populating either the **Document Header Text (MKPF-BKTXT)** or the **Item Text (MSEG-SGTXT)** field with the reference material document number.
- This reference can be **automatically maintained by customers** using an **exit in standard transactions**.
- Without maintaining this reference, it is **not possible to identify referenced documents** in cases where the material document has **no order reference**.
- In the context of the **RDP Integration Accelerator**, we recommend that customers maintain the **MSEG-SGTXT field** with the reference material document number using a custom **user exit**.
- If this approach is implemented, the accelerator will be able to **extract reference documents** reliably based on the maintained text field.

- **Handling Movement Types 301, 303, 305 in Material Documents**
  During the loading of **material documents**, the Integration Accelerator retrieves relevant data from the source system and processes it according to predefined conditions. Standard filters and mapping rules are applied to ensure accurate and consistent data replication.

  Default Behavior for Movement Types

  - By default, the Integration Accelerator does not use movement types as selection criteria.
  - However, **movement types 301, 303, and 305** are included in the standard logic to ensure correct data selection and replication.
  - If a customer has defined **custom movement types (Z-types)** corresponding to any of these standard movement types (**301, 303, or 305**), they must be explicitly specified in the BAdI implementation using the method:

    /VPCOE/IF_ADJ_DATA_RETRIEVAL~DEFINE_MOVEMENT_TYPE

    When to Use This Method:

    - **Custom Movement Types Exist** – If custom movement types replace standard movement types (**301, 303, or 305**) and need to be considered during data selection.
    - **Ensuring Correct Data Selection** – To apply the appropriate selection conditions for these custom movement types.

    No Implementation of this method is required if:

    - No custom movement types exist for **301, 303, and 305**.

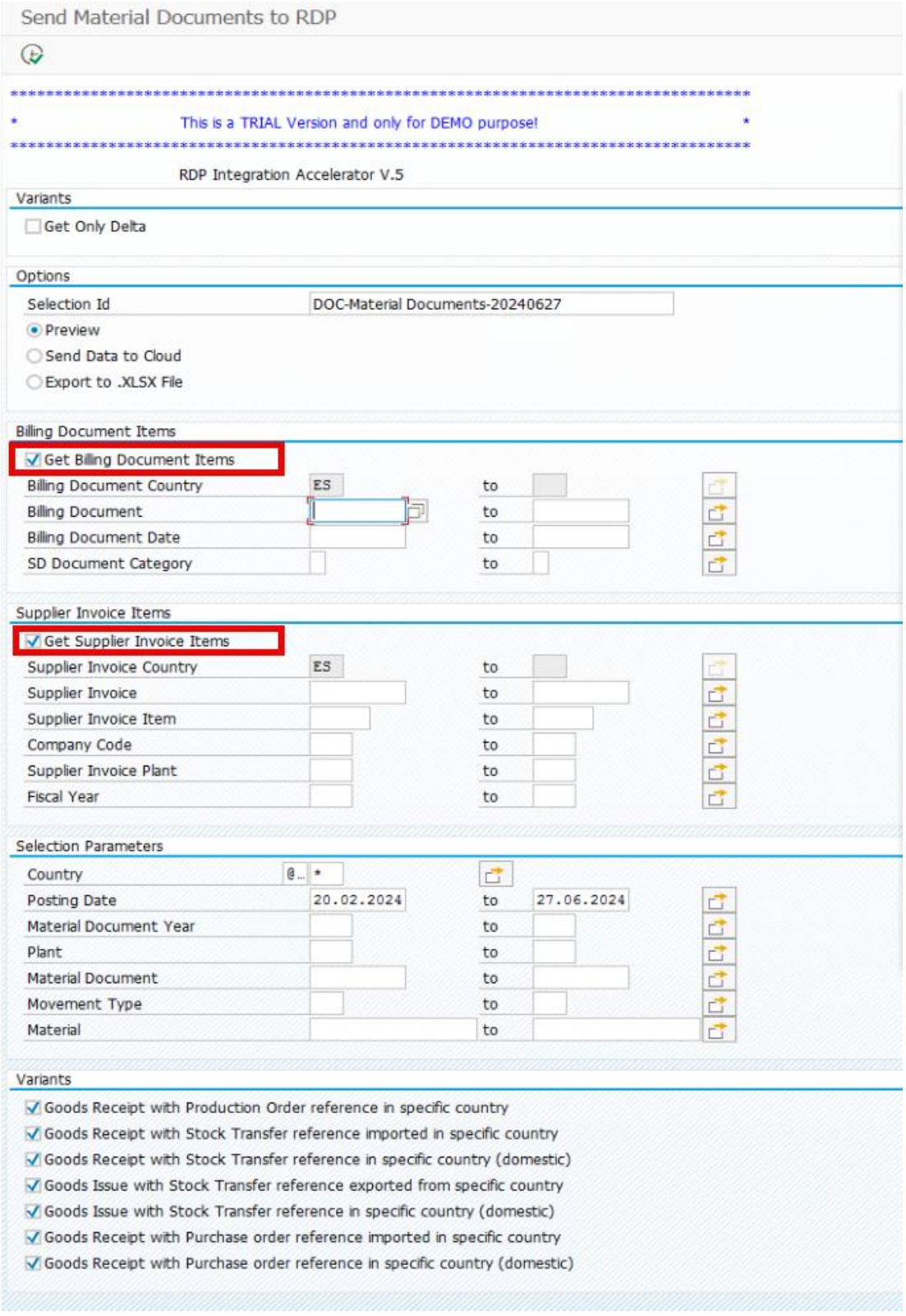
For more details on badi implementation example please have a look in the method itself of sample class implementation and here.

Loading Billing Document Items and Supplier Invoice Items

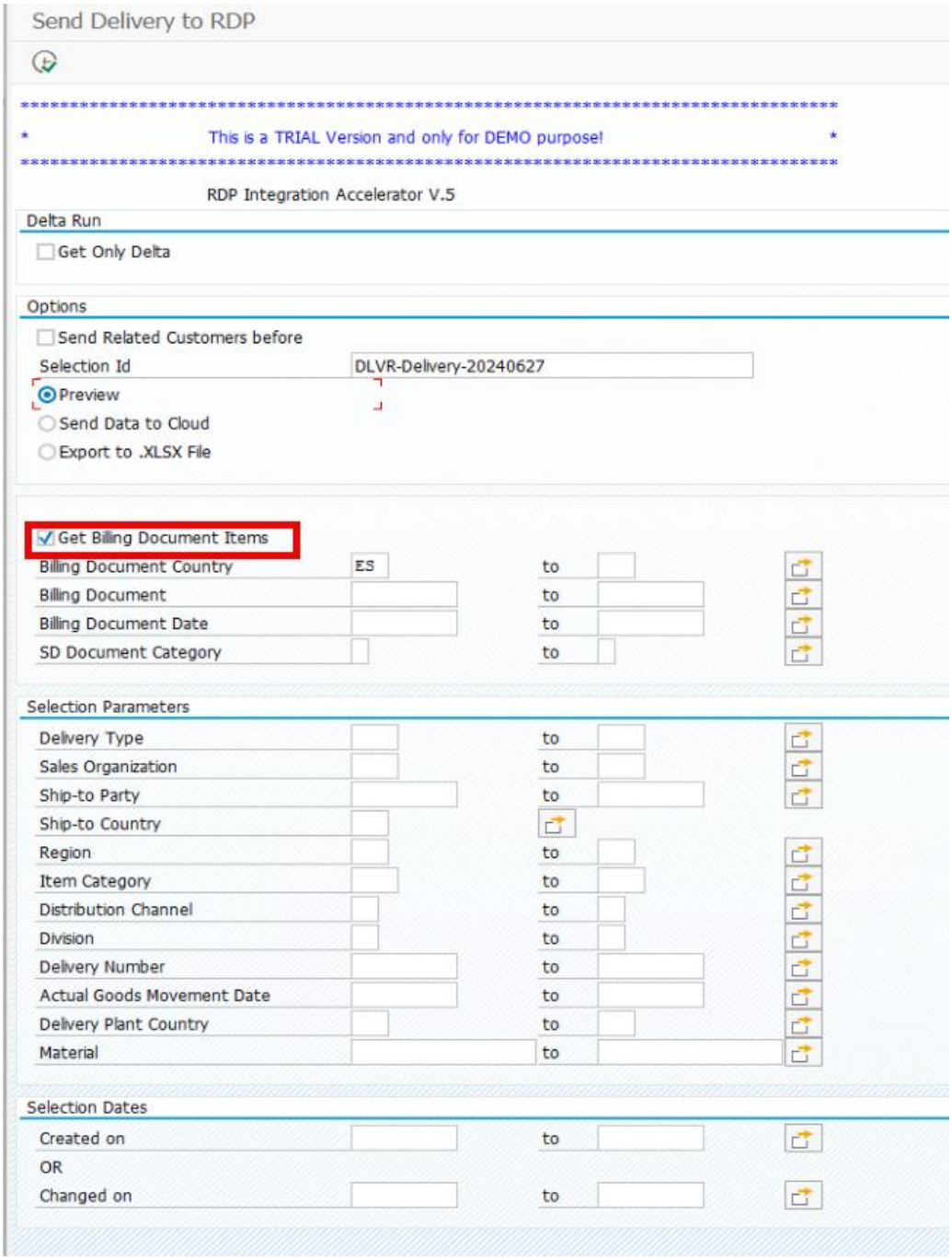
For loading billing document items and supplier invoice items, the existing reports for material documents and delivery replication should be used.

User needs to select the respective checkboxes available to select and send billing documents and/or invoice document items for respective material documents selected. Only the matching objects will be replicated.

Material documents:


Send Material Documents to RDP
This is a TRIAL Version and only for DEMO purpose!
RDP Integration Accelerator V.5
Variants
Get Only Delta
Options
Selection Id
DOC-Material Documents-20240627
Preview
Send Data to Cloud
Export to .XLSX File
Billing Document Items
Get Billing Document Items
Billing Document Country
ES
Billing Document
Billing Document Date
SD Document Category
Supplier Invoice Items
Get Supplier Invoice Items
Supplier Invoice Country
ES
Supplier Invoice
Supplier Invoice Item
Company Code
Supplier Invoice Plant
Fiscal Year
Selection Parameters
Country
Posting Date
20.02.2024
27.06.2024
Material Document Year
Plant
Material Document
Movement Type
Material
Variants
Goods Receipt with Production Order reference in specific country
Goods Receipt with Stock Transfer reference imported in specific country
Goods Receipt with Stock Transfer reference in specific country (domestic)
Goods Issue with Stock Transfer reference exported from specific country
Goods Issue with Stock Transfer reference in specific country (domestic)
Goods Receipt with Purchase order reference imported in specific country
Goods Receipt with Purchase order reference in specific country (domestic)

Deliveries:



This replication for additional Billing documents/Supplier Invoices is required **only for Spanish customers** therefore replication for billing documents/invoices are restricted to only documents related to Spain (therefore Billing Document Country and Supplier Invoice Country pre-defaulted with 'ES' value) and only for documents with status Posted.

Supplier invoices and billing documents information for the transactional documents will be selected accordingly as below:

- A reference to **Supplier Invoice** with link to Material document for:
    - *Goods Receipts with Purchase Order reference imported in Specific country* where
        - Country <> ES and ShipToCountry = ES
    - *Goods Receipts with Purchase Order reference in specific country (domestic)* where
        - ShipFromCountry = ES and ShipToCountry = ES
- A reference to B**illing Document** with link to material document for:
    - *Goods Receipts with Stock Transfer reference imported in specific country* where

        - ShipFromCountry <> ES and ShipToCountry = ES
        - ReferenceSDDocumentCategory = Delivery (J) , Reversal (T)
        - billings document posting status VBRK-RFBSK = C, H (C-“Posting document has been created'; H- “Posted via invoice list”)

    - *Goods Issue with Stock Transfer exported from Specific country* where
        - ShipFromCountry = ES and ShipToCountry <> ES
        - ReferenceSDDocumentCategory = Delivery (J) , Reversal (T)
        - billings document posting status VBRK-RFBSK = C, H (C-“Posting document has been created'; H- “Posted via invoice list”)

- For domestic stock transfer - no billing documents is required for
    - *Goods Receipts with Stock Transfer reference in specific country (domestic)*
    - *Goods Issues with Stock Transfer reference in specific country (domestic)*

- For deliveries as exports
    - A reference to billing documents with link to Material document where
        - ShipFromCountry = ES and ShipToCountry <> ES
        - ReferenceSDDocumentCategory = C, G, H, I, K, L, J, T
        - billings document posting status VBRK-RFBSK = C, H (C - “Posting document has been created'; H- “Posted via invoice list”)

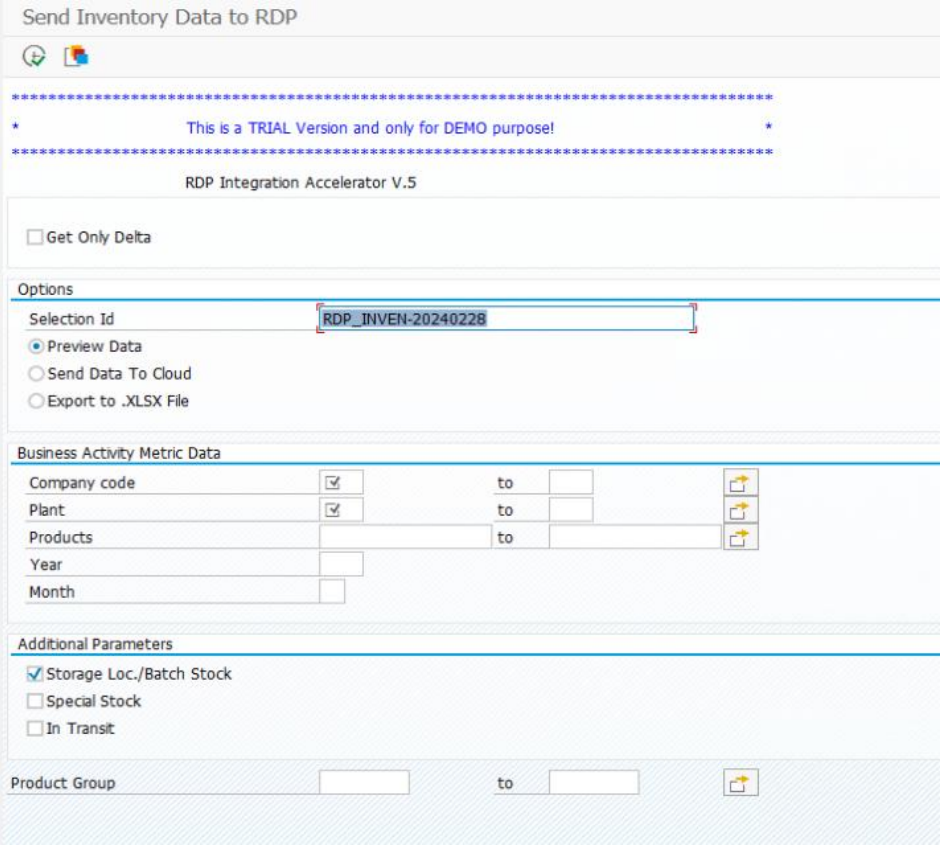
- Cancellations are excluded from reporting therefore no reference documents need to be sent for these.
- Returns are not excluded from reporting, therefore reference to account documents are replicated for:
    - Material documents with PO return - domestic
    - Material documents with PO return - not domestic
    - Material documents with Stock Transfer return - not domestic

- No replication of Supplier Invoices/Billing Documents is required for the variant *Goods Receipt with Production Order reference*.

Additional details for attribute - field mapping for billing and supplier invoice items can be found on Billing Document Item | SAP Help Portal and Supplier Invoice Item | SAP Help Portal.

Loading Inventory

Open transaction code VPCOE/RDP and select service ‘Inventory’ (alternatively you can run the report /VPCOE/INVENTORY_TO_RDP):



Actual stock Inventory data will be extracted per material and plant for a specific key data (e.g. end of month).

Initial load approach for Inventory report:

- all the objects selected based on filters criteria will be sent to RDP
- data will be extracted **for specified document year/month** only (inventory on last day of the month)

Delta load approach for Inventory report:

- all the objects selected based on filters criteria will be sent to RDP
- All objects (Inventory stock) **for the specified month only** will be selected
- Year and Month are predefined with current year and month in delta load

Each month it will display the closing stock based on ***material and company code/plant combination.*** Every monthly extract contains all combinations of materials and company code/plant, even if the inventory quantity of that specific month is “0”. Negative stocks are not sent to RDP.

Note: Zero or negative stocks are not sent to RDP when using SDF API.

Also, please note that all stock type/categories are being considered.

Note: Company code, Year and Month are selection parameters that can be set on initial load and delta load also. All the other parameters are read-only on delta load.

Integration accelerator is using standard report MB5B to extract inventory data. If there are performance issues when selecting a large number of materials with storage location stocks (without batch or special stocks) it is recommended to implement sap note 2459529 - MB5B: Performance optimization during the selection of the storage location stock - SAP for Me

and SAP note 1516684 - Enhancing MSEG with MKPF fields - performance optimization - SAP for Me

Extensibility for RDP ERP Integration reports

The RDP Integration Accelerator provides extensibility through the BAdI **/VPCOE/ADJUST_DATA_RETRIEVAL**, enabling customer-specific adaptations for various integration scenarios.

1. Customer-Specific Adaptations

The BAdI **/VPCOE/ADJUST_DATA_RETRIEVAL** should be implemented to enhance RDP integration scenarios by:

- Adapting the **reading of source data** from other objects.
- Modifying the **mapping of data attributes** to interface attributes.
- Mapping custom values to ensure accurate data replication.

This BAdI is relevant if:

- Different source data are required for entity selection.
- Custom attributes need to be added during replication.

2. Applicable Reports

The following reports use the **/VPCOE/ADJUST_DATA_RETRIEVAL** BAdI:

Report Name	Description
/VPCOE/BATCH_TO_RDP	Send Batch data to RDP.
/VPCOE/CONFIGURATION_TO_RDP	Send Configuration data to RDP.
/VPCOE/CUSTOMER_TO_RDP	Send Customer data to RDP.
/VPCOE/DELIVERY_TO_RDP	Send Delivery data to RDP.
/VPCOE/INCOTERMS_TO_RDP	Send Incoterms data to RDP.
/VPCOE/INVENTORY_TO_RDP	Send Inventory data to RDP.
/VPCOE/MAT_DOC_TO_RDP	Send Material document data to RDP.
/VPCOE/ORG_DATA_TO_RDP	Send Organization data to RDP.
/VPCOE/PRODUCT_TO_RDP	Send Product data to RDP.
/VPCOE/SUPPLIER_TO_RDP	Send Supplier data to RDP.

Note: When any of these reports are executed, the BAdI implementation is triggered to apply customer-specific enhancements.

3. Interface and Methods in BAdI Implementation

The BAdI implements the interface **/VPCOE/IF_ADJ_DATA_RETRIEVAL** which contains the following methods:

Method Name	Description
ADJUST_DATA_RETRIEVAL	Adjust API data retrieval logic.
ADJUST_JSON	Modify the JSON payload based on customer-specific requirements.
ADJUST_MAPPING	Adapt or overwrite standard VPCOE mapping.
ADJUST_TEXT_MAPPING	Adjust mapping for text-based attributes.

GET_EXT_DATA	Retrieve extension data for Product and/or Customer scenarios.
SKIP_SELECTION	Skip standard data selection and apply custom selection logic.
ADJUST_BUILD_WHERE_FOR_VARIANT	Define custom WHERE conditions for selection logic.
DEFINE_MOVEMENT_TYPE	Implement custom movement type logic (needed only for movement types 301, 303, 305 and their reversals).

4. Method Descriptions and Usage

4.1 SKIP_SELECTION

Allows skipping standard data selection logic.

If this method is implemented, the data retrieval logic must be fully customized using ADJUST_DATA_RETRIEVAL.

4.2 ADJUST_DATA_RETRIEVAL

Enables customer-specific data selection and adjustments before transmitting data.

4.3 ADJUST_JSON

Allows modifications to the JSON payload before sending data to RDP.

4.4 ADJUST_MAPPING

Supports custom mapping of attributes from the SAP backend to the RDP JSON structure.

4.5 GET_EXT_DATA

Retrieves additional extension data for Product and Customer replication.

4.6 ADJUST_TEXT_MAPPING

Provides customization for text-based attribute mappings.

4.7 ADJUST_BUILD_WHERE_FOR_VARIANT

Allows customization of WHERE conditions for selection logic, applicable for Material Document replication.

4.8 DEFINE_MOVEMENT_TYPE

- Specifies custom Z movement types corresponding to standard movement types **301, 303,** and **305**.

Ensures correct data selection for custom movement types aligned with standard accelerator logic.

5. Handling Custom Movement Types in Material Documents

Background

- The Integration Accelerator does **not use movement types** as selection criteria by default.
- However, for movement types **301, 303, and 305**, additional logic is applied in the standard accelerator to ensure correct data selection.
- If **custom movement types (Z-types)** exist that correspond to these standard movement types, they must be defined explicitly in the BAdI using the method:

```
METHOD /VPCOE/IF_ADJ_DATA_RETRIEVAL~DEFINE_MOVEMENT_TYPE.
  CASE iv_code.
    WHEN sc_variants-gr_st.
```

```
      mt_movement_type = VALUE #( ( sign = 'I' option = 'EQ' low = 'Z305' )
                                  ( sign = 'I' option = 'EQ' low = 'Z303' )
                                  ( sign = 'I' option = 'EQ' low = 'Z301' ) ).
      mt_stock_change_cat = VALUE #( ( sign = 'I' option = 'EQ' low = 'Z305' ) ).

    WHEN sc_variants-gi_st.
      mt_movement_type = VALUE #( ( sign = 'I' option = 'EQ' low = 'Z303' )
                                  ( sign = 'I' option = 'EQ' low = 'Z301' ) ).
  ENDCASE.
ENDMETHOD.
```

Usage

- Use this method to map **custom movement types** to the standard types **301, 303 and/or 305** to ensure correct data retrieval and mapping.
- If no custom movement types are defined, this method does not require implementation.

6. Sample BAdI Implementation

A **sample BAdI implementation** is provided in class:

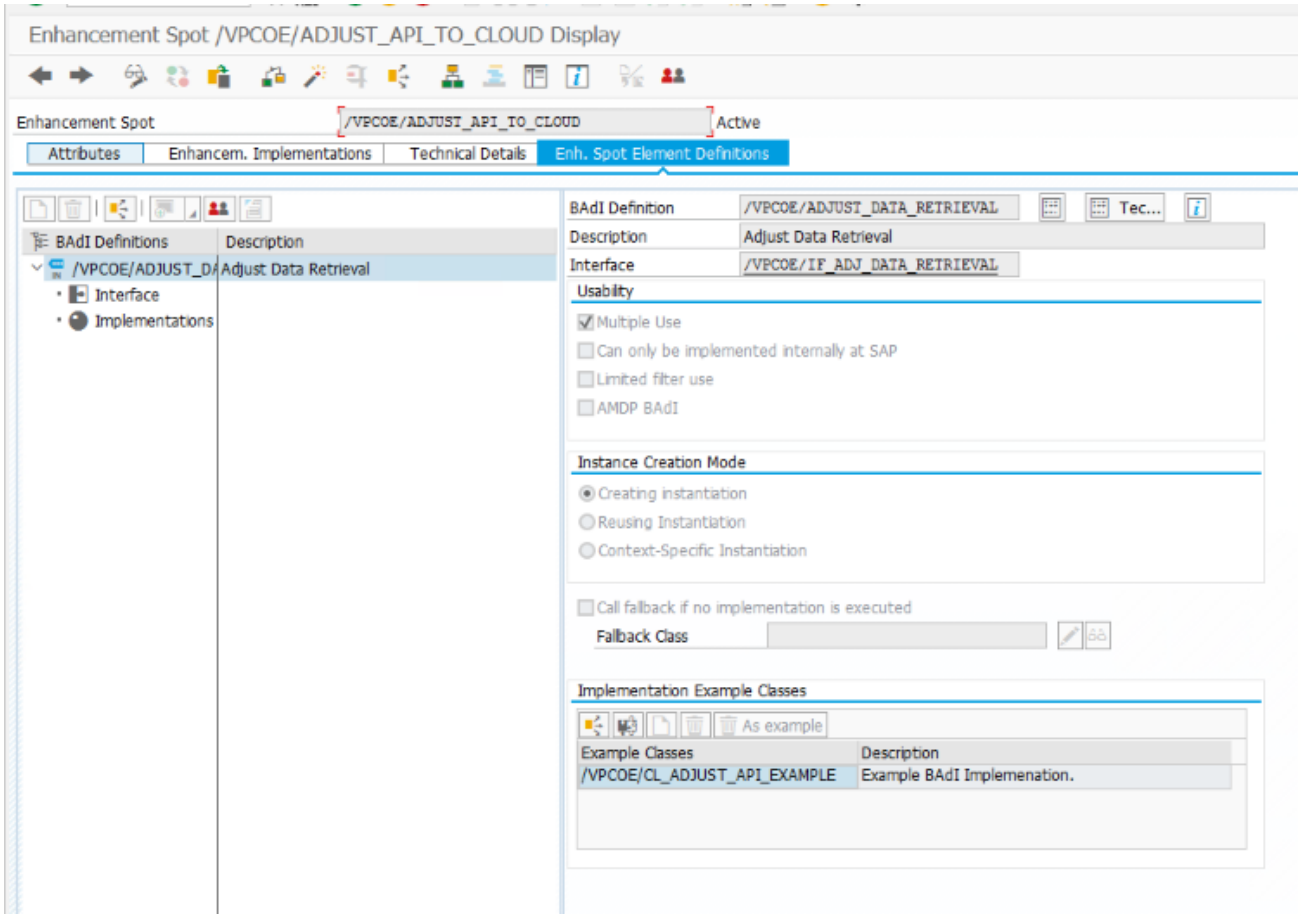
- **Class Name:** /VPCOE/CL_ADJUST_API_EXAMPLE

BAdI Implementation Guidelines:

- To **skip standard data selection**, set the parameter CV_SKIP = 'true' in **SKIP_SELECTION**. This ensures that the standard integration accelerator selection logic is bypassed, allowing full customization via /VPCOE/IF_ADJ_DATA_RETRIEVAL~ADJUST_DATA_RETRIEVAL
- If custom mapping of root nodes or subnodes is required, implement **ADJUST_MAPPING**.
- For direct JSON modifications, use **ADJUST_JSON**.
- In Material Document replication BAdI implementations, after defining custom selection logic in **ADJUST_DATA_RETRIEVAL**, it is mandatory to call adjust_header to ensure that ship-to and ship-from countries are correctly derived.
    - For further customization of country determination, implement **DEFINE_COUNTRY** accordingly.

7. Accessing BAdI Implementation

- To view or modify the sample implementation, use transaction **SE18** to explore the BAdI definition.
- For details of the parameters used in each method, open transaction **SE24**, enter the interface **/VPCOE/IF_ADJ_DATA_RETRIEVAL**, and execute it.

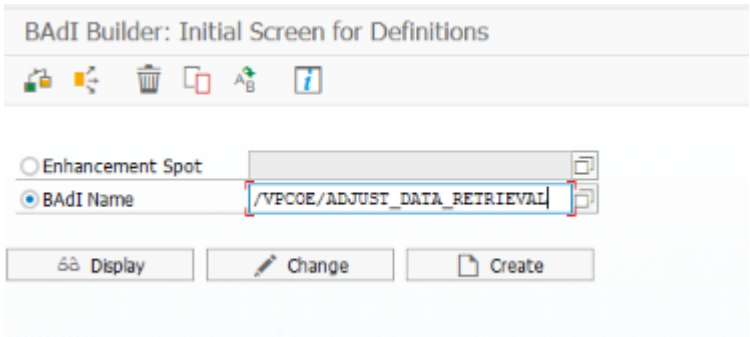


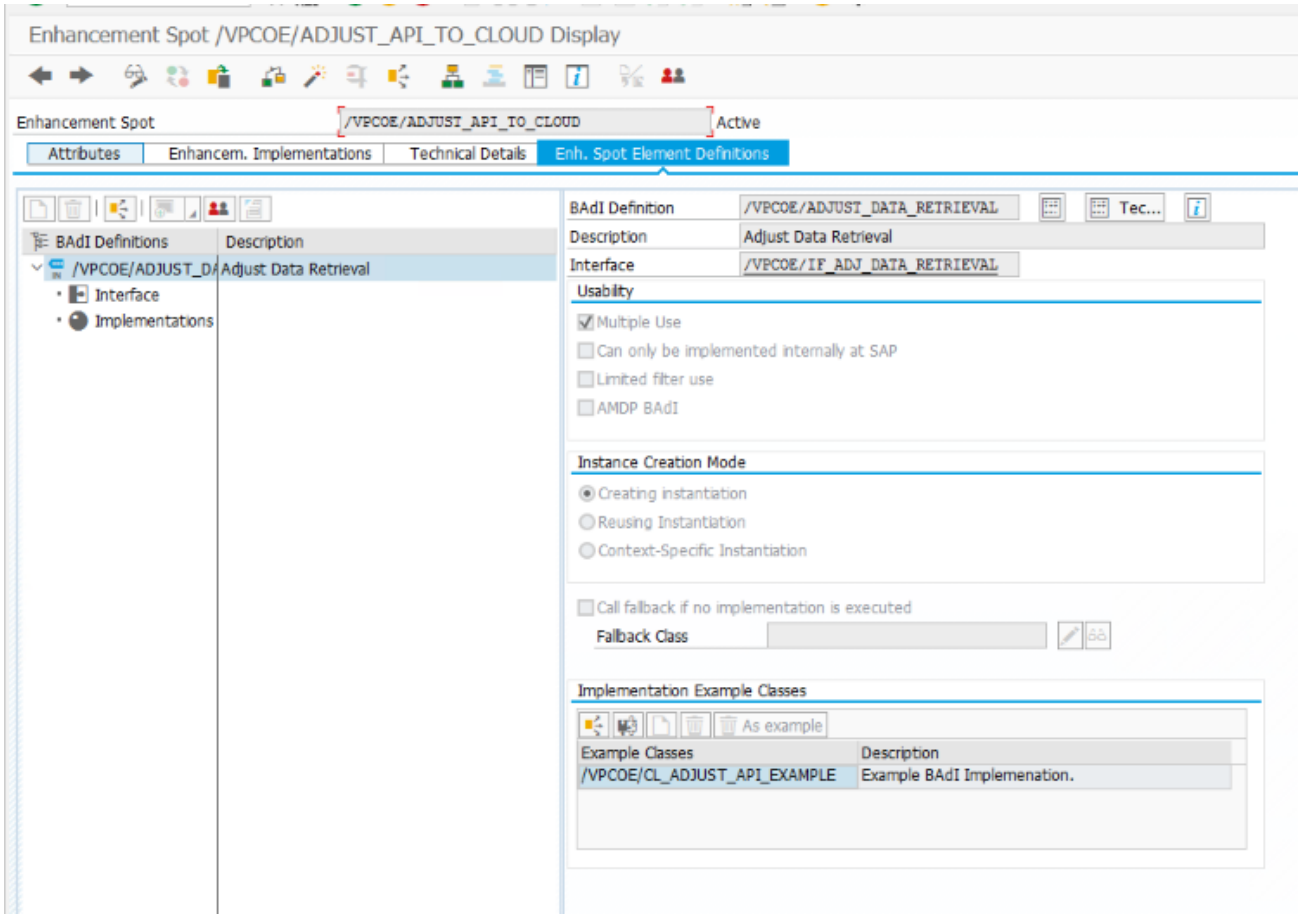
Sample Implementation

A sample BAdI implementation is provided in class /VPCOE/CL_ADJUST_API_EXAMPLE as a reference for custom development.

For more details, execute transaction SE18 to access the sample BAdI implementation.

Execute transaction SE18 to get more details regarding sample BAdI implementation.



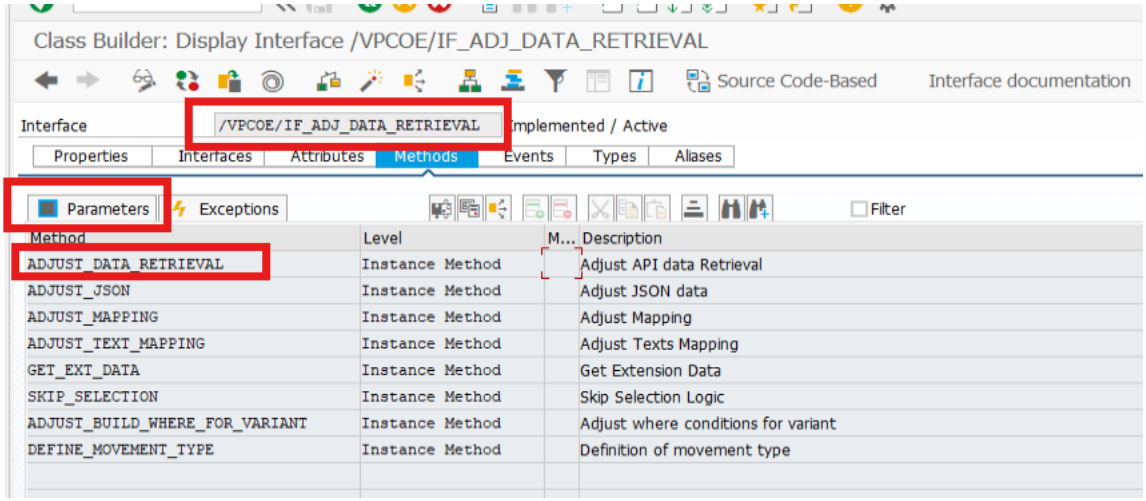


Class Name: /VPCOE/CL_ADJUST_API_EXAMPLE

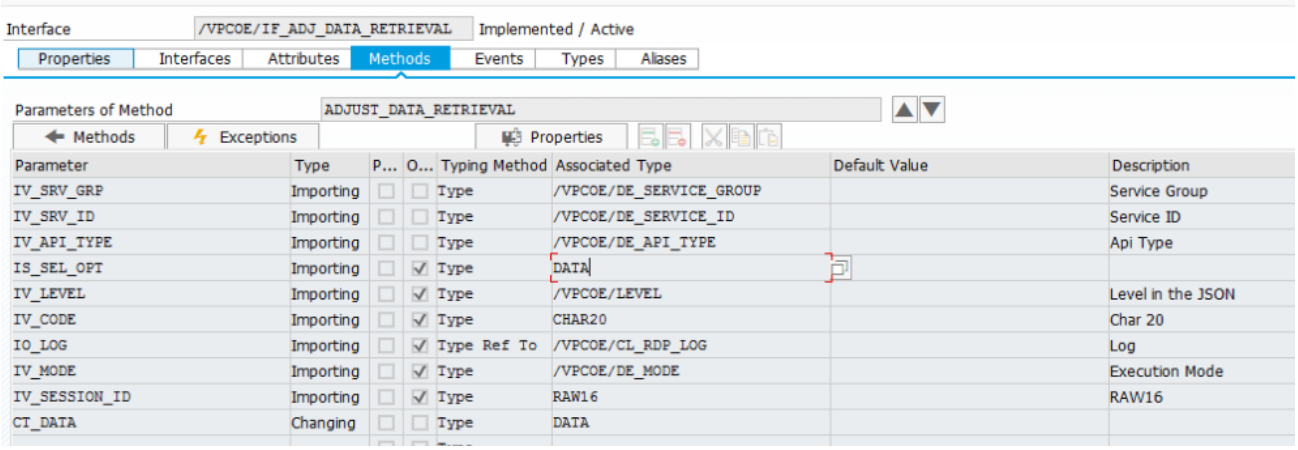
Badi Implementation Guidelines

- To **skip standard data selection**, set the parameter **CV_SKIP = 'true'** within /VPCOE/IF_ADJ_DATA_RETRIEVAL~SKIP_SELECTION. This ensures that the standard VPCOE selection logic is bypassed, allowing full customization via /VPCOE/IF_ADJ_DATA_RETRIEVAL~ADJUST_DATA_RETRIEVAL.
- If **custom mapping** of root nodes or subnodes is required, use /VPCOE/IF_ADJ_DATA_RETRIEVAL~ADJUST_MAPPING.
- For direct JSON modifications, implement /VPCOE/IF_ADJ_DATA_RETRIEVAL~ADJUST_JSON.
- In Material Document replication BAdI implementations, after defining custom selection logic in /VPCOE/IF_ADJ_DATA_RETRIEVAL~ADJUST_DATA_RETRIEVAL, it is **mandatory** to call **adjust_header**. This ensures that **ship-to** and **ship-from** countries are correctly derived for logistics processing.
    - If further customization of country determination is needed, implement DEFINE_COUNTRY accordingly.

To get further details of the parameters of each method, open transaction SE24, give the interface details /VPCOE/IF_ADJ_DATA_RETRIEVAL and execute it.



Select the method for which you want to see the parameter details and click on parameters, you will get all parameters details for that method.



Standard settings

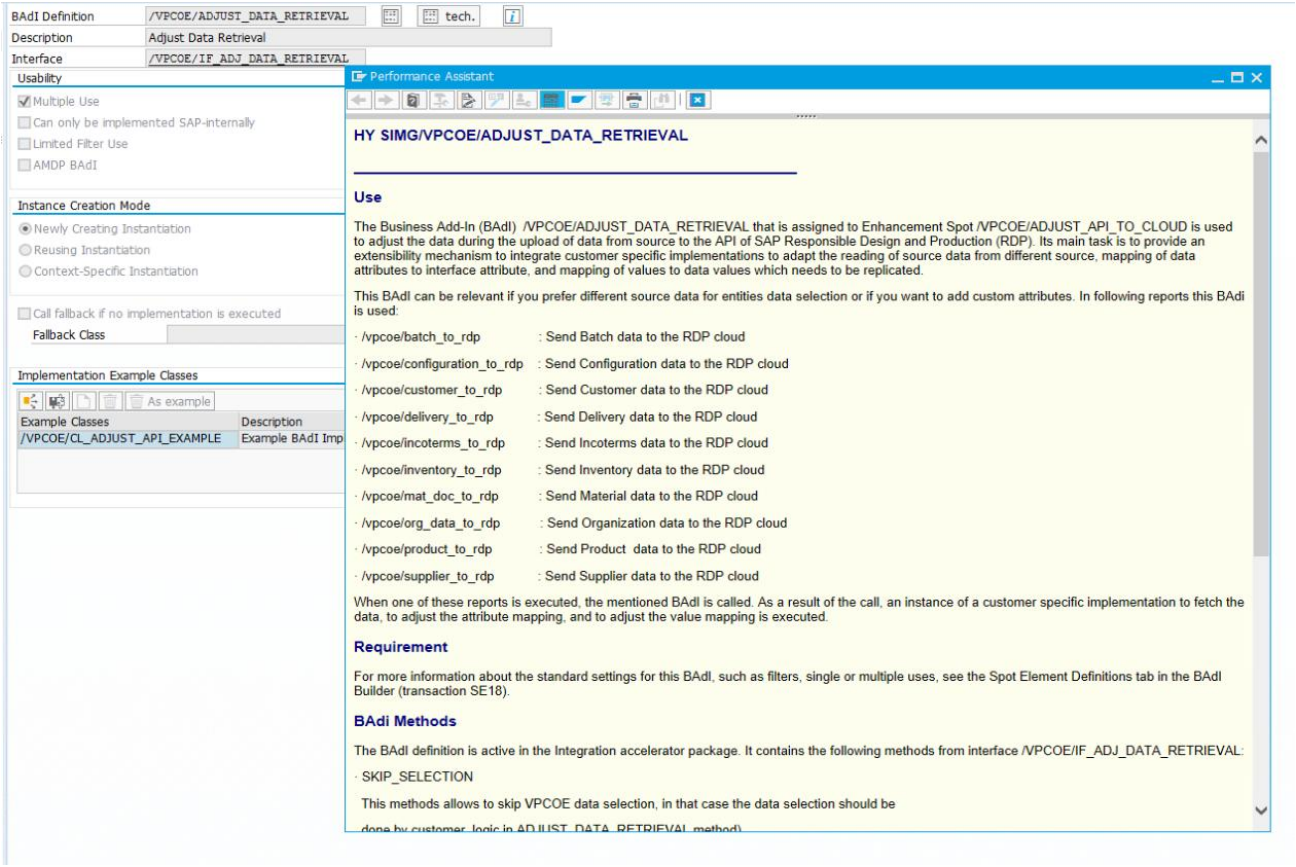
For more information about the standard settings (filters, single or multiple uses), see the *Enh. Spot Element Definitions* tab in the *BAdI Builder* (transaction SE18).

See also

This BAdI uses the interface `/VPCOE/IF_ADJ_DATA_RETRIEVAL`.

For more information, display the interface in the Class Builder (Transaction SE24).

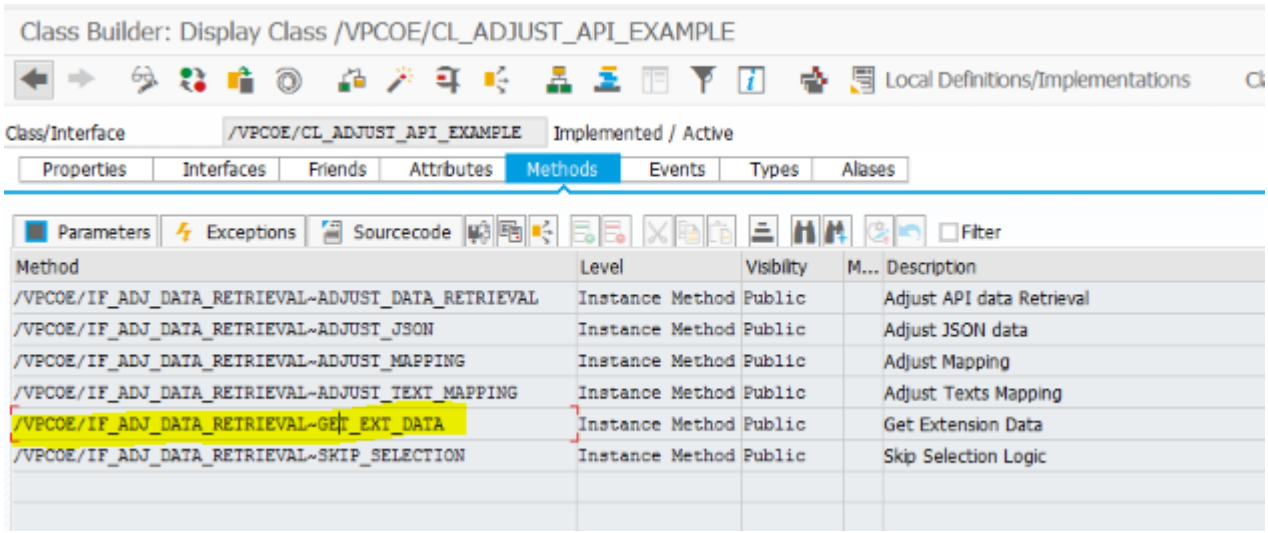
Additional BAdI documentation is available for the Badi:



Product and Customer extensions

Class /VPCOE/CL_ADJUST_API_EXAMPLE is sample implementation for your reference to implement the actual BAdI /VPCOE/ADJUST_DATA_RETRIEVAL for Product/Customer extension reports as per your requirement.

The method to be implemented for extension data is
/VPCOE/IF_ADJ_DATA_RETRIEVAL~GET_EXT_DATA



For more information on datatypes for extension attributes please have a look at ‘RDP Fields Mapping v6.xsls’ file.

Also, additional details on all available attributes and its values for extension data are available here:

[Product Extension | SAP Help Portal]

Product Extension:

<table>
<tr><th>Column</th><th>API Element</th><th>SAP

ECC/ S/4HANA

Attribute</th><th>Description</th></tr>
<tr><td>Product Number</td><td>id</td><td>MATNR</td><td>ID of the product</td></tr>
<tr><td>Reference Quantity</td><td>referenceQuantity</td><td>DEC10</td><td>Reference quantity for fee in base unit of measure.

The reference quantity is used during fee calculation to provide enough decimals for an accurate calculation. The default value is 10,000 and the attribute refers to unit of measure of the product.</td></tr>
<tr><td>Product Type harmonized</td><td>productTypeHarmonized</td><td>CHAR10</td><td>Possible values:

<table><tr><th>Value</th><th>Description</th></tr><tr><td>FIN_GOOD</td><td>Finished good</td></tr><tr><td>RAWMATPACK</td><td>Raw material for packaging</td></tr><tr><td>PACKMAT</td><td>Packaging material</td></tr><tr><td>NOTAPPL</td><td>Not relevant</td></tr><tr><td>SEMIPACKM</td><td>Semifinished Packaging Material</td></tr></table></td></tr>
<tr><td>Product Content</td><td>productContent</td><td>CHAR10</td><td>Possible values:

<table><tr><th>Value</th><th>Description</th></tr><tr><td>BEV_ALC</td><td>Alcoholic beverages</td></tr><tr><td>BEV_ALC_BE</td><td>Beer</td></tr><tr><td>BEV_OTHER</td><td>Other Beverages</td></tr><tr><td>BEV_WATER</td><td>Potable water</td></tr><tr><td>BEV_SOFT</td><td>Soft drinks</td></tr><tr><td>CHEMB2B</td><td>Chemicals in a B2B context (for example for industrial or agricultural use)</td></tr><tr><td>CLEANING</td><td>Cleaning products</td></tr><tr><td>CLOTHING</td><td>Clothing and apparel</td></tr><tr><td>COSM_CARE</td><td>Cosmetics and personal care</td></tr><tr><td>ELECTRONIC</td><td>Electronics</td></tr><tr><td>FOOD</td><td>Food and nutrition</td></tr><tr><td>FOOD_MED</td><td>Medical food</td></tr></table></td></tr>
</table>

			<table><tr><td>FOOD_DIR</td><td>Food for direct consumption</td></tr><tr><td>MED_DEVICE or DEVICE_MED</td><td>Medical Electrical Device</td></tr><tr><td>FURNITURE</td><td>Furniture, furnishings, and household items</td></tr><tr><td>GAS</td><td>Gas, aerosol</td></tr><tr><td>HOME</td><td>Home improvement and gardening</td></tr><tr><td>NONFOOD</td><td>Other non-food</td></tr><tr><td>PAINT</td><td>Paint</td></tr><tr><td>PET</td><td>Pet food and animal supplies</td></tr><tr><td>PHARMA</td><td>Pharmaceutical and medical goods (only for human medicine)</td></tr><tr><td>SANI_WASTE</td><td>Intended to capture Hazardous Sanitary Waste</td></tr><tr><td>TOBACCO</td><td>Tobacco</td></tr><tr><td>TOYS</td><td>Toys</td></tr><tr><td>VET_PHARMA</td><td>Veterinary pharmaceutical products (only for animal application)</td></tr><tr><td>AGRI</td><td>Products designed for agricultural or livestock consumption</td></tr></table>
Brand ID	brand	CHAR30	Brands consist of a name and an ID, they can be assigned to a product. You can create brands in the **Configuration Cockpit** application and assign brands to products in the **Manage Products** application.
Hazardous Product CLPEU	isHazardousClpEu	CHAR5	Indicates if the product is classified as hazardous according to the CLP regulation. The CLP (Classification, Labeling, and Packaging) regulation is a European Union regulation that aligns the European Union legislation with the GHS (Globally Harmonized System of Classification and Labeling of Chemicals), which is managed by the United Nations.
Product comes without packaging	hasNoPackaging	CHAR5	The product comes without packaging. This can apply to bulk products in a container without further packaging or the product is filled into returnable packaging brought by the end consumer, for example, a milk dispenser. These products do not need a packaging composition and can be ignored by most EPR

			reports. Packaging for products in this category is not distributed, collected, cleaned, or reused by the producer or retailer. Instead, they are fully under the control of the end-consumer and therefore would not require a packaging composition on the seller side. This model of packaging aims to avoid single-use packaging entirely.
Product Countries	productCountries		<table><tr><th>API Element</th><th>ECC/S4H attribute</th></tr><tr><td>country</td><td>LAND1</td></tr><tr><td>validFrom</td><td>DATS</td></tr><tr><td>tag</td><td>CHAR10</td></tr><tr><td>isNotEprRelevant</td><td>CHAR5</td></tr><tr><td>eprProductFamily</td><td>CHAR10</td></tr><tr><td>inhouseProductionPercent</td><td>DEC3</td></tr></table>

Example of payload for Product extension:

```
{
    "source": "ECC",
    "elements": [
        {
            "id": "AFR-11-ART-001_____00011",
            "productTypeHarmonized": "PACKMAT",
            "productContent": "BEV_WATER",
            "isHazardousClpEu": true,
            "isDgProductEu": true,
            "hasNoPackaging": true,
            "brand": "1001",
            "referenceQuantity": "10000",
            "productCountries": [
                {
                    "country": "DE",
                    "validFrom": "2021-04-25",
                    "tag": "IYSIUY765H",
                    "isNotEprRelevant": true,
                    "eprProductFamily": "id3",
                    "inhouseProductionPercent": 13
                }
            ]
        }
    ]
}
```

Customer Extension:

<table>
<tr><th>Column</th><th>API Element</th><th>SAP ECC/S/4HANA Attribute</th><th>Description</th></tr>
<tr><td>Customer Number</td><td>id</td><td>KUNNR</td><td>ID of the customer</td></tr>
<tr><td>Tag</td><td>tag</td><td>Customer specific</td><td>Tag that can be assigned to customers. Customer tags can be created in the Configuration Cockpit application. Tags can be assigned to customers in the Manage Customers application.</td></tr>
<tr><td>Role</td><td>role</td><td>Customer specific</td><td>A role that can be assigned to a customer.
Possible values:
<table>
<tr><th>Value</th><th>Description</th></tr>
<tr><td>END_CONSUM</td><td>End-consumer</td></tr>
<tr><td>RETAILER</td><td>Retailer</td></tr>
<tr><td>PROF_CUSTM</td><td>Professional customer</td></tr>
<tr><td>NEXT_PACKP</td><td>Next level packaging producer</td></tr>
<tr><td>HOSPITAL</td><td>Hospital</td></tr>
</table>
</td></tr>
</table>

Example of payload for Customer extension:

```
{
  "source": "ECC",
  "elements": [
    {
      "id": "1000000001",
      "tag": "TAG",
      "role": "RETAILER"
    }
  ]
}
```

Please note, in case you don't have any value for some of the extension attributes, you need to implement the logic in BAdI method '/VPCOE/IF_ADJ_DATA_RETRIEVAL~ADJUST_JSON' to replace empty values with NULL. Below is small example of the code you may introduce for this matter (a similar example it's also on the sample BAdI implementation we are providing):

In BAdI method '/VPCOE/IF_ADJ_DATA_RETRIEVAL~ADJUST_JSON', introduce code below:

```
IF     iv_api_type = /vpcoe/cl_customizing_helper=>sc_api_type-rdp
    AND iv_srv_grp  = /vpcoe/cl_customizing_helper=>sc_grp_id-product
    AND iv_srv_id   = /vpcoe/cl_customizing_helper=>sc_service_id-PRODUCT_EXT.

    LOOP AT ct_json ASSIGNING <ls_json>.
      REPLACE ALL OCCURRENCES OF ':""' IN <ls_json>-elements WITH ': null'.
    ENDLOOP.
  ENDIF.
```

For customer extension *role* attribute, the sample badi impelmentation contains the logic based on RDP recomendation on class and characteristics:

Class	ZCCL_RDP_CUST_EXT
Description	RDP specific Customer Attributes
Class Type	011

Characteristics	Description	Data Type	Length	Decimal	UoM / Values
ZRDP_CUST_EXT_ROLE	Customer Role	CHAR	10	0	Code List
ZRDP_PROD_EXT_TAG	Customer Tag	CHAR	10	0	Customer defined code list

Email notification adaptation

Whenever replication is failing for an entity, if customer maintained email addresses configuration settings, an email will be sent to all maintained email addresses.

BAdI /VPCOE/NOTIFICATION_PROCESSING is available for adapting the email sender / email body content or for overwriting error notification processing.

For more details and example implementation you can have a look at sample BAdI implementation /VPCOE/CL_NOTIFICATION_EXAMPLE.

Excel documents processing adaptation

BAdI /VPCOE/XLS_PROCESSING is available for excel documents processing adjustments.

/VPCOE/XLS_PROCESSING – technical documentation

The Business Add-In (BAdI) /VPCOE/XLS_PROCESSING that is assigned to Enhancement Spot /VPCOE/ADJUST_API_TO_CLOUD, this allows to override the logic for saving the Excel file. With this BADI, you can add your own logic for saving files both in the background and in real time, or completely change it. It also provides the option to change the save path of the file, if necessary.

This BAdI is used in the "/VPCOE/CL_XLS_HANDLER" class in the "SAVE_XLS_FILE" method.

Excel file download is available in the following reports:

- /vpcoe/batch_to_rdp Send Batch data to the RDP cloud
- /vpcoe/configuration_to_rdp Send Configuration data to the RDP cloud
- /vpcoe/customer_to_rdp Send Customer data to the RDP cloud
- /vpcoe/delivery_to_rdp Send Delivery data to the RDP cloud
- /vpcoe/incoterms_to_rdp Send Incoterms data to the RDP cloud
- /vpcoe/inventory_to_rdp Send Inventory data to the RDP cloud
- /vpcoe/mat_doc_to_rdp Send Material data to the RDP cloud
- /vpcoe/org_data_to_rdp Send Organization data to the RDP cloud
- /vpcoe/product_to_rdp Send Product data to the RDP cloud
- /vpcoe/supplier_to_rdp Send Supplier data to the RDP cloud

For more information about the standard settings for this BAdI see the Spot Element Definitions tab in the BAdI Builder (transaction SE18).

BAdI Methods

The BAdI definition is active in the Integration accelerator package. It contains the following method from interface /VPCOE/IF_XLS_PROCESSING:

· ADJUST_FILE_SAVING

As described above, this method allows to rewrite the logic for saving the Excel file or change the path that was specified on the selection screen.

The method uses the following parameters:

- IV_BACKGROUND : An import parameter that is used to determine in which mode the file will be saved
- IO_LOG : An import parameter for writing messages to the LOG
- EV_SKIP_STANDARD_PROCESS : The export parameter. If the value of the parameter is "true", then we skip the standard implementation of the "SAVE_XLS_FILE" method.
- CV_FILE_PATH : Changing parameter. This parameter is passed the path where the file is saved, as specified in the selection screen. Since the type of this parameter is "Change", you can change it.
- CT_XLS_DATA : Change parameter. This is a table that stores all the information of an Excel file in the Binary type

BAdi Implementations

The BAdI implementation must be defined by the user and is not provided by SAP.

Sample implementation

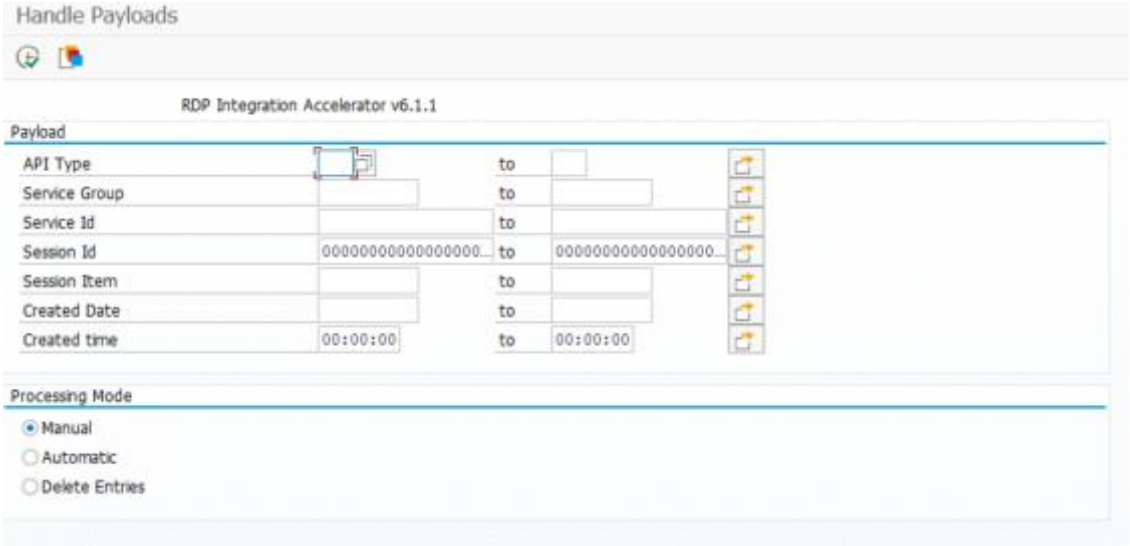
As example see class /VPCOE/CL_XLS_PROCESSING_EXMPL.

Resending failed payloads

If, during data replication, certain payloads fail for various reasons, they are automatically recorded for later analysis and reprocessing.

> ⚠ **Important Note:**
> If data inconsistencies or missing information exist in the data objects in system, simply resending the payload will not resolve the issue. In such cases, the data must first be corrected, and the relevant data extraction report should be executed again to reselect and resend the data.

You can reprocess failed payloads by running the report **/VPCOE/REPROCESS_PAYLOAD** (transaction **SE38**).

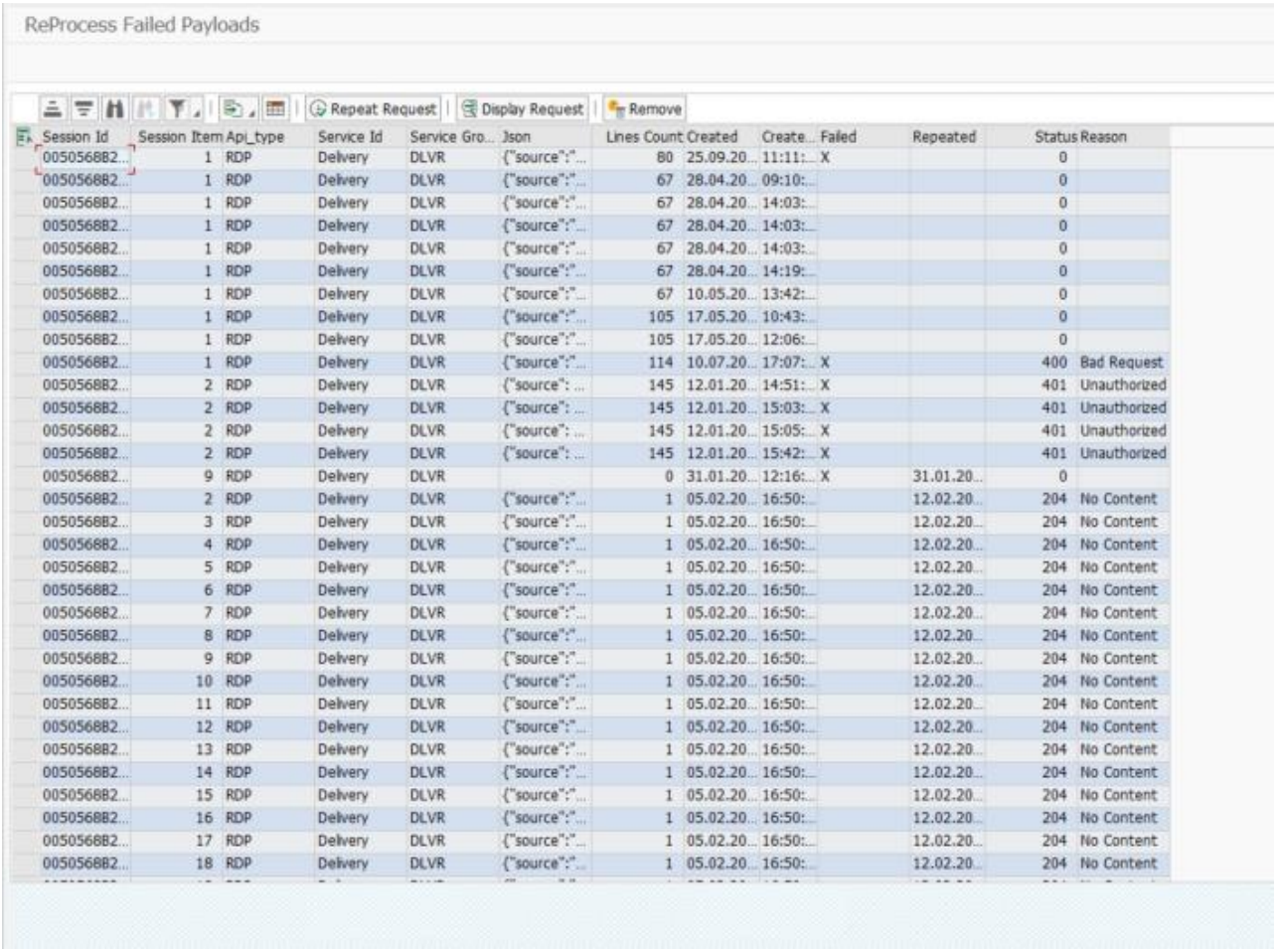


Processing Modes

You can choose between three processing modes:

- **Manual Mode:**
  Allows you to manually review the list of failed payloads, select specific requests, and resend them.
- **Automatic Mode:**
  Automatically resends all failed payloads that match the filter criteria defined on the selection screen. This mode is typically used when scheduling the report as a background job.
- **Delete Entries:**
  This option removes obsolete or irrelevant payload entries from the internal table.
  It is recommended to use this option periodically to clean up old entries, as maintaining a smaller number of records in this table can improve replication performance—especially during JSON payload processing.

When executing the report in Manual Mode, if a failed payload requires additional investigation, you can use the **Display Request** option to view, copy, or export the payload content. This can be helpful for further analysis, for example when testing the payload manually using tools such as **Postman**.



Session Id	Session Item	Api_type	Service Id	Service Gro...	Json	Lines Count	Created	Create...	Failed	Repeated	Status	Reason
005056882...	1	RDP	Delivery	DLVR	{"source":"...	80	25.09.20...	11:11:...	X		0	
005056882...	1	RDP	Delivery	DLVR	{"source":"...	67	28.04.20...	09:10:...			0	
005056882...	1	RDP	Delivery	DLVR	{"source":"...	67	28.04.20...	14:03:...			0	
005056882...	1	RDP	Delivery	DLVR	{"source":"...	67	28.04.20...	14:03:...			0	
005056882...	1	RDP	Delivery	DLVR	{"source":"...	67	28.04.20...	14:03:...			0	
005056882...	1	RDP	Delivery	DLVR	{"source":"...	67	28.04.20...	14:19:...			0	
005056882...	1	RDP	Delivery	DLVR	{"source":"...	67	10.05.20...	13:42:...			0	
005056882...	1	RDP	Delivery	DLVR	{"source":"...	105	17.05.20...	10:43:...			0	
005056882...	1	RDP	Delivery	DLVR	{"source":"...	105	17.05.20...	12:06:...			0	
005056882...	1	RDP	Delivery	DLVR	{"source":"...	114	10.07.20...	17:07:...	X		400	Bad Request
005056882...	2	RDP	Delivery	DLVR	{"source": ...	145	12.01.20...	14:51:...	X		401	Unauthorized
005056882...	2	RDP	Delivery	DLVR	{"source": ...	145	12.01.20...	15:03:...	X		401	Unauthorized
005056882...	2	RDP	Delivery	DLVR	{"source": ...	145	12.01.20...	15:05:...	X		401	Unauthorized
005056882...	2	RDP	Delivery	DLVR	{"source": ...	145	12.01.20...	15:42:...	X		401	Unauthorized
005056882...	9	RDP	Delivery	DLVR		0	31.01.20...	12:16:...	X	31.01.20...	0	
005056882...	2	RDP	Delivery	DLVR	{"source":"...	1	05.02.20...	16:50:...		12.02.20...	204	No Content
005056882...	3	RDP	Delivery	DLVR	{"source":"...	1	05.02.20...	16:50:...		12.02.20...	204	No Content
005056882...	4	RDP	Delivery	DLVR	{"source":"...	1	05.02.20...	16:50:...		12.02.20...	204	No Content
005056882...	5	RDP	Delivery	DLVR	{"source":"...	1	05.02.20...	16:50:...		12.02.20...	204	No Content
005056882...	6	RDP	Delivery	DLVR	{"source":"...	1	05.02.20...	16:50:...		12.02.20...	204	No Content
005056882...	7	RDP	Delivery	DLVR	{"source":"...	1	05.02.20...	16:50:...		12.02.20...	204	No Content
005056882...	8	RDP	Delivery	DLVR	{"source":"...	1	05.02.20...	16:50:...		12.02.20...	204	No Content
005056882...	9	RDP	Delivery	DLVR	{"source":"...	1	05.02.20...	16:50:...		12.02.20...	204	No Content
005056882...	10	RDP	Delivery	DLVR	{"source":"...	1	05.02.20...	16:50:...		12.02.20...	204	No Content
005056882...	11	RDP	Delivery	DLVR	{"source":"...	1	05.02.20...	16:50:...		12.02.20...	204	No Content
005056882...	12	RDP	Delivery	DLVR	{"source":"...	1	05.02.20...	16:50:...		12.02.20...	204	No Content
005056882...	13	RDP	Delivery	DLVR	{"source":"...	1	05.02.20...	16:50:...		12.02.20...	204	No Content
005056882...	14	RDP	Delivery	DLVR	{"source":"...	1	05.02.20...	16:50:...		12.02.20...	204	No Content
005056882...	15	RDP	Delivery	DLVR	{"source":"...	1	05.02.20...	16:50:...		12.02.20...	204	No Content
005056882...	16	RDP	Delivery	DLVR	{"source":"...	1	05.02.20...	16:50:...		12.02.20...	204	No Content
005056882...	17	RDP	Delivery	DLVR	{"source":"...	1	05.02.20...	16:50:...		12.02.20...	204	No Content
005056882...	18	RDP	Delivery	DLVR	{"source":"...	1	05.02.20...	16:50:...		12.02.20...	204	No Content

Trace Maintenance Report

A new trace functionality has been introduced to help users monitor the status of replicated objects more effectively.

⚠ **Important**: Tracing is only active if the following application parameter is maintained:

- ENABLE_TRACE = X

This setting can be adjusted in the **Application Parameters** section of the RDP configuration. If the parameter is not set, no trace data will be recorded.

Purpose

The report **/VPCOE/TRACE_MAINTENANCE** enables logging and review of all objects that have been selected for replication—whether successfully sent or not sent due to various reasons (e.g. missing data, configuration issues).

For each payload created to be sent to RDP, the report saves trace details to allows tracking of both:

- **Successfully replicated objects**
- **Objects selected but not sent**

Trace Data Captured

The following fields are recorded for each relevant object:

- **Trace ID** – Unique identifier for the trace session.
- **API Type** – Type of outbound API used (e.g., CUSTOMER, MATERIAL, DELIVERY).
- **Date/Time** – Timestamp of when the replication run occurred.
- **User** – User ID that triggered the replication.
- **Object ID** – Business object identifier (e.g., Customer Number, Material Number).

User Functionality

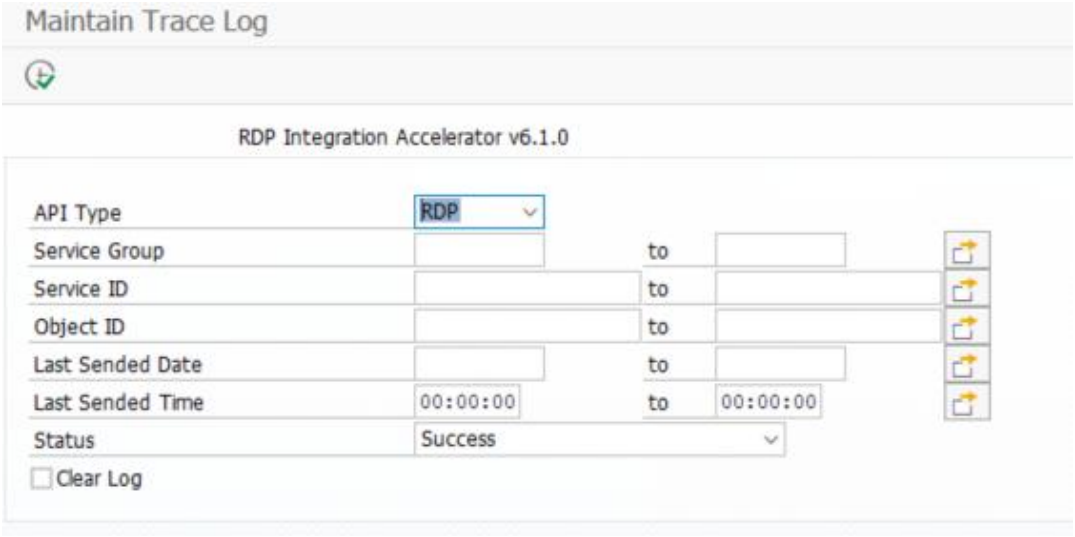
The report provides two key functions:

1. **Display Mode**
   Users can search and display trace data using the following selection criteria:
   - Trace ID
   - API Type
   - Date/Time Range
   - Object ID

This helps with analyzing what was sent, skipped, and why.

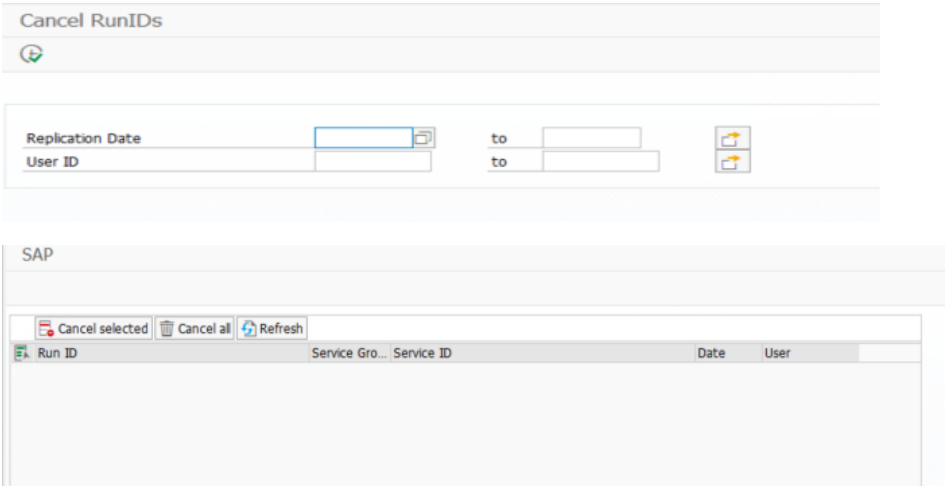
2. **Cleanup Mode**
   A deletion feature allows users to **clean up the trace data** based on:
   - Date range
   - Specific API types
   - Or complete flush of the table if needed

⚠️ **Note**: The trace table is intended for short-term diagnostic use. Regular cleanup is recommended to prevent data volume issues.



Report for RunId maintenance

Release **V7** introduces the **RunID Maintenance** report to allow users to view, filter, and cancel RunIDs directly in ECC/S4, without requiring RDP application intervention.



When using the **SAP Sustainability Data Foundation (SDF)** API for replication, each replication execution creates a **RunID**.
A RunID represents a replication session and SDF requires that each session is properly closed before a new replication for the same entity can be started.

If a replication ends in error, is interrupted, and never receives a closing confirmation, the RunID may remain **open**.
In such cases, SDF will **reject new replication attempts** for the same entity until the previous RunID is explicitly cancelled.

RDP Packaging Data Integration reports

Packaging data, such as packaging compositions, packaging elements, and basic fractions that are uploaded and evaluated in SAP Responsible Design and Production can be stored in SAP ECC or SAP S/4HANA in different locations, for example, in SAP Product Lifecycle Management (PLM),

SAP Recipe Management (RM) or Bill of Material (BOM). Therefore, this integration feature offers only basic functionality to extract the packaging data out of these storage locations. Customer specific program extensions and a BAdI implementation, where these program extensions are referenced, are required. If no customer specific program extensions are implemented, the upload reports will not upload any packaging data to SAP Responsible Design and Production.

Reports included:

- Report for Uploading Packaging Elements from PLM: /VPCOE/R_UPH_PCKG_DATA_LOAD
- Report for uploading Packaging Compositions from PLM: /VPCOE/R_UPH_PCKG_CMP_PLM_LOAD
- Report for Uploading Packaging Compositions from Recipe Management: /VPCOE/R_UPH_PCKG_CMP_RCP_LOAD
- Report for Uploading Packaging Composition from BoM /VPCOE/R_UPH_PCKG_CMP_BOM_LOAD
- Report for Uploading Packaging elements from Material Classification /VPCOE/R_UPH_PCKG_MCL_LOAD

- The BAdI provided for Upload Entity processor implementation: /VPCOE/BADI_UPH_CUSTOM (use enh. spot /VPCOE/ES_UPH_CUSTOM)

To upload or push data out of the ECC/S4HANA to the SAP RDP solution, the following software parts are necessary:

1. Report(s) to start the upload of data (specification database data or other) to RDP endpoint
2. Implementation of Upload Entity
    - Gathering of current relevant data (full or delta data) of the Upload Entity (if necessary, provide and use BADIs for customer extensibility)
    - Packaging the relevant data
    - Map the gathered data to internal data structures
    - Transmit the data stored in internal data structures to the corresponding RDP-Push API
    - For Initial/delta upload, store the information about uploaded data

An **Upload Entity** represents an RDP business object (like Packaging Element) with all its defined child nodes (like Packaging Fractions, Product assignments). The structure of an Upload Entity must fit to the provided RDP OData Entity structure for writing data.

Here you can find more details on extensibility and how to implement entity processor class.

Creating recommended classes and characteristics

To facilitate effective packaging data management within SAP RDP, a specialized report has been included to generate recommended classes and characteristics. The report is creating the classes and characteristics for:

- Packaging Elements in Material Classification
- Packaging Compositions in BOM, classifying Pckg. Elmnts. into Main/Subordinate
- Packaging Compositions and Main/Subordinate Packaging Elements in PLM SpecDB
- Packaging fee data classification
    - Class: ZRDP_ES_PLASTICTAX
    - Characteristics:
        - ZRDP_ES_PLASTIC
        - ZRDP_ES_NONRECYCLED_PLASTIC

- ZRDP_ES_EXEMPTION
- ZRDP_ES_DECREFNO
- ZRDP_ES_CERTREFNO
- ZRDP_ES_PRODORDER_CHECK
- ZRDP_ES_PACKMAT_TAX

- Customer extension data
  - Class: ZRDP_RDP_CUST_EXT
  - Characteristics:
    - ZRDP_CUST_EXT_ROLE
    - ZRDP_PROD_EXT_TAG

The report name is `/VPCOE/RDP_CLAS_DATA_CREATOR`.

The transaction code `/VPCOE/DATA_CREATOR` belongs to this report execution.

There are two parameter sections: Set Prefix, and Set Operation Scope.

In the Set Prefix section, set prefixes for classifications and characteristics. Prefixes must not be empty and can't be longer than four characters.

In the Set Operation Scope section, choose between Material Classification and Specification Workbench. The prefixes will be reset automatically to default values that match the default implementation of class /`VPCOE/CL_PCKELEM_MCL_EXMPL`.

After the execution the program will show the results of created and not created classifications and characteristics.

To modify classifications and characteristics that the program creates, you need to create own Z-objects and copy the code of the report and class /`VPCOE/CL_CL_CHAR_DATA_CREATE`, which the report calls upon execution.

In your own Z-class you have the following options:

- Changing the classifications and characteristics for packaging element **fractions**: Go to the setup_packelem_frac_dataset() method and modify `lt_pckfrac_classi_params` and `lt_pckfrac_charac_params`.
- Changing the classifications and characteristics for packaging element **attributes**: Go to the `setup_packelem_attr_dataset()` method and modify `lt_pckattr_classi_params` and `lt_pckattr_charac_params`.

Please note that your modifications must match modifications in example class `/VPCOE/CL_PCKELEM_MCL_EXMPL`.

More details on RDP data model for classifications and characteristics: [Data Model | SAP Help Portal]

Upload Packaging Elements data from PLM

The upload of packaging element data from PLM (Product Lifecycle Management) requires that the according data exists in the Environment, Health, and Safety specification database. In addition, an Upload Entity Processor implementation for the respective Upload Entity PE_PLM exists in the system and is returned by the BAdI implementation of /VPCOE/BADI_UPH_CUSTOM.

The Upload Entity: Packaging data (PLM) retrieves the packaging related data out of the Specification Management.

The report name is `/VPCOE/R_UPH_PCKG_DATA_LOAD`.

The transaction code /VPCOE/UPH_PE_PLM belongs to this report execution.

The target internal data structures for the retrieved data are implementations of interface /VPCOE/IF_UPH_ENTITY_DATA.

Processor classes are implementations of interface /VPCOE/IF_UPH_ENTITY_PROC and in addition for the PLM case of interface /VPCOE/IF_UPH_ENTITY_PLM_PROC. The base class /VPCOE/CL_UPH_PROC_BASE_PLM provides a basic implementation of these processor interfaces and must be extended for specific adoptions.

Class overview

Interface/Class	Description
/VPCOE/IF_UPH_ENTITY_PROC	*Interface for Upload Entity Processors*
/VPCOE/IF_UPH_ENTITY_PLM_PROC	*Interface for Entity Processors based on PLM data*
/VPCOE/CL_UPH_PROC_BASE	*Abstract base processor implementation. Concrete data extraction implementations must inherit from this base class.*
/VPCOE/CL_UPH_PROC_BASE_PLM	*Base processor implementation for Upload Entity Packaging PLM. Concrete PLM data extraction implementations must inherit from this base class.*
/VPCOE/CL_UPH_WRAP_SPEC	*Wrapper Class for a Specification*
/VPCOE/CL_UPH_WRAP_VA	*Wrapper Class for Value Assignment data (with Characteristics) of a Specification*
/VPCOE/IF_UPH_ENTITY_DATA	*Interface for Upload Entity data*
/VPCOE/CL_UPH_ENT_PCKG_ELEMENT	*Entity data for Packaging Element*
/VPCOE/CL_UPH_ENT_PCKG_FRCTN	*Entity data for Basic Material Fraction*
/VPCOE/CL_UPH_ENT_PCKG_PRODUCT /VPCOE/IF_UPH_ENTITY_HU_PROC	*Entity data for Packaging Element to Product* *Interface for Upload Handling Units*
/VPCOE/CL_UPH_ENT_HANDL_UNIT	*Entity data for Packaging Composition Items (Handling Units)*

Extensibility

Implementing the Upload Entity Processor

An Upload Entity Processor implementation for the Upload Entity PE_PLM should inherit from base class /VPCOE/CL_UPH_PROC_BASE_PLM. This base class is already implementing the methods of the interface /VPCOE/IF_UPH_ENTITY_PROC and only the mapping of the specification data must be provided by implementing the method /VPCOE/IF_UPH_ENTITY_PLM_PROC~map_spec_data.

See the **Technical Documentation** of BAdI /VPCOE/BADI_UPH_CUSTOM for an example implementation of such an Upload Entity Processor.

The implementation of custom entity processor can be realized by the following steps:

1. Create a processor implementation by implementing the interface /VPCOE/IF_UPH_ENTITY_PROC. If the processor is used to read out of the specification management, derive the implementation from the basis class /VPCOE/CL_UPH_PROC_BASE_PLM.

    - implement method /VPCOE/IF_UPH_ENTITY_PROC~prepare_process to determine which **specifications are relevant**. Based on data input on the selection screen, the relevant specifications for the data upload are determined. When running in delta processing mode, here the relevant specifications must be

filtered to only the changed one (since the last processing).

- implement all **relevant Value Assignment Types** which contain Packaging Element or Fraction data in method `/VPCOE/IF_UPH_ENTITY_PLM_PROC~get_relevant_vat`
- All **mappings** between specification data and the target Entity Data (Packaging Element, Packaging Fraction, ...) must be implemented in the method `/VPCOE/IF_UPH_ENTITY_PLM_PROC~map_spec_data`

2. Exchange the default BADI implementation for `/VPCOE/BADI_UPH_CUSTOM` and deliver the new implementation.

Last change information

There is no CDS views (internal API) that used in the S/4 HANA implementation to get some of the required data. That's why separate class has been implemented for ECC solution - `/VPCOE/CL_PLM_DB_ACCESS`.

Upload Packaging Compositions from PLM

The upload of packaging composition data from PLM requires that the accompanying data exists in the Environment, Health, and Safety specification database. In addition, an Upload Entity Processor implementation for the respective Upload Entity PC_PLM exists in the system and is returned by the BAdI implementation of VPCOE/BADI_UPH_CUSTOM. The Report name is /VPCOE/R_UPH_PCKG_CMP_PLM_LOAD.

The transaction code `/VPCOE/UPH_PC_PLM` belongs to this report execution.

Implementing the Upload Entity Processor

An Upload Entity Processor implementation for the Upload Entity PC_PLM should inherit from base class /VPCOE/CL_UPH_PROC_BASE_PLM. This base class is already implementing the methods of the interface /VPCOE/IF_UPH_ENTITY_PROC and only the mapping of the specification data must be provided by implementing the method /VPCOE/IF_UPH_ENTITY_PLM_PROC~MAP_SPEC_DATA from interface VPCOE/IF_UPH_ENTITY_PLM_PROC.

See the **Technical Documentation** of BAdI VPCOE/BADI_UPH_CUSTOM for an example implementation of such an entity processor.

Class overview

The objects created for packaging elements could be reused for the extraction of packaging compositions. The list below supports the statement:

Report	/VPCOE/R_UPH_PCKG_CMP_PLM_LOAD	Extract Packaging Composition from Spec DB
Factory Class	SAME - Add new transfer mapper for the composition	
BADI	No Change Needed - Call right upload entity from the entity processor	
BASE	No Change Needed	
BASE PLM	No Change Needed	
WRAP_SPEC	No Change Needed	

WRAP_VA	No Change Needed	
Example Class Implementation	/VPCOE/ZCL_UPH_CUSTOM_IMPL	Call the right example implementation;
	/VPCOE/CL_UPH_PROC_PLM_EXMPL1	Custom/Example Implementation for Extraction of Packaging Composition from Spec DB
Logger class	No change	

Access the Specifications data

There is no CDS views (internal API) that used in the S/4 HANA implementation to get some of the required data. So, separate class is implemented for ECC solution - `/VPCOE/CL_PLM_DB_ACCESS`.

To get Specification data the following methods are used:

- `GET_SPECIFICATIONFORKEYDATE`
- `GET_SPECASSIGNEDMATLFORKEYDATE`

Upload Packaging Compositions from Recipe Management

Retrieves the packaging composition related data out of the Recipe Development.

The Report is `/VPCOE/R_UPH_PCKG_CMP_RCP_LOAD`.

The transaction code `/VPCOE/UPH_PC_RCP` belongs to this report execution.

Class overview

The objects created for packaging elements and packaging components from Spec DB could be reused for the extraction of packaging compositions from Recipe development. The list below supports the statement:

Report	/`VPCOE/R_UPH_PCKG_CMP_RCP_LOAD`	Extract Packaging Composition from Recipe Development; Report screen fields are mentioned in the table below
Factory Class	No Change Needed	
BADI	No Change Needed	
BASE	No Change Needed	
BASE PLM	`/VPCOE/CL_UPH_PROC_BASE_RCP`	Base Class to retrieve packaging composition data from Recipe development
WRAP_SPEC	`/VPCOE/CL_UPH_WRAP_CMP_RCP`	Wrapper class to retrieve packaging composition data from recipe
WRAP_VA		

Example class Implementation	/VPCOE/ZCL_UPH_CUSTOM_IMPL	Call right example implementation
	/VPCOE/CL_UPH_CMP_RCP_EXAMPLE	Custom/Example Implementation for Extraction of Packaging Composition from Recipe development
Class for Comp_header	/VPCOE/CL_UPH_ENT_PCKG_CMP_HDR	All of these Packaging Composition elements from Spec DB would be reused
Class for Comp_Item	/VPCOE/CL_UPH_ENT_PCKG_CMP_ITM	
Transfer Mapper class	/VPCOE/CL_UPH_ENT_PCKG_CMP_ITM	
Logger class	No change needed	Adapt for messages specific to recipe development

Access the Recipe Management data

There is no CDS views (internal API) that used in the S/4 HANA implementation to get some of the required data. So, separate class is implemented for ECC solution - /VPCOE/CL_PLM_DB_ACCESS.

To get Specification data the following methods are used:

- GET_RECIPESTATUSTEXT
- GET_SINGLE_RECIPETYPE
- GET_RECIPEFORKEYDATE
- GET_RECIPEFORMULAITEM

Uploading Packaging Compositions from Bills of Material

The upload of packaging composition data from Bills of Material (BOMs) requires that the according data exist in the bill of material. In addition, an Upload Entity Processor implementation for the respective Upload Entity PC_BOM exists in the system and is returned by the BAdI implementation of /VPCOE/BADI_UPH_CUSTOM.

The Report is /VPCOE/R_UPH_PCKG_CMP_BOM_LOAD.

The transaction code /VPCOE/UPH_PC_BOM belongs to this report execution.

Implement the Upload Entity Processor

An Upload Entity Processor implementation for the Upload Entity PC_BOM should inherit from base class /VPCOE/CL_UPH_PROC_BASE_BOM. This base class is already implementing the methods of the interface /VPCOE/IF_UPH_ENTITY_PROC and only the mapping of the BOM must be provided by implementing the method /VPCOE/IF_UPH_ENTITY_BOM_PROC~MAP_BOM_DATA.

See the **Technical Documentation** of BAdI /VPCOE/BADI_UPH_CUSTOM for an example implementation of such an entity processor.

Report parameters:

- **Initial Load / Delta Load** (Radio Buttons):

If the Initial Load radio button is selected, then the report selects the BOMs based on the selection screen parameters.

If the Delta Load radio-button is selected, then the selection screen parameters of the previous successful initial load are used and therefore all selection screen parameters except the Test Mode are hidden. In addition to that, during delta load, only those BOMs that have been updated since the last successful initial or delta load will be extracted and transferred.

- **Test Mode** (Check Box):

  If the Test Mode check box is selected, then the entire processing in terms of extraction and mapping of BOM data remains same only the transferring of the data to the SAP Responsible Design and Production API is skipped. Test Mode is supported in initial load or delta load.

- **Source ID** (Input Parameter):

  Source ID is a mandatory input parameter which is transferred to SAP Responsible Design and Production. It identifies from which source system the data is coming.

- **RFC Destination** (Input Parameter):

  This parameter is mandatory and contains the customer implemented target destination for the SAP Responsible Design and Production API (defined in transaction sm59). The default value VPCOE_RDP_PLM can be changed here.

- **Use Filter Criteria**:

  The following input parameters are enabled for data filtering.

  - Material (include ranges, patterns, multi selection): Filter for BOMs that are assigned to this material
  - Material type (include ranges, patterns, multi selection): Filter for BOMs that are assigned to materials of this material type
  - Plant (include ranges, patterns, multi selection): Filter for BOMs that are assigned to this plant
  - BOM Usage (include ranges, patterns, multi selection): Filter for BOMs that are defined for this usage area
  - Alternative BOM (include ranges, patterns, multi selection): Filter for BOMs with this alternative number
  - BOM Status (single selection): Filter for BOMs with this status

- BOM Valid Before Date (Valid From): This field is used to filter the BOM validity. All BOMs that are valid before or on this date will be selected for processing. By default, this field is hidden and has the value '31.12.9999' which means that every version will be selected for processing.

  Use the DISPLAY_VALIDITY function code to see this field. You can change the value but consider that changing the BOM Valid Before Date can lead to missing links between packaging elements and packaging compositions when not all versions are uploaded.

  BOM Valid From Date:

  Following scenarios shall be supported:

  1. Valid From date is provided: All BOM that are valid for this key date shall be replicated. Apply this logic for both, initial load and delta load.
  2. Valid From date is visible and is left empty: Empty from-date means that from-date = current date. All BOM that are valid for this rolling key date shall be replicated. Apply this logic for both, initial load and delta load.
  3. Valid From date is hidden: the default value for this field is '31.12.9999' which means that every version will be selected for processing.

- Change Date of BOM (include ranges, multi selection): Filter for BOMs with this (overall) change date

Additional Settings: The following additional settings configure the search.

- Explosion Level (Input parameter): A mandatory parameter that defines the maximum number of levels that are used to explode the BOM. This is relevant for nested BOMs only.

Change Date of entire BOM

Following scenarios are supported:

1. Provide only from-date, only BOM header and components changed on that specific date are extracted (whole BOM is extracted)
2. Provide from-date and to-date, all BOM header and components changed between (>= from-date and <= end-date) are extracted

Default Values for Selection Parameters:

- The BOM Valid From Date is always defaulted to ‘’31.12.9999’’
- Radio button for Initial load is defaulted.

Mandatory Selection Parameters for Filtering Packaging Compositions

If extracting by Filter Criteria:

- The field "Maximum explosion level" shall be mandatory with default value 10.

The upload of packaging composition data from Bills of Material (BOMs) requires that the according data exist in the bill of material. In addition, an Upload Entity Processor implementation for the

respective Upload Entity PC_BOM exists in the system and is returned by the BAdI implementation of /VPCOE/BADI_UPH_CUSTOM.

Use report /VPCOE/R_UPH_PCKG_CMP_BOM_LOAD to extract packaging composition data from BOMs and to push it to the SAP Responsible Design and Production.

An Upload Entity Processor implementation for the Upload Entity PC_BOM should inherit from base class /VPCOE/CL_SURDP_UPH_PROC_BASE_BOM. This base class is already implementing the methods of the interface /VPCOE/IF_UPH_ENTITY_PROC and only the mapping of the BOM must be provided by implementing the method /VPCOE/IF_UPH_ENTITY_BOM_PROC~MAP_BOM_DATA.

Class Overview

Report	/VPCOE/R_UPH_PCKG_CMP_BOM_LOAD	Extract Packaging Composition from BoM
Factory Class	SAME - Add new transfer mapper for the composition	
BADI	No Change Needed - Call right upload entity from the entity processor	
WRAP_SPEC	No Change Needed	
WRAP_VA	No Change Needed	
Implementing Class	/VPCOE/ZCL_UPH_CUSTOM_IMPL	Call the right example implementation;
Example Implementation	/VPCOE/CL_UPH_CMP_BOM_EXAMPLE	Custom/Example Implementation for Extraction of Packaging Composition from BoM
Logger class	No change	

Sample BAdI Implementations for V1 and V2 RDP Packaging data APIs
Sample BAdI implementations are provided for both **V1** and **V2** of the Packaging Data API. These implementations offer a comprehensive framework for processing and transferring BOM (Bill of Materials) data as part of the RDP Integration Accelerator.

Key Features Included in the Sample BAdI Implementation

1. **BOM Extraction Based on Filters**
   - Extracts BOMs based on the filter criteria defined in the report.
2. **Material Type Filtering**
   - Filters BOM positions by materials of **material type VERP** (packaging materials).
3. **Classification and Assembly BOM Checks**
   - For each BOM position:
     - Checks if a **material classification** for packaging data exists.
     - Validates the **valid-from date** of the BOM position.
     - Checks if the BOM position is an **assembly BOM** with an existing batch classification as the main element.
4. **Transfer as Packaging Composition Item**
   - If a **packaging data classification** exists, the BOM position is transferred as a **packaging composition item**.
5. **Combination of VERP Positions in Assembly BOMs**
   - Searches through all assembly BOMs for positions of **type VERP**.

- Combines the **VERP positions** into a single **packaging composition** for the top-level BOM header (Product).

6. **Handling Assembly BOMs with Batch Classification**
   - In case of an assembly BOM with an **existing batch classification** as the main element:
     - Positions within the assembly are considered as **subordinate elements**.
     - The presence of **packaging data material classification** and **batch classification** for subordinate elements is required.
7. **Exclusion of BOM Positions without Classification**
   - BOM positions are **not transferred** if no classification exists.

Additional Considerations:

- Both **V1** and **V2** implementations allow for further customization if required by the business.
- The sample BAdI implementations can be adapted to meet specific requirements by modifying the standard logic provided in the default classes.

Uploading Packaging elements from Material Classifications

The upload of packaging element data from Material Classification requires that the according data exists in the 'Material'. In addition, an Upload Entity Processor implementation for the respective Upload Entity `PE_MCL` exists in the system and is returned by the BAdI implementation of `/VPCOE/BADI_UPH_CUSTOM`.

The Report name is `/VPCOE/R_UPH_PCKG_MCL_LOAD`.

The transaction code `/VPCOE/UPH_PCKG_MCL` belongs to this report execution.

Implementing the Upload Entity Processor

An Upload Entity Processor implementation for the Upload Entity PE_MCL should inherit from base class /VPCOE/CL_UPH_PROC_BASE_MCL. This base class is already implementing the methods of the interface /VPCOE/IF_UPH_ENTITY_PROC.

The mapping of the material classification must be provided by implementing the method /VPCOE/IF_UPH_ENTITY_MCL_PROC~MAP_MAT_CLAS_DATA to determine which classification is relevant for the business case.
Method /VPCOE/IF_UPH_ENTITY_MCL_PROC~GET_RELEVANT_MAT_CLAS must be also implemented.

For an example implementation of this entity processor, see class /VPCOE/CL_UPH_PCKELEM_MCL_EXMPL and the contained ABAP documentation.

Report parameters

- Initial Load / Delta Load (Radio buttons):

  If the Initial Load radio button is selected, then the report selects the material classifications based on the selection screen parameters.

  When the Delta Load radio-button is selected then the previous successful initial load of the selection screen parameters is used and therefore all selection screen parameters except the Test Mode are hidden. In addition to that, during delta load, only those specifications

that have been updated since the last successful initial or delta load will be extracted and transferred.

- Test Mode (Check box):

  If the Test Mode check box is selected, then the entire processing in terms of extraction and mapping of material classifications remains same, only the transferring of the data to the SAP Responsible Design and Production API is skipped. Test Mode is supported in initial load and delta load.

- Source ID (Input parameter):

  Source ID is a mandatory input parameter that is transferred to SAP Responsible Design and Production. It identifies from which source system the data is coming.

- RFC Destination (Input parameter):

  This parameter is mandatory and contains the customer implemented target destination for the SAP Responsible Design and Production API (defined in transaction SM59). The default value RDP_REPL_API can be changed here.

- Use Filter Criteria: The following input parameters are input enabled for data filtering:
  - Material (include ranges, patterns, multi selection): Filter for material classification that are assigned to this material
  - Material type (include ranges, patterns, multi selection): Filter for material classifications that are assigned to materials of this material type
  - Change Date of Whole Spec (include ranges, multi selection): Filter for materials that were last changes on these dates

Handling of Multiple Versions

Material Classification supports multiple versions with date and time related validity via Change Numbers. However, the report always extracts a packaging element out of the Material Classification as it is valid on the day of execution of the replication.

Change Date of Material

Following scenarios are supported:

1. If only from-date is provided, only materials changed **on** that specific date are extracted. Also, the change of a characteristic value from the material classification is considered as change.
2. If from-date and to-date are provided, all materials changed between (>= from-date and <= end-date) are extracted´. Also, the change of a characteristic value from the material classification is considered as change.

Default Values for target RFC destination Parameters

- The parameter should have the customer implemented target destination as default. If the customer does not have custom implementation for target RFC destination, then VPCOE_RDP_PLM_PKG_ELMNT as default value but can be adjusted by the user. This gives customers more flexibility in choosing a destination name and avoids potential name conflicts.

Class Overview

Report	`/VPCOE/R_UPH_PCKG_MCL_LOAD`	Extract Packaging elements from Mat. Classification
Factory Class	`SAME - Add new transfer mapper for the composition`	
BADI	`Call right upload entity from the entity processor`	
WRAP_SPEC	`No Change Needed`	
WRAP_VA	`No Change Needed`	
Implementing Class	`/VPCOE/ZCL_UPH_CUSTOM_IMPL`	Call the right example implementation;
Example Implementation	`/VPCOE/CL_UPH_PROC_MCL_EXAMPLE`	Custom/Example Implementation for Extraction of Packaging Element from Material Classification
Logger class	`No change`	

Uploading Packaging Composition Items from Handling Units

This report is used when it is necessary to **add additional packaging elements** to an existing packaging composition in SAP Responsible Design and Production (RDP).

Use Case

- Products often require **tertiary packaging materials** (e.g., pallets, stretch foil) to protect and secure finished products during transport.
- Such packaging materials are typically bundled as **handling units (HUs)**. For example:
  - A **pallet** with a dozen cardboard boxes.
  - Each box containing individually packed finished goods.
  - The pallet and boxes are wrapped with a few meters of stretch foil.

How the Report Works

- The report looks up the **effectively used packaging material** from handling units in **deliveries** within a specified timeframe.
- It adds the corresponding **packaging elements** to the existing **packaging compositions** in RDP, each element with its **proportionate quantity**.

- This enables RDP to identify how much of each packaging material is used per finished product unit.

Handling Variations in Packaging Material Quantities

- It's possible that deliveries with the same packaging composition may have slightly different packaging material quantities.
- To account for such variations, the report **calculates the average** over the defined period to provide a more accurate estimate.

Extensibility and Custom Implementation

Due to the variety of handling unit configurations, the report offers extensibility options for:

- **Extracting handling units** using custom logic.
- **Updating packaging compositions** with additional items.

Upload Entity Processor for PCI_HU

- An **Upload Entity Processor** implementation for the respective entity PCI_HU must be present in the system.
- This implementation is returned by the BAdI **/VPCOE/BADI_UPH_CUSTOM**.

Report for Uploading Handling Unit Data

- **Report Name:** /VPCOE/R_PCKG_CMP_HU_LOAD
- **Purpose:** Replicate packaging composition items derived from handling units to RDP.

Prerequisites and Limitations

Prerequisites

- To use the report for uploading packaging composition items:
    - Create a **Replication Configuration** in RDP through the **Configuration Cockpit** for the object type **Packaging Composition – Additional Items**.

Limitations and Considerations

When implementing the BAdI for uploading additional packaging composition items, consider the following limitations:

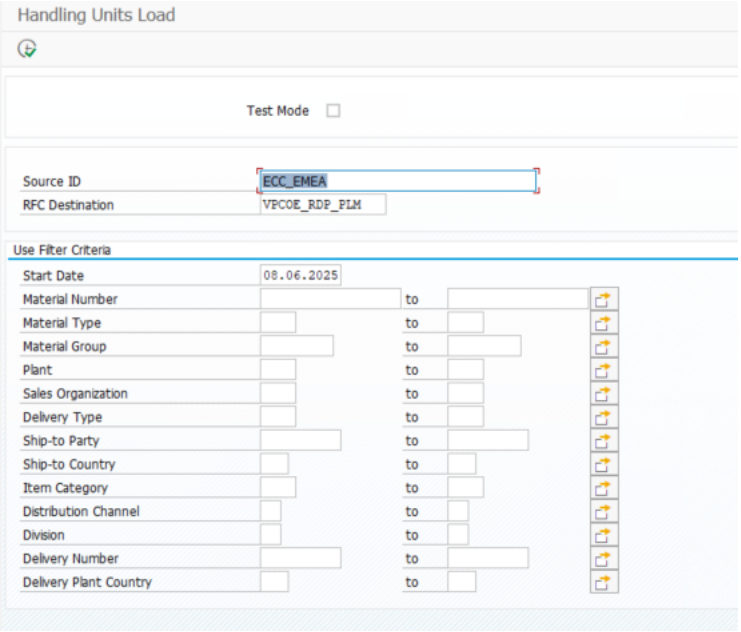
- **Separate Source System Requirement:**
    - Replication configurations for **Packaging Composition** and **Packaging Composition – Additional Items cannot** be created in the same source system.
    - A replication configuration for **Packaging Composition – Additional Items** must refer to a system **not yet used**.
- **Handling Data from the Same System:**
    - If data for **Packaging Composition** is pushed from the same system:
        - Create a replication configuration with a **different Source System ID**.
        - Use this alternative name only when calling the API for **packaging composition items**.

Implement the Upload Entity Processor

An Upload Entity Processor implementation for the Upload Entity PC_HU should inherit from base class /VPCOE/CL_UPH_PROC_BASE_HU. This base class is already implementing the methods of the interface /VPCOE/IF_UPH_ENTITY_PROC and only the mapping of the HU must be provided by implementing the method /VPCOE/IF_UPH_ENTITY_HU_PROC~MAP_HU_DATA.

Sample BAdi implementation is provided with entity processor class /VPCOE/CL_PCKCOMP_HU_EXMPL.

The transaction code `/VPCOE/HU` belongs to this report execution.



Report Parameters

Please consider the following when using report selection parameters:

- Test Mode (Check box):

If the Test Mode check box is selected, then the entire processing in terms of extraction and mapping of HU data remains same only the transferring of the data to the **SAP Responsible Design and Production** API is skipped. **Test Mode** is supported in **Initial Load** or **Delta Load**.

- **Source ID** (Input parameter):

This mandatory input parameter is transferred to SAP Responsible Design and Production. It identifies from which source system the data is coming. It is required that the source ID of the handling unit is different from the source system of the packaging composition that should be enriched.

- **RFC Destination** (Input parameter):

This parameter is mandatory and contains the customer implemented target destination for the SAP Responsible Design and Production API (defined in transaction **SM59**). The default value VPCOE_RDP_PLM can be changed here.

- Use Filter Criteria:

- **Start Date**: Defines the start date of the lookback period. All deliveries that have a goods movement posting date between start date and the end date (default value is current date) are selected for processing.
- **End Date** (hidden by default): Defines the end date of the lookback period. The default value is today. This field can be displayed by entering the **DISPLAY_VALIDITY OK** code.

The following filter criteria can be entered including ranges, patterns, and multi selection:

- **Material Number**: Filter the materials of the delivery items.
- **Material Type**: Filter the material types of the delivery items.
- **Material Group**: Filter the material groups of the delivery items.
- **Plant**: Filter the plants of the delivery items.
- **Sales Organization**: Filter deliveries which belongs to the given sales organization.
- **Delivery Type**: Filter deliveries for the given delivery type.
- **Ship-to Party**: Filter deliveries for the given ship to party.
- **Ship-to Country**: Filter deliveries for the given ship to country/region.
- **Item Category**: Filter the item category of the delivery items.
- **Distribution Channel**: Filter deliveries for the given distribution channel.
- **Division**: Filter deliveries for the given division.
- **Delivery Number**: Filter deliveries for the given delivery number.
- **Delivery Plant Country**: Filter the plant country of the delivery items.

Because of the varieties of customizing on HU, the report is offering a sample Badi implementation covering:

- Extracting data for additional packaging composition items from handling units.

  In this step, the report finds all the deliveries issued within the specified Lookback Period, which were packed using Handling Units. Please, note that in standard logic only HU items with item category HUPM and LEIH are considered. Please, adjust the logic adding your custom item categories if any. Then, packaging materials are allocated to their respective products and average use of packaging per product is calculated.

- Identifying the packaging composition from RDP that need to be enriched.

  In this step, all existing packaging compositions for the products which were found in deliveries from the above step will be identified. If no packaging composition for a product is found in RDP, however, additional packaging composition items are calculated from HU data, no new packaging composition will be created in RDP, but the corresponding message will be added to the log output of the report.

- Replicating additional items to RDP for specific Packaging Composition.

In this step, Packaging Compositions are updated on RDP with additional packaging composition items and calculated quantities.

The distribution algorithm considers the handling units in all found deliveries. When it calculates how much of packaging material X was used for finished product Y, it only considers handling units, that contain at least one line item of finished product Y. Other handling units that used packaging material X, but didn't contain finished product Y, are not considered.

The following generic formula is used to calculate the average usage of any packaging material (PM) per finished product (FP):

$$\text{Average PM per FP} = \frac{\sum_{\text{Handling Units (HU)}} \left(\text{PM used in HU} \times \frac{\text{Quantity of FP in HU}}{\text{Total Quantity of Products in HU}}\right)}{\text{Total Quantity of FP in all HUs using PM}}$$

Class Overview

Report	/VPCOE/R_PCKG_CMP_HU_LOAD	Extract Packaging Composition Items from Handling Units
Factory Class	SAME - Add new transfer mapper	
BADI	Call right upload entity from the entity processor	GC_UE_PCI_HU
WRAP_SPEC	No Change Needed	
WRAP_VA	No Change Needed	
Example Implementation	/VPCOE/CL_PCKCOMP_HU_EXMPL	Custom/Example Implementation for Extraction of Packaging Element from HU
Logger class	No change	

Extensibility in packaging data reports

The upload of packaging data from PLM (Product Lifecycle Management) requires that the according data exists in the Environment, Health, and Safety specification database. In addition, an Upload Entity Processor implementation for the respective Upload Entity PE_PLM and PC_PLM exist in the system and is returned by the BAdI implementation of /VPCOE/BADI_UPH_CUSTOM.

The upload of packaging composition data from recipes requires that the accompanying data exists in Recipe Management. In addition, an Upload Entity Processor implementation for the Upload Entity PC_RCP exists in the system and is returned by the BAdI implementation of /VPCOE/BADI_UPH_CUSTOM.

The upload of packaging composition data from BoM requires that the according data exist in the bill of material. In addition, an Upload Entity Processor implementation for the respective Upload Entity PC_BOM exists in the system and is returned by the BAdI implementation of /VPCOE/BADI_UPH_CUSTOM.

The upload of packaging elements data from Material Classifications requires that the according data exist in the material classification. In addition, an Upload Entity Processor implementation for the respective Upload Entity PC_MCL exists in the system and is returned by the BAdI implementation of /VPCOE/BADI_UPH_CUSTOM.

The upload of packaging composition data from Handling Units requires that the according data exist in handling units. In addition, an Upload Entity Processor implementation for the respective Upload Entity PCI_HU exists in the system and is returned by the BAdI implementation of /VPCOE/BADI_UPH_CUSTOM.

Which entity processors are used for the corresponding Upload Entity is controlled by the BAdI /VPCOE/BADI_UPH_CUSTOM (Enhancement Spot /VPCOE/ES_UPH_CUSTOM). The BAdI mechanism allows to exchange a processor if needed, e.g. if customer specific data structures must be read.

Further, the BAdI implementation controls the used destination configuration. Destinations are possible for:

1. The source of the Packaging data (RFC destination to another system)
2. The target endpoint of the RDP Push API

VPCOE/BADI_UPH_CUSTOM – technical documentation

The Business Add-In (BAdI) /VPCOE/BADI_UPH_CUSTOM that is assigned to Enhancement Spot /VPCOE/BADI_UPH_CUSTOM is used during the upload of packaging related data to the API of SAP Responsible Design and Production (RDP). Its main task is to provide an extensibility mechanism to integrate customer specific implementations for extracting and processing of packaging related data.

This packaging related data is in the following called **Upload Entity**. An Upload Entity consists of:

- An RDP business object, for example, a packaging element with all its defined child nodes, such as, packaging fractions, product assignments and so on.
- The area of extraction, for example, Plant Maintenance (PLM) or Recipe Development (RD).

The following Upload Entities are currently defined in the system:

- **PE_PLM**: Packaging elements from PLM
- **PC_PLM**: Packaging compositions from PLM
- **PC_RCP**: Packaging compositions from Recipe Development
- **PC_BOM**: Packaging compositions from Bill of Material (BOM)
- **PE_MCL**: Packaging elements from Material Classification
- **PE_HU**: Packaging Compositions Items from Handling Units

For each Upload Entity there is a corresponding report to start the data upload.

When one of these reports is executed, the mentioned BAdI is called. As a result of the call, an instance of a customer specific implementation for the extraction and processing of the Upload Entity data must be returned. This instance is called Upload Entity Processor.

Handling Multi-Level BOMs in Packaging Composition Upload

Supported Depth for Subordinate Packaging Elements

- **SAP Responsible Design and Production (RDP)** currently supports only **one level** of subordinate packaging elements.

- As a result, the **example entity processor mapping implementation** used in the **Packaging Composition Upload Report for BOM** processes **only the first level** of the assembly BOM.
- Deeper levels beyond **level 1** are **not processed** by the standard implementation.

Handling Multi-Level BOMs (Level 1+N)

- If a **multi-level BOM** contains relevant subordinate packaging elements beyond the first level (**level 1+N**), this scenario is **not handled** by the standard logic.
- To accommodate such cases, a **custom mapping implementation** should be developed to:
    - Traverse deeper levels of the BOM hierarchy.
    - Extract and map relevant subordinate packaging elements beyond the first level.
    - Ensure accurate transfer of all necessary data to RDP.

Custom Mapping Considerations

- When implementing a **custom mapping logic**:
    - Ensure that all relevant subordinate elements beyond the first level are captured.
    - Adapt the mapping logic to correctly represent the additional levels of subordinate packaging elements in the JSON payload.
    - Test the implementation thoroughly to validate the correct transfer of data to RDP.

Prerequisites for Uploading Packaging Element and Packaging Composition Data

To successfully upload packaging element or packaging composition data from different sources to SAP Responsible Design and Production (RDP), the following prerequisites must be met:

1. **Data Availability in Source Systems**

Ensure that the required data exists in the respective source systems before initiating the upload:

- PLM (Product Lifecycle Management):
    - For uploading packaging composition data from PLM, the corresponding data must exist in the Environment, Health, and Safety (EHS) specification database.
- Recipe Development:
    - For uploading packaging composition data from Recipe Development, the corresponding data must be available in the Recipe Development system.
- Bill of Materials (BOM):
    - For uploading packaging composition data from a BOM, the relevant data must exist in the Bill of Materials maintained in the system.
- Material Classification:
    - For uploading packaging element data from Material Classification, the required data must be classified in the material master data.

2. **Upload Entity Processor Implementation**

- An Upload Entity Processor implementation must exist in the system for the respective Upload Entity.
- This implementation ensures that data is correctly processed and transferred to RDP.

3. **BAdI Implementation for Upload Entity Processor**
   - The BAdI responsible for providing the Upload Entity Processor instance must be implemented.
   - A sample BAdI implementation is provided for reference, which demonstrates how to configure and extend the processor logic.
4. **RFC Destination and OAuth Configuration**
   - The RFC destination and OAuth configuration must be properly maintained in the system to enable communication with the RDP Push API.
5. **BAdI Standard Settings and Configuration**
   - For more information on standard settings related to this BAdI (such as filters, single or multiple usage scenarios), refer to the Spot Element Definitions tab in the BAdI Builder using transaction SE18.

BAdi Methods

The BAdI definition is active in the Integration accelerator package. It contains the following methods from interface `/VPCOE/IF_UPH_CUSTOM`:

· `get_entity_processor`:

This method returns which entity processor is used for the corresponding upload entity. The BAdI mechanism allows to exchange a processor if needed, for example, if customer specific data structures must be read.

In general, the implementation of this method will return a customer specific Upload Entity Processor implementation (that implements the `IF_SURDP_UPH_ENTITY_PROC` interface) for each Upload Entity type.

· `get_target_endpoint_dest`:

This method will return the end point destination of the target system. If this method is defined, the returned value will be displayed as the default RFC destination in the report selection screens for all the upload entities. If no specific value is returned, the default destination RDP_REPL_API is used.

BAdi Implementations

The BAdI implementation must be defined by the user and is not provided by SAP. As example see class `/VPCOE/CL_EXMPL_UPH_CUSTOM`.

The custom Upload Entity Processors must implement the /vpcoe/if_uph_entity_proc interface. To ease the definition of custom entity processors, there are base classes available which provide already necessary functionality:

- `/VPCOE/CL_UPH_PROC_BASE_PLM` for the extraction out of PLM
- `/VPCOE/CL_UPH_PROC_BASE_RCP` for the extraction out of Recipe Development
- `/VPCOE/CL_UPH_PROC_BASE_BOM` for the extraction out of Bill of Material
- `/VPCOE/CL_UPH_PROC_BASE_MCL` for the extraction out of Material Classification
- `/VPCOE/CL_UPH_PROC_BASE_HU` for the extraction from Handling Units

Therefore, custom Upload Entity Processors should inherit from these base classes. For examples of the custom Upload Entity Processors please have a look at the following classes:

- /vpcoe/lcl_exmpl_cust_cmp_plm for packaging composition extraction out of PLM
- /vpcoe/lcl_exmpl_cust_cmp_rcp for packaging composition extraction out of Recipe Development
- /vpcoe/lcl_exmpl_cust_cmp_bom for packaging composition extraction out of Bill of Material
- /vpcoe/lcl_exmpl_cust_pe_plm for packaging element extraction out of PLM
- /vpcoe/lcl_exmpl_cust_pe_mcl for packaging element extraction out of Material Classification
- /vpcoe/cl_hu_exmpl for packaging composition items extraction from handling units

Common class overview

Interface/Class	Description
/VPCOE/CL_UPH_EXEC_LOAD_WRAP	Wrapper Class for the Execution of an Entity Upload (used by the report)
/VPCOE/IF_UPH_ENTITY_PROC	Interface for Upload Entity Processors
/VPCOE/CL_UPH_FACTORY	Static Factory class, which provides the Upload Entity Processor implementations per Upload Entity
/VPCOE/IF_UPH_CUSTOM	BADI Interface for custom aspects of the data upload
/VPCOE/BADI_UPH_CUSTOM	BADI Definition for custom aspects of the data upload.
/VPCOE/IF_UPH_LOGGER	General Logger Interface
/VPCOE/CL_UPH_LOGGER	Default Logger implementation
/VPCOE/CL_UPH_PROC_BASE	Base (abstract) Upload Entity Processor class. Concrete Entity Processors implementations should inherit from this base class.

Messages

The Upload report shall output the following messages (if present: when the condition defined in column rule is met).

Message class: /VPCOE/PLM

API Version Selection for Packaging Data Integration

The RDP Integration Accelerator supports both V1 and V2 versions of the Packaging Data API. The version to be used can be explicitly defined in the entity processor class within the BAdI implementation.

Key Considerations:

- For V1 API: The processor class must be specified directly.
- For V2 API: The mapper handler class should be specified using the GET_ENTITY_MAPPER method.

Configuration Details:

API Version	Example Class for Implementation
V1 API	/VPCOE/CL_UPH_TM_PCKG_CMP_V1 (Packaging Composition)
	/VPCOE/CL_UPH_TM_PCKG_ELEM_V1 (Packaging Elements)
	/VPCOE/CL_UPH_TM_HU (Handling Units)
V2 API	/VPCOE/CL_TM_PCKG_CMP_V2 (Packaging Composition)

	/VPCOE/CL_UPH_TM_PCKG_ELEM_V2 (Packaging Elements)

Customization and Adjustments:

- A sample processor class is provided by default, but it can be replaced if further adjustments are required.
- If customization is needed, the processor class can be substituted with a custom implementation tailored to specific business needs.
- In the `GET_ENTITY_MAPPER` method, users can configure the appropriate class based on the API version selected.

This flexibility ensures seamless integration with both API versions while allowing for business-specific adjustments as needed.

Packaging fee replication and posting

SAP Responsible Design and Production (RDP) is calculating packaging fees, which needs to be considered in the pricing procedures and invoices in the SAP ERP system. Therefore, RDP provide the possibility to transfer packaging fees as pricing condition records and material classification data in ERP system.

To be able to create these invoices, data needs to be replicated regularly from SAP Responsible Design and Production to your source system. The data extraction for this purpose is possible through Packaging Fees Retrieval report. Packaging Fee Posting report allows you to fill the according data into Pricing Conditions and Material Classification.

The RDP Packaging Fees Integration:

The RDP Packaging Fees Integration facilitates the retrieval and processing of packaging fee data from SAP Responsible Design and Production (RDP) into SAP ECC/S4H. It includes multiple reports and a BAdI framework for mapping and processing data according to business needs.

1. Reports for Packaging Fee Processing

Report Name	Description	Tcode
/VPCOE/R_PCKF_RETRIEVAL	Retrieves packaging fees from RDP.	/VPCOE/PCKF_RETRIEVE
/VPCOE/R_PCKF_POSTING	Converts transferred packaging fees into pricing condition records and/or material classifications.	/VPCOE/PCKF_POSTING
/VPCOE/R_PCKF_CUSTOM_EXEMPT	Retrieves customer exemptions.	
/VPCOE/R_PCKF_COMP_CODE_EXEM	Retrieves company code exemptions.	
/VPCOE/R_PCKF_CUSTOMER_ROLE	Retrieves customer roles.	

2. BAdI Implementation for Packaging Fee Processing

- A sample BAdI implementation is provided for converting transferred packaging fees into pricing condition records and material classifications.
- The sample implementation covers:

- Customer-dependent and customer-independent SD (Sales and Distribution) and MM (Materials Management) pricing conditions.
- Flexible mapping of data for pricing and classification structures.

3. Customization and Mapping

- When converting transferred packaging fees into pricing conditions and material classification data, a BAdI implementation is required for mapping.
- Due to the high degree of flexibility in pricing configurations and material classifications, SAP does not provide a default implementation.
- Users can customize the mapping logic as per their business-specific pricing and classification requirements.

Prerequisites to use the reports:

- The BAdI is implemented and provides the Entity Processor instance (custom class or base class) for the Entity Types PCKG_FEE and APL_BUSP.
- The RFC destination and the OAuth configuration are maintained in the system to access the RDP API.
- Definition of condition tables, condition types, access sequence, pricing procedures

When one of these reports is executed, the BAdI implementation is called. As a result of the call, an instance of an (customer specific) Entity Processor implementation must be returned.

Example implementation class: /VPCOE/CL_EXMPL_PCKF_CUSTOM

The example implementation class of the BAdI contains an entity processor implementation (commented out) for the Packaging Fee entity provided by a local class (lcl_exmpl_pckg_fee).

For logging in custom entity processor implementation, use application Logging Object /`VPCOE/RDP`, Sub-object `PCKG_FEE`.

Packaging fee retrieval

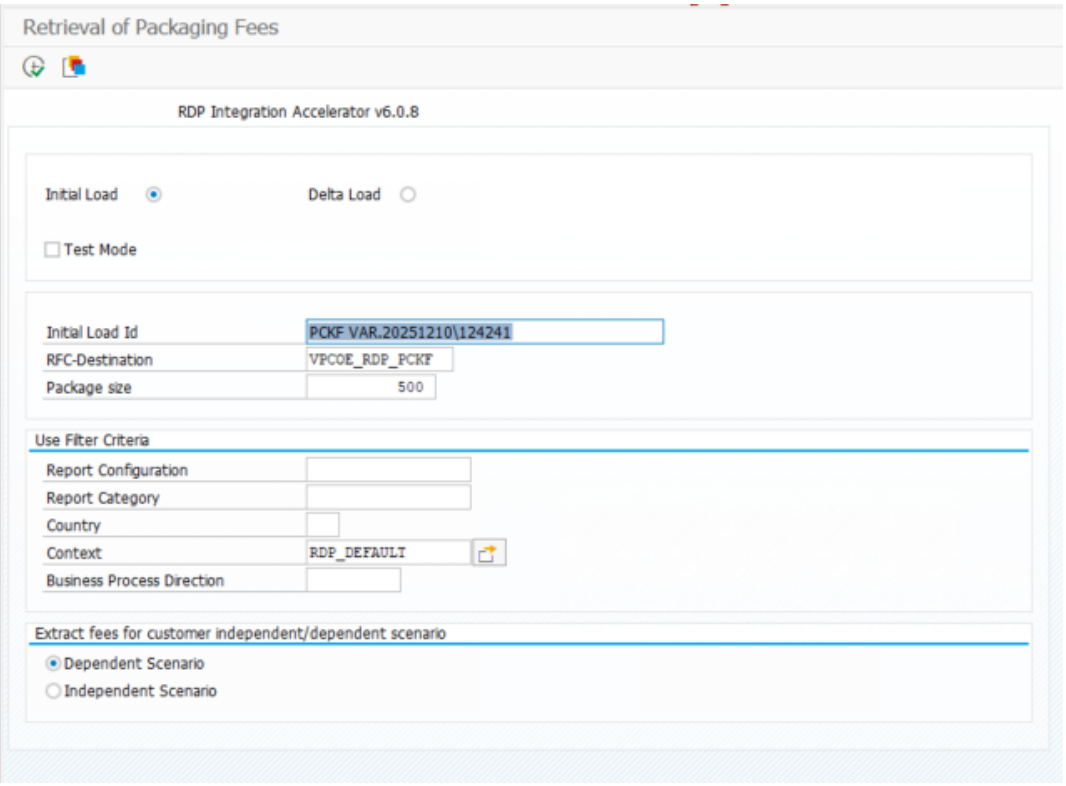
The Packaging Fee Retrieval Report /VPCOE/R_PCKF_RETRIEVAL handles the retrieval of entity data provided by the RDP Packaging Fee API.

Packaging fee retrieval is not supported in SDF API.

Both **API V1** and **API V2** are supported. Customers can select the API version by adjusting the **API path** in the RFC destination configuration.

When **RDP API V2** is used, the **Context** field is removed from the selection screen (as it is obsolete in API V2).

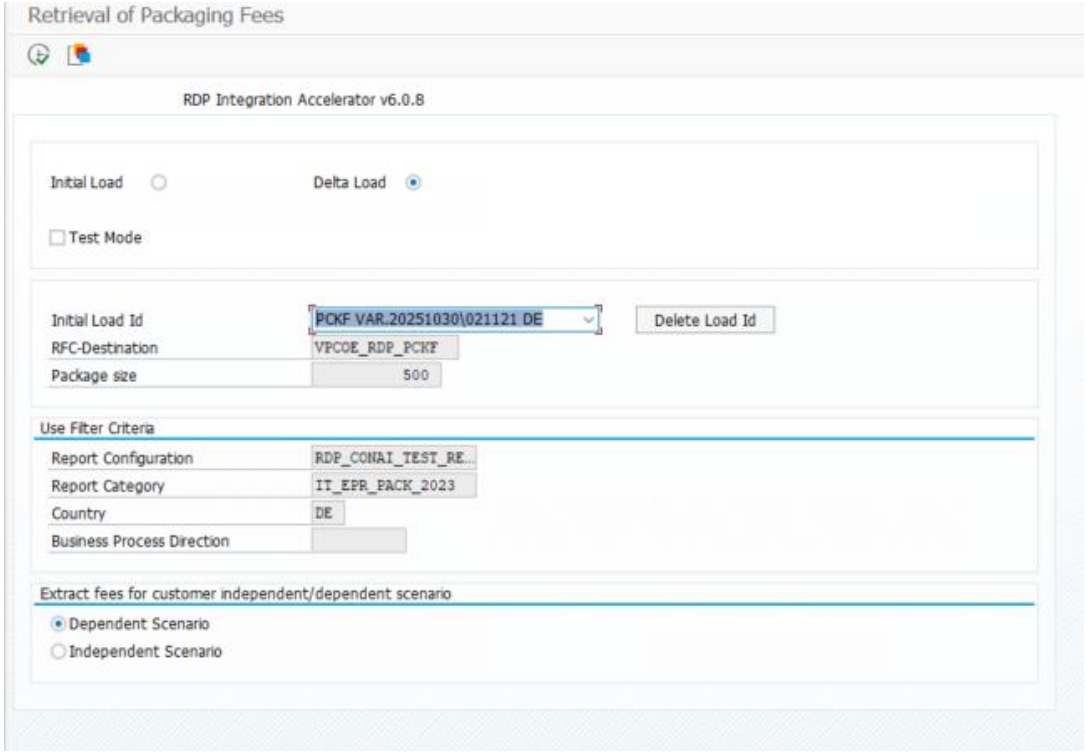
Initial Load ID has been introduced to enhance data extraction processes during **Delta Load,** supporting multiple delta tokens. During the *Initial Load,* selection parameters are saved and associated with a unique *Initial Load ID*.



To use the support for **customer role–dependent fees** in the API, it is necessary to select the *Dependent Scenario* option. This sets the includeCustomerRoleDependentFees parameter in the request to *true*.

- If includeCustomerRoleDependentFees is set to false, it means the attribute *ShipToPartyRole* is empty, and only customer role–independent fees are returned.
- If it is set to true, *ShipToPartyRole* is populated with the relevant customer role or left empty if no role is specified.
  For more information about available customer roles, see Customer Role.

When the **Delta Load** radio button is selected, then based on selected initial load Id (from the drop-down) all the selection parameters will be reused. In this mode, all selection parameters except *Test Mode* are not editable. During delta load, only those fees that have been updated since the last successful Initial or Delta Load of selected Initial Load ID will be extracted and transferred.



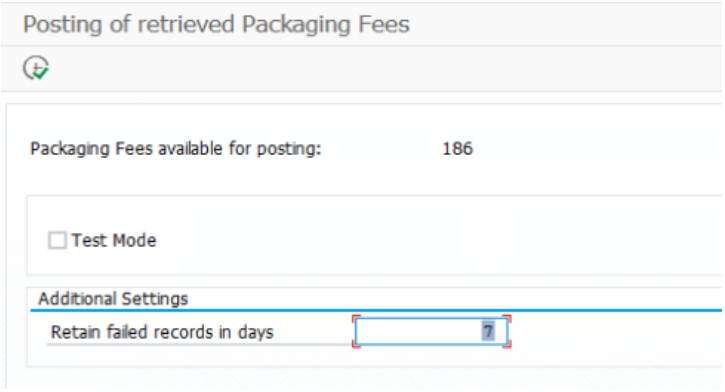
Example implementations for creating pricing conditions for customer-dependent and customer-independent scenarios are provided in the sample BAdI class `/VPCOE/CL_EXMPL_PCKF_CUSTOM`.

For the API documentation, see API Reference.

For details on the packaging fee extracted data please have a look here: Extracted Data | SAP Help Portal.

Packaging fee posting

The Packaging Fee Posting report /`VPCOE/R_PCKF_POSTING` is responsible to use the entities (calculated tax data) from the RDP for creating the pricing conditions and convert the product data to Material Classification for the product the individual record belongs to.



When executing Packaging Fee Posting report:

- The *Posting* report attempts to post all retrieved packaging fees (previously saved in cache table)
- If any errors occurred during the posting, failed records remain in the cache for the number of days specified in the *Retain failed records in days* field. This allows users to review and analyze errors.
- If this field is set to *0 days*, the failed records are deleted immediately from the cache/temporary table.

For packaging fee posting, customer is responsible to:

- Definition of condition tables, condition types, access sequence, pricing procedures
  - Define concept for SD & MM Conditions for Packaging Fees
  - Create Condition Table(s)
  - Add Condition Table(s) to Access Sequence
  - Assign Access Sequence to Pricing Condition Type
  - Add Condition Type to Pricing Procedure
- Implement available BAdI (Copy Sample BAdI provided) and adjust the filling for packaging fees into condition records
  - Define filters based on which the data shall be replicated
  - Extend the mapping with the exact pricing condition tables
  - Logic to split the data to multiple ERP systems (optional)

Replication of Customer Exemptions

The Customer exemptions retrieval report /VPCOE/R_PCKF_CUSTOM_EXEMPT is handling replication of customer exemptions which can be used for the creation of customer and report fraction-based sales and distribution pricing conditions for exemptions. Those replications are currently applicable only for Extended Producer Responsibility (EPR) reporting solution for Italy (CONAI). For more information about maintaining customer and company code exemptions, see Report Category Extension.

Note: If the Customer is blocked, it will not be returned in the response, even if it is applicable to the respective PackagingComposition. If the blocked Customer is the only one applicable to the PackagingComposition, then the fee with this packaging composition will be skipped from the replication.

Replication of CompanyCode Exemptions

The Company Code exemptions retrieval report /VPCOE/R_PCKF_COMP_CODE_EXEM is handling replication of company code exemptions which can be used for the creation of company code-based material management pricing conditions for exemptions. Those replications are currently applicable only for Extended Producer Responsibility (EPR) reporting solution for Italy (CONAI). For more information about maintaining customer and company code exemptions, see Report Category Extension.

Replication of Customer Roles

The Customer Roles retrieval report /VPCOE/R_PCKF_CUSTOMER_ROLE allows the extraction of customer roles. Replication of customer roles to the SAP ERP systems is used for the identification and creation of the correct customer role and product-based sales and distribution pricing condition.

Extensibility in Packaging fee reports

The packaging fee retrieval and posting reports and also the exemptions replications and customer roles dependent fees replications are using the BAdI /VPCOE/BADI_PCKF_CUSTOM. The BAdI enables the customer to implement customer specific extensions.

Interface/Class/Report	Description
/VPCOE/ES_PCKF_CUSTOM	Enhancement Spot for BAdI
/VPCOE/BADI_PCKF_CUSTOM	BAdI Definition
/VPCOE/IF_PCKF_CUSTOM	BAdI Interface
/VPCOE/CL_EXMPL_PCKF_CUSTOM	Example Implementation for BAdI Interface

Interface /VPCOE/IF_PCKF_CUSTOM:

Method	Description
GET_ENTITY_PROCESSOR	Returns the entity processor instance for the given entity type.

GET_TARGET_ENDPOINT_DEST	Gets the default destination name. This method can return the destination name which defines a http target to the RDP Test Client.

For logging in custom entity processor implementation, use application Logging Object /VPCOE/RDP, Sub-object PCKG_FEE.

/VPCOE/BADI_PCKF_CUSTOM – technical documentation

The Business Add-In (BAdI) /VPCOE/BADI_PCKF_CUSTOM that is assigned to enhancement spot /VPCOE/ES_PCKF_CUSTOM is used during the download and processing of packaging fee data from the API of SAP Responsible Design and Production (RDP). Its main task is to provide an extensibility mechanism to integrate customer specific implementations for the retrieval and posting of packaging fee data.

Before describing this extension mechanism in more detail, it is necessary to understand the concept of Entity Types and their respective Entity Processor implementations.

Entity Type and Entity Processor

The Packaging Fee API of SAP Responsible Design and Production (RDP) is providing different entities:

- Packaging Fee (PCKG_FEE)
- Applicable Business Partners (APL_BUSP)

These entities are named in the following as Entity Type. The processing of the corresponding entity data of an Entity Type is implemented by a so-called Entity Processor class. An Entity Processor class must implement the interface /VPCOE/IF_PCKF_ENTITY_PROC that contains the following important methods:

- INIT_PROCESSOR: This method is used to initialize the Entity Processor
- PREPARE_RETRIEVE: The preparation steps for the entity data retrieval are implemented in this method.
- RETRIEVE_PACKAGE: The implementation of the entity data retrieval is implemented in this method. For example, the retrieval of the entity data from the corresponding RDP API must be implemented in this method.
- TRANSFER_PACKAGE: The transfer of the entity data to its target is implemented in this method. For example, the conversion of Packaging Fee entity data to ERP pricing conditions must be implemented here in the Entity Processor class for Entity Type PCKG_FEE.

For each Entity Type there is already a base class available in the system that must be derived for a custom logic:

- /VPCOE/CL_PCKF_PROC_PCKG_FEE: base class for Entity Type PCKG_FEE
- /VPCOE/CL_PCKF_PROC_APPL_PART: base class for Entity Type APL_BUSP

The data retrieval from the RDP API (method PREPARE_RETRIEVE and RETRIEVE_PACKAGE) is already implemented in these base classes.

But for the implementation of the customer specific conversion logic of Packaging Fee entity data to ERP pricing conditions, a custom implementation must be provided by deriving from class /VPCOE/CL_PCKF_PROC_PCKG_FEE and overriding the method TRANSFER.

BAdI Methods

The BAdI definition is active in the standard system. It contains the following methods from interface /VPCOE/IF_PCKF_CUSTOM:

- get_entity_processor:
  This method returns which entity processor is used for the corresponding entity type. The BAdI mechanism allows to exchange a processor, for example, when a customer specific conversion of Packaging Fee data to ERP Pricing Conditions is needed.
  If no custom implementation is returned, the Entity Processor base class is used instead.

- get_target_endpoint_dest:
  This method will return the end point destination of the RDP source system. If this method is defined, the returned value will be displayed as the default RFC source in the retrieval report selection screen. If no specific value is returned, the default destination VPCOE_RDP_PCKF is used.

BAdI Implementations

The BAdI implementation must be defined by the user and is not provided by SAP. As an example, class /VPCOE/CL_EXMPL_PCKF_CUSTOM is linked as a BAdI example implementation.

For more information about settings for a BAdI, such as filters, single or multiple uses, see the Spot Element Definitions tab in the BAdI Builder (transaction SE18).

Example implementation class: /VPCOE/CL_EXMPL_PCKF_CUSTOM

The example implementation class of the BAdI contains an entity processor implementation (commented out) for the Packaging Fee entity provided by a local class (lcl_exmpl_pckg_fee).

During the posting, the packaging fee records are converted to pricing conditions of Condition Type ZRDP. Since this condition type currently holds 3 different condition tables (identified by separate keys) the according condition table (key) is selected with regards to the available organizational data in the packaging fee.

HOW TO CHECK REPLICATION RESULTS

When running the RDP reports, the results are displayed on the screen and also are logged in SLG Object **'/VPCOE/RDP'.**

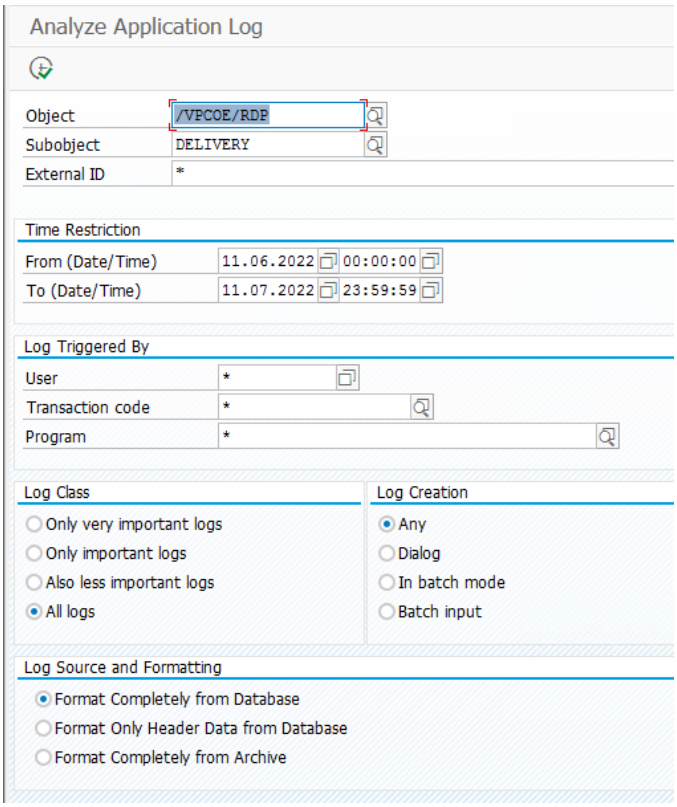
Results on the screen

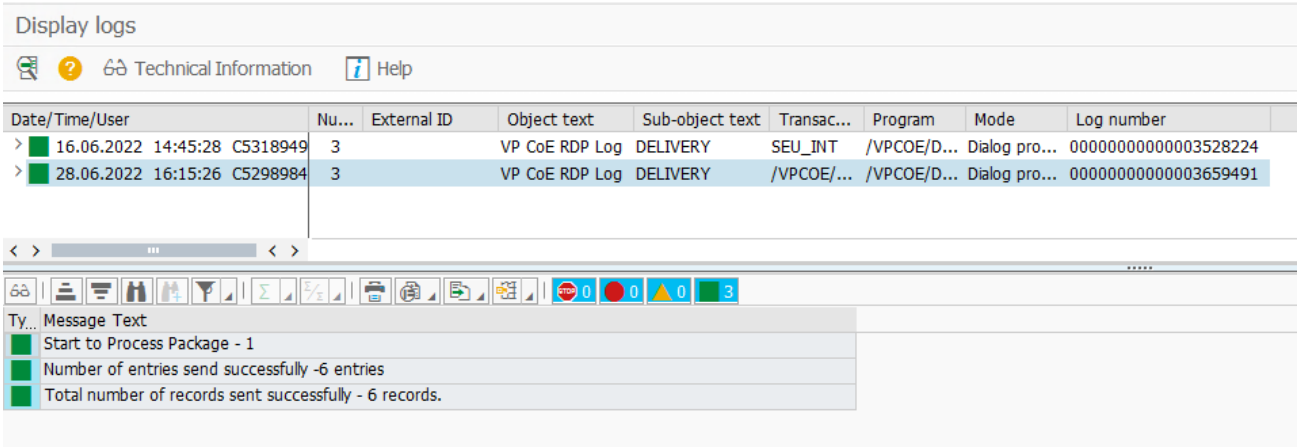
When running any RDP integration report in foreground mode the replication results are visible on the screen in a similar way as below:



Application log results

- After running the RDP Integration reports you can see the results on the logged object (even when running in background mode):
  1. Open transaction '*SLG1*'. Use object **'/VPCOE/RDP'.**
  2. For 'Subobject' use the required service ID or '*'. For packaging fee replication logs, use the sub-object '`PCKG_FEE`' .





In case of failed replication that can be resent to RDP (e.g. technical issues during replication) the payloads can be resent using Resending Failed replication report.

In case user needs to trace replicated objects, the Trace Maintenance Report should be used.

KNOWN TECHNICAL ISSUES AND TROUBLESHOOTING

Troubleshooting Guide – RDP Integration Accelerator

This section provides guidance on how to analyze and resolve common issues during data replication between ECC/S/4HANA and RDP.

1. Check Results on the Execution Screen

After executing a replication report:

- Review the **result** displayed on the screen.
- Verify:
    - Number of selected objects
    - Number of successfully sent objects
    - Skipped entries
    - Error messages (if any)

If inconsistencies are detected:

- Check selection parameters used.
- Confirm whether Initial or Delta Load was executed.

2. Application Log (Transaction SLG1)

For detailed technical messages:

1. Open transaction **SLG1**
2. Use:
    - Object: /VPCOE/RDP
    - Subobject: relevant service (e.g., DELIVERY, PRODUCT, DOC)
3. Execute and review log entries.

Use SLG1 to:

- Analyze HTTP errors (400, 401, 403, 405, etc.)
- Identify authentication/token issues
- Review payload-level validation errors returned from RDP

3. Trace Maintenance Report

If the parameter **ENABLE_TRACE = X** is active:

- Run report /VPCOE/TRACE_MAINTENANCE
- Review:
    - Selected objects
    - Skipped objects
    - Technical replication issues/error messages returned by RDP

Trace is useful for:

- Validating selection logic
- Understanding why and what objects were skipped

⚠ Recommendation:
Clean up old entries regularly to avoid performance degradation.

4. Failed Payload Monitoring

If payload transmission fails:

Run report:
/VPCOE/REPROCESS_PAYLOAD

You can:

- **Display Request** – view JSON payload for analysis
- **Repeat Request** – resend selected payloads manually
- **Automatic Mode** – resend all failed entries via background job
- **Delete Entries** – remove obsolete or irrelevant failed payloads

Important:
If the failure is caused by incorrect or missing master data, simply resending the payload will not resolve the issue. Correct the data first, then re-execute the original replication report.

For advanced debugging:

- Copy the JSON payload and test it using tools such as Postman/Bruno

5. Background Job Monitoring

If replication runs in background:

1. Check transaction **SM37**
2. Review:
    - Job status
    - Spool output
    - Short dumps (if any)

Spool logs typically contain:

- Number of selected records
- Error summaries

- API call results

6. Common Error Categories

Authentication Errors (401 / 403)

- Verify OAuth RFC destination
- Check client ID / secret
- Validate SSL certificates

Method Not Allowed (405)

- Verify API path prefix
- Confirm API version (v1 vs v2)
- Check correct HTTP method

Bad Request (400)

- Review JSON payload
- Check mandatory fields
- Validate field mappings in BAdI

7. Performance Issues

If performance is degraded:

- Check number of entries in failed payload table
- Clean up old payload entries
- Review trace table size and perform cleanup
- Verify package size configuration
- Confirm parallel work process configuration (WPCount)

8. When to Re-run Initial Load

A new Initial Load is recommended if:

- Data correction was performed
- Structural mapping was changed
- Custom BAdI logic was modified

Note: A new initial load overwrites data in RDP.

Recommended Troubleshooting Flow

1. Check execution screen results (if foreground)
2. Check job spool (if background)

3. Review SLG1 logs
4. Analyze failed payloads (in ReProcess Failed Payloads report)
5. Review replication trace (if activated and if deeper analysis needed)
6. Correct data or mapping in the BAdI implementation (if any)
7. Re-run replication

Known Technical Issues

Technical Issue/ restriction	Solution	Valid from / to SAP Software Components
Errors when installing the general object CR, due to missed objects in standard code (ex. MEINS_D being not available in the system)	SAP Note 2526405 - /UI2/CL_JSON Corrections released in 2018 should be implemented	SAP_UI 750/751/752 UI_700 200
Multilanguage capabilities are not supported (only English language is supported for now)		

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